



# MY PROGRAMING PROJECT



Oscar Winter

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## Analysis

### My idea

a short summary of what my idea is. My idea is to do a game that will open with a login screen where the user first has to either sign in or register. Once signed in, I'm going to have two game modes: either the user has to pick the platformer, which is more of a friendly experience where the user progresses through levels facing smart enemy AI, and I'll be using games to get inspiration. Then for the other game mode it will be a more intense speed run where you're up against an AI model I trained myself to solve mazes, and there will be several difficulties to select to match the players skill level and requirements. There will be a scoreboard for the time taken as well as the coins collected along the maze. I will be basing my maze design after Pac Man, but it will only have a single route that the player must solve. I want to do this because I'm very passionate about getting the younger generation interested in AI and just how intelligent it can be. The genre of my game will be a mix of a puzzle game and a platformer, which includes some action and fighting.

### Stakeholders

A stakeholder is a person with an interest in or concern in something, typically a business.

My stake holders will be people who are interested in playing and even helping to develop or test my game as I develop it. My game can be for all ages as it is simple to play and caters to all user abilities; however, my main target audience is due to the friendly graphics. My main audience is anyone under the age of 18, as my goal is to interest the younger generation in AI and its uses. And the younger generation can relate to and gain a large amount of interest in games.

My first stakeholder will be called Tester 1. He is 17 years old and, through his childhood, has played many platformer games as well as puzzle games, which match my game perfectly. So, my goal will be to create a game that the tester finds enjoyable to play and sparks an interest in him for how the AI works.

My second stakeholder will be called Tester 2. He is 15 years old and has only ever played puzzle games but hasn't played games online. He said he has played some of Pac Man before and has done many maze puzzles. So, my aim with tester two is to try and get tester 2 to solve the maze against AI, hopefully also getting him to really enjoy beating the AI. I intend for my tester to help me iteratively improve and change the game so that it improves over time and is able to meet all my criteria. For example, I will ask them questions throughout the development process about the current features I'm adding, asking what they like, what they don't, and what they'd like to see added. And I will take actions based on stakeholder decisions.

### Computational thinking

### Thinking abstractly

When it comes thinking abstractly that means the difference between the game and reality for example one place where I have used abstraction is that in reality someone solving a maze won't be picking up coins as they go through the maze another example of this could be that in real life the walls in the maze are often hedges but I'll be simplifying this to just some walls . my last use of abstraction is in the dimensions and how in person doing a maze is in three dimensions, but I'll be reducing this to 2 dimensions, so the user is able to see a bird's eye view of the map.

### Thinking ahead

The computational method of Thinking ahead refers to the developer being able to see potential issue with the inputs and output in certain areas of the game before they have happened. In my case I'll be using thinking ahead in a few places I'll be using validation like in the sign in so that when the user enters there details it makes sure to check there are in the system. And when creating their account, I am going to apply checks to the password field in order to make user the passwords secure. the first check I'm using is a length check to make sure the password is at least 8 characters long. the second check I'm using is a presence check to make sure the password includes numbers, symbols and capital letters

There is another meaning to thinking ahead as well which the developer is thinking about how code will operate and what knowledge will be needed. and that if the code and knowledge required is beyond the scope of the developer, they will identify this and come up with a plan to learn what they need. An area of my game I'm not sure how ill code is the login system. but I have experimented with the use of file handling in code before. but to solve this I plan of doing some experimenting in unity with taking inputs of fields and using them in visual studio. I would like to develop some basic ground for my sign in system after the practice.

### Thinking procedurally

- What is thinking procedurally? Thinking procedurally is using decomposition and subroutines to organise and break down tasks into small more manageable sections, so things don't get unorganised and complicated to develop.
- When coding scripts for the game I will separate a lot of the scripts for instance the player movement code and the triggers caused by the player will be stored in separate scripts which I will link together in addition to this I will also be using sub routines this way long chunks of code won't become unorganised for example in the triggers and collisions I will have in separate sub routines which are referenced in main
- Another case where I'll be using decomposition is when creating the scripts for the log in system to break down a task, I'm not completed prepared for into small tasks making it more manageable
- Some of the tasks I'll be splitting up:
  - Login system
  - Leader board
  - Ai gameplay
  - Player movement
  - second game mode platformer

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- as these are all of what I consider my larger tasks

### Thinking logically

- Thinking logically is when your program (in my case game) gets to any point where a decision must be made an example of this could be in blackjack where the player has to make the decision to either stand or hit
- Where this will be seen in my game is in a few different places to start while playing through the maze the player will of course encounter when the map splits off into several different sections and at that point the player must make a choice to go in one of the directions the other case will be in my game menus the first where the user has to pick between signing in or creating an account then after this having to choose to start to game or to look at the leader board and the last example is after player death weather or not they want to play again or go back to main menu

### Thinking concurrently

- thinking concurrently is when parts of the program are able to be solved at the same time an example could be in a two-player game both users' actions must be solved at the same time
- this will be implemented first in my maze where the player is racing the AI meaning their inputs have to be managed concurrently in addition to this each of these times and scores must also be all calculated at the same time
- when the player finishes there match the system must recall past scores then semiseriously work out each of these places in the hierarchy for the leader board system

### Research item 1

- I'll be testing a game for my research of which being the google PacMan for its 30<sup>th</sup> anniversary as it doesn't require purchasing however still retains all the functions of the original game where the goal is for the player to collect all of the points around the map without being caught by the ghosts the player gets three lives and after the player loses all three lives there score will be displayed on screen. I'll be marking and assessing what I do and don't like from the game based off of the following criteria (enemies, power ups, map layout, UI, controls)
- Enemies in PacMan the main enemies are 4 ghosts red, yellow, blue and pink these 4 chase the player around the map with the goal of not letting the player collect the dots and once a power ups consumed it now allows the player to get their own back on the ghosts by eating them giving the player a large amount of points

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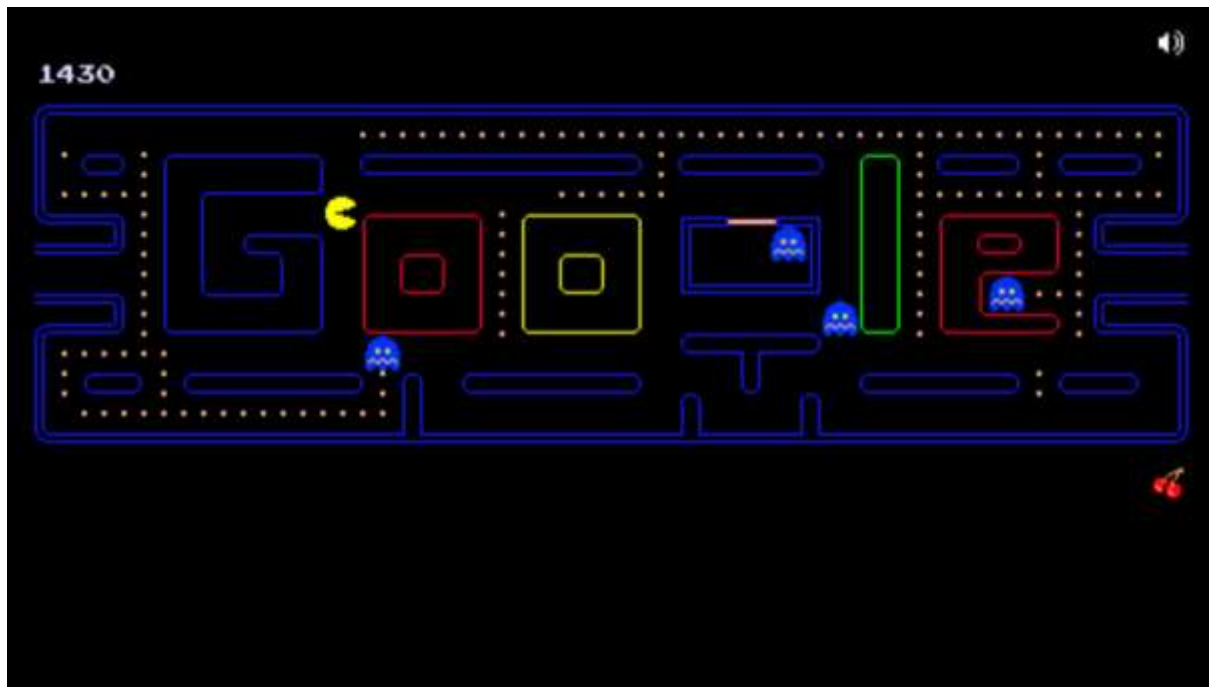
- controls the player in paceman is controlled when he comes to a bend by using the up, down, left and right keys on the keyboard he turns in the direction you indicate however beyond this there is no other controls as all power ups are used instantly



- Power ups in PacMan the power ups are limited and consist of a larger dot placed at 5 points throughout the map each of which at the corners and one near the centre after consumption the play is enabled to be able to eat the ghosts indicated by their visual change in colour to a dark blue however there is a arguable power up and that's cherry and strawberry in this version of the game they only act as a way to get more points than your typical dot

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- UI during game play the players current score is displayed in the top left-hand corner and player starts with 3 lives shown in the bottom left-hand corner by the number of pacmen on screen

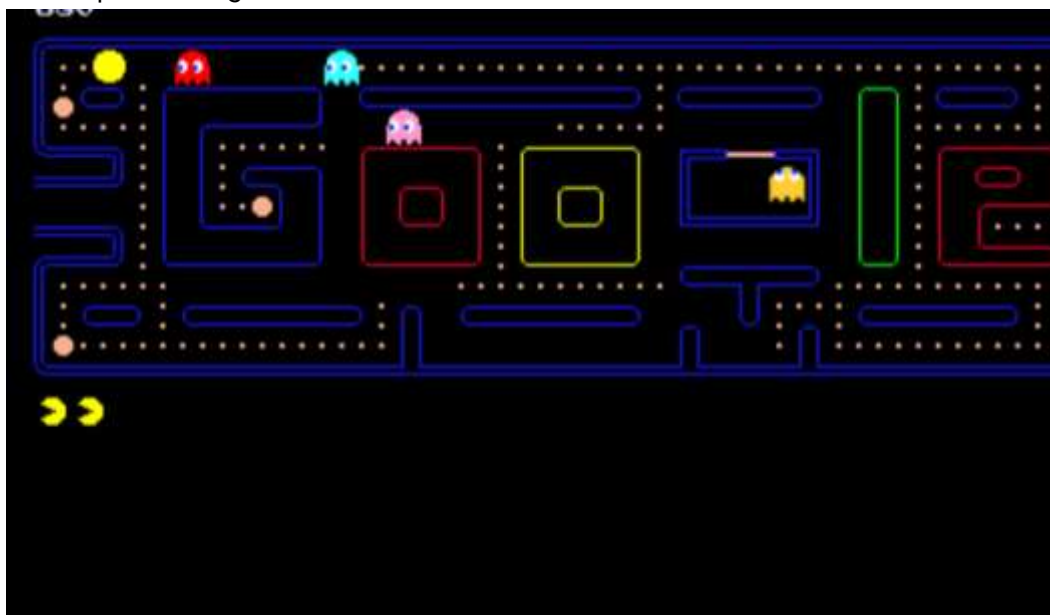


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### MY THOUGHTS AS WELL AS THE TESTERS

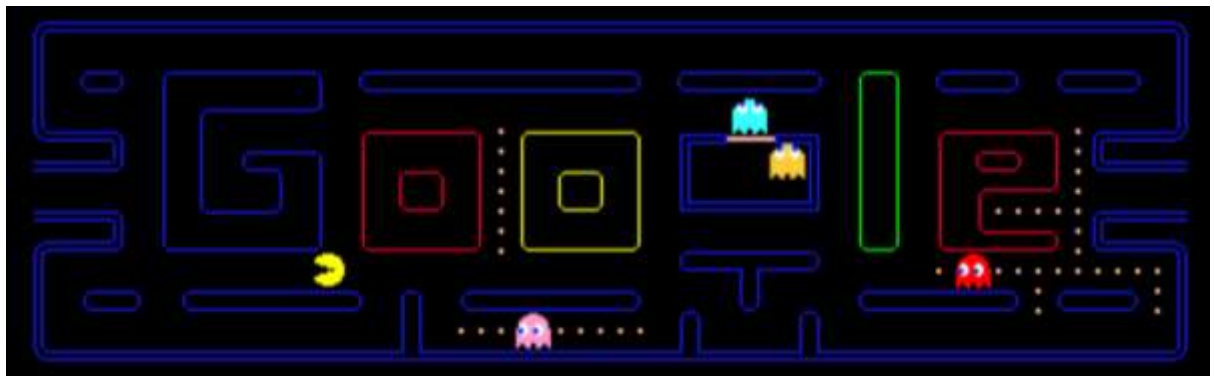
- For the enemies in the google PacMan they are very intelligent for example while playing id get in the view of red and red would chase me, I did enjoy this added pressure it gave the game a sense of urgency which I did enjoy and I would like to integrate this into my game perhaps in my alternate game mode where my characters will use pathfinding



- As see in photo above
- As for the power ups within mac man 30<sup>th</sup> anniversary tester 1 said they are limited to the larger dots you see on the screen shot above and that they think it's a nice addition to the game how the tables are reversed on the enemies as after consumption the player is now able to eat the ghosts indicated by their changed appearance .however tester1 is not a fan of the lack of diversity in the power up's . something I'd like to change for my game it makes it appeal more to tester 1(shown in the image bellow)



- Now for the next criteria the map this goes for the google 30<sup>th</sup> anniversary version as well as the original the map is a birds eye view of the whole map this meant the learning curve of the game was much easier as I could see the paths in my game I'd like to have at least 50% of the map visible and have the camera follow the player smoothly .the full view of the map is shown bellow



- My last criteria was the UI the in Pac man I thought was really good and there was nothing I would change about it while playing me as well as tester 1 and 2 found it was easy to visually see your score and how many lives you had left so I'll be listening to their input and creating a UI inspired by this .you are able to see this in the top left the score and bottom left the number of lives

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it's good for me to take responses from my tester after they have played the game as they fit my games demographic and are my target audience allowing me to improve and develop my game to cater them which should be representative of the majority of my target demographic

| Things that the testers liked | Why they liked it?                                                                                                                                                                                          |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Power ups                     | Tester 1 said "they enjoyed the idea of power ups and being able to get their own back on the ghosts" however there are some negative comments they will be discussed in the thins they didn't like section |
| UI                            | Here both testers 1 and 2 said the UI was easy to navigate and clearly showed all player metrics such as score meaning I'll definitely be using inspiration from this                                       |
| Map                           | I like the map layout being a bird's eye view because it meant you didn't get lost in the maze id like my game to not show the whole map to make it a bit challenging                                       |
|                               |                                                                                                                                                                                                             |

| Things the testers like | Why didn't they like it?                                                                                                                                                                                                                        |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Enemies                 | Tester 2 said the enemies at points where too difficult to evade making the game frustrating and unenjoyable to play. I'll be careful not to make the game too difficult as I want to get the younger generation interested rather than annoyed |

|           |                                                                                                                                                                                                                    |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Power ups | Tester 1 was saying they did like power ups but then they went on to say that the power ups were very limited and would like to see a larger range to make the game more interesting. which I plan on implementing |
| controls  | Tester 2 didn't like the arrow keys as the keys as they were too cramped together and in an awkward position on the keyboard, so I'll be moving this to the more preferred WASD controls                           |
|           |                                                                                                                                                                                                                    |

## Research item 2

For my second research item I've chosen to play a game called hollow knight this is because I found that it uses pathfinding algorithms as well as reinforcement learning in its AI as well as this already are in the game and have played a fair amount of it the game's goal is for a character called the Knight to move throughout the map defeating bosses and other enemies the player gets 3 lives at the start of the game but can expand on this as the game progresses for this research item I'll be focusing mainly on how the AI works within the game as well as a few of the mechanics more specially (Pathfinding AI, reinforcement learning AI and then game mechanics as a whole) I will be testing the both forms of AI just myself then I'll have tester one and two help with the reviewing of the game mechanics

### Pathfinding AI



As displayed above you're able to see the enemy following the player this appears to be in a fluid motion and is done using something called A\* pathfinding where G score and F score are calculated and combined create an F score which indicates the cost of the path to get to the player throughout a grid of nodes the enemy will then start moving towards the nodes which has the lowest cost I believe that this advanced method of enemy following will be very useful to implement into my own project specifically in the platform section as it would be good for my enemies to follow my player a path influenced by the current game state rather than making the enemy's actions predictable and instead unique

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## Reinforcement AI

As hollow knight doesn't directly use reinforcement learning in the source code for this I'll be researching a repository which can be used in the game to have the player navigate through the game using reinforcement learning the repository is called hollow knight RL reinforcement learning is where an AI starts off by triggering random actions in the hope it will achieve something correctly also known as reward as a I gained more and more rewards it will start to learn patterns and certain actions which give it rewards allowing it to progress.

I believe I want to implement this kind of learning from my maize gaming AI as I can't just have a single type of AI from trying to educate, I need to show several different cases of AI and what it's capable of this could also make it easier for the younger audience to understand as is similar to how we learn.

## Game mechanics

After both test 1 and test 2 testing the game both of them thought that the in-game mechanic of being able to regain health. This happens when the player defeats an enemy, they drop something called soul which then allows the players we're going to health I think I'd like to incorporate this into my game by using some kind of hidden power up that will increase the players remaining lives by 1. Comments by Tester1" I personally like the fact that when I got low on health, I was able to collect soul to regain my health allowing me to play the game for longer". comments by tester two "I also enjoyed being able to regain my health points I'd also like to add that I enjoyed the puzzle so solving involved with certain sections of the game". My response: ill plan to make a new power up that will allow the player to regain health and also take on the advice for the game design process (health and soul bar shown below)



## My solution

- when opening my game, you will first be greeted by a sign in menu giving the user to option to sign in using a password and if I get time also a username if the user doesn't have an account, they will be able to create one and then sign in after this.
- on the next screen the user will be presented with 3 options either the leader board or the maze racing game mode or the platform game mode
- when selecting the leader board option, the user will be presented with the top 10 player scores of all time on the racing game mode.

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- when selecting the maze racing game mode, the user will be dropped into a start screen where they will be presented with the option of either playing or going back.
- if the user clicks play the game will start
- when the game starts the timer will start
- so will the players score
- the player will collect coins to increase their score.
- the player when entering the game has the goal of collecting as many coins as possible in as short a time as possible with power ups to collect along the way giving them an advantage over the AI.
- The power ups I intend to use are:
- Speed boost increasing the speed of the player hopefully allow them to reduce their finish time however it will be difficult to control so there's a high-risk high reward for the player
- ill also include a power up which doubles the players current score and because this is such a big power up ill make it hard to reach and once again a risky play
- and last of all ill add a de-buff for a reduction in speed if the player hit it will make this easier to avoid keeping the game balanced
- There will be a UI displaying the users current time as well as point score in each of the left-hand corners this allows the user to easily visually see their progress as they faster while the games happening
- after either the player or the ai finishes, the game will finish giving the player a score and a time and then sent back to the play or back screen.
- then if the player selected the platformer game mode, then they will be places in the platformer with enemies to navigate through while collecting coins until they reach the end where there greeted with the finish screen and the option to go back to the main screen
- as well as on the racing in the platformer ill also have a UI which displays the users current score and their number of lives remaining

#### hardware and software specification

| Minimum software                                       | justification                                                                                                                                      |
|--------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Visual studio editor 2.0.11 or newer                   | This is needed in order to fun the actual scripts in the game                                                                                      |
| Unity 2019.4 or newer                                  | As this is the oldest version of unity compatible with visual studio editor 2.0.11 used to build and compile the game scripts into a visual output |
| Windows 7, windows 10, windows 11 64-bit versions only | These are the operating systems unity is supported on as well as visual studio editor                                                              |
| Graphics Api DirectX 10, DirectX 11, DirectX 12        | As this is required by unity to give a graphical output                                                                                            |
| Minimum hardware                                       |                                                                                                                                                    |

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|                                                                    |                                                                                                                     |
|--------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| CPU:4 cores at 2+ GHz                                              | Required as the game is very graphically intensive and is required to run both the players character and the AI     |
| RAM:8 GB                                                           | Required to run widows OS then also have 4 spare to run the game                                                    |
| Graphics capable of DirectX 10 to 12 and above 4gb of video memory | This is because unity requires DirectX 10-12 and needs 4GB of video memory to support the graphical load from unity |
| Storage: 3GB                                                       | 3GB is required to store the game on the specified device                                                           |
| keyboard                                                           | Required for the user to give movement inputs as well as character inputs                                           |
| Monitor                                                            | Used to output the games graphics to the user                                                                       |
| Mouse                                                              | Required to give inputs on buttons which on mouse press is activated                                                |

| Achievable success                                                                                                                    | justification                                                                                                                                                                                                       |
|---------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The player must be able to log in using the login screen                                                                              | A log in is required for the player to be able to have their score up on the leader board as a username is required for display                                                                                     |
| The player must be able to use their mouse to navigate the UI menus of which being 3 either leader board or one of the two game modes | As a mouse is common and sensible input that everyone has to then access each of the menus                                                                                                                          |
| The platformer game mode must have intelligent enemies                                                                                | As stated on page 9 the testers didn't like how challenging the game was to beat but I don't want to. Ake it too easy as there needs to be some drive behind the game so I'll make the enemies somewhat intelligent |
| The controls in the platformer will use A and D to move left and right and c to jump                                                  | This is common for most basic platformers, so I'll be using a similar approach                                                                                                                                      |
| There will be at least 3 power ups in the platformer                                                                                  | As tester one said they enjoyed however they didn't like the lack of power ups therefore I'll be using 3 to give tester one this added diversity                                                                    |
| The player will be able to collect coins as they progress through the platformer                                                      | This is so the player can get scores which they can aim to beat each time they play                                                                                                                                 |
| The platformer must have 3 levels minimum                                                                                             | Depending on the time constraints ill at least build 3 levels to the game as I think this is a good number to start with                                                                                            |
| The platformer must have a live UI updating score and lives                                                                           | This is as both tester 1 and tester 2 said they liked seeing the live UI as it helped                                                                                                                               |

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|                                                                                         |                                                                                                                                                                                                                                                                                      |
|-----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                         | them keep track of their current lives and score                                                                                                                                                                                                                                     |
| The game must have an easy learning curve                                               | Games that have a high learning curve are often unenjoyable to play so I'll have my game have an easy learning curve to make it appeal to the younger ages                                                                                                                           |
| The player must have 3 lives                                                            | Testers 1 and 2 had no complaints or comments on the number of lives so I'm going to use 3 lives as it this seems to be a good indicator this is a good number                                                                                                                       |
| The leader board must have the top 10 scores along with the username of the player      | I asked tester 2 what they would like the leader board to look like and they requested it to have the top 10 peoples scores along with their username                                                                                                                                |
| There must be exit buttons on every page                                                | The player needs a way to get out of menus or even quit the game if they need to therefore require a back button to do so                                                                                                                                                            |
| The player and the AI must have the ability to race one another at the same time        | This is so the player can see in real time weather they are beating the AI or not                                                                                                                                                                                                    |
| The player must get at least three power ups / debuffs where the AI gets a maximum of 2 | As tester one said they enjoyed however they didn't like the lack of power ups therefore I'll be using 3 to give tester one this added diversity but the reason why the AI gets less powerups is to keep the game balanced as the AI smarter than the user (depending on difficulty) |
| There must be a UI system to display the players time and score in real time            | This is as both tester 1 and tester 2 said they liked seeing the live UI. This would also allow the player in this game mode to see if there on track to beating any of the top 10 scores                                                                                            |
| There must be a visual indicator the player has gotten a power up / debuff              | As my target audience is younger, they may not have noticed they have gotten a powerup / debuff the visual indicator makes it clear and easy for them to see                                                                                                                         |
| The AI must have 3 levels of difficulty (noob, intermediate, expert)                    | This is so that my game caters to all ages of my target stakeholders to different levels of ability                                                                                                                                                                                  |
| The players game data must save                                                         | I spoke with tester 2 and they said they wanted a game save so they can enter the same account and improve on their own personal scores                                                                                                                                              |
| Controlling the player in maze racing game mode                                         | This will be done as specified before using the WASD to relate to the x and y positions of the player as tester 2 said they didn't like how cramped the arrow                                                                                                                        |

|                                            |                                                                                                                                               |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
|                                            | keys are as well as the unnatural position on the keyboard                                                                                    |
| My game must be suitable for ages under 18 | This is because this is my specified age range for my stakeholders and my goal is to have my game be appropriate and appeal to this age range |
|                                            |                                                                                                                                               |
|                                            |                                                                                                                                               |

## Limitations of proposed solution

### A\* Pathfinding algorithm

Limitation – the coding of the A\* pathfinding algorithm

Description - this is the algorithm used to generate a list of the nodes that form the most optimal route through the graph from the enemy to the player.

Justification – as in the current A level specification the A\* Pathfinding algorithm is only described with what it does and how to perform the operations of the algorithm this means that coding the algorithm is not covered For this reason I will need to research how the A\* algorithm works in more depth as well as some code I wouldn't have used before an example of this is to set values to infinity like in the operations show in the specification and more

### Training Reinforcement AI

Limitation – the development of the reinforcement AI

Description – this will be what's used to train my true AI that will be used to solve the maze training for different periods of time will give the effect of different levels of difficulty

Justification - The training of my reinforcement AI is a limitation as in the current a level specification only the concept of general AI is introduced. Therefore, I current don't have the skills to develop I reinforcement AI so I'll have to research how to develop one as it would require me to program large neural networks will have to search for a plugin that would allow me to continue the development. After this it would then require further research into the commands within the plugin as this will be code ive not see before.

### Complex file handling in C#

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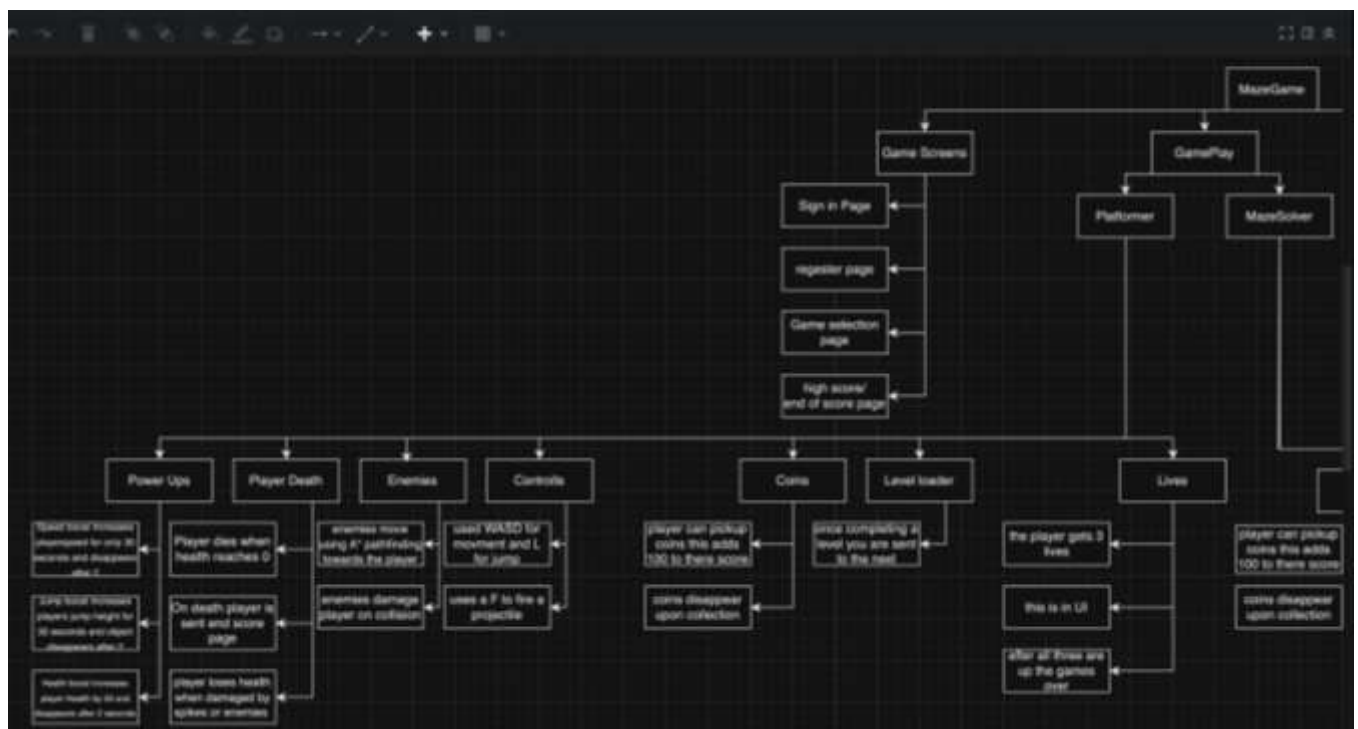
Limitations - the use of complex file handling for development in the login/registration system and the leader board system.

Description – this will be used in the storing of current player details and comparing to inputs as well as used to add new player details at registration and the displaying of the top scores

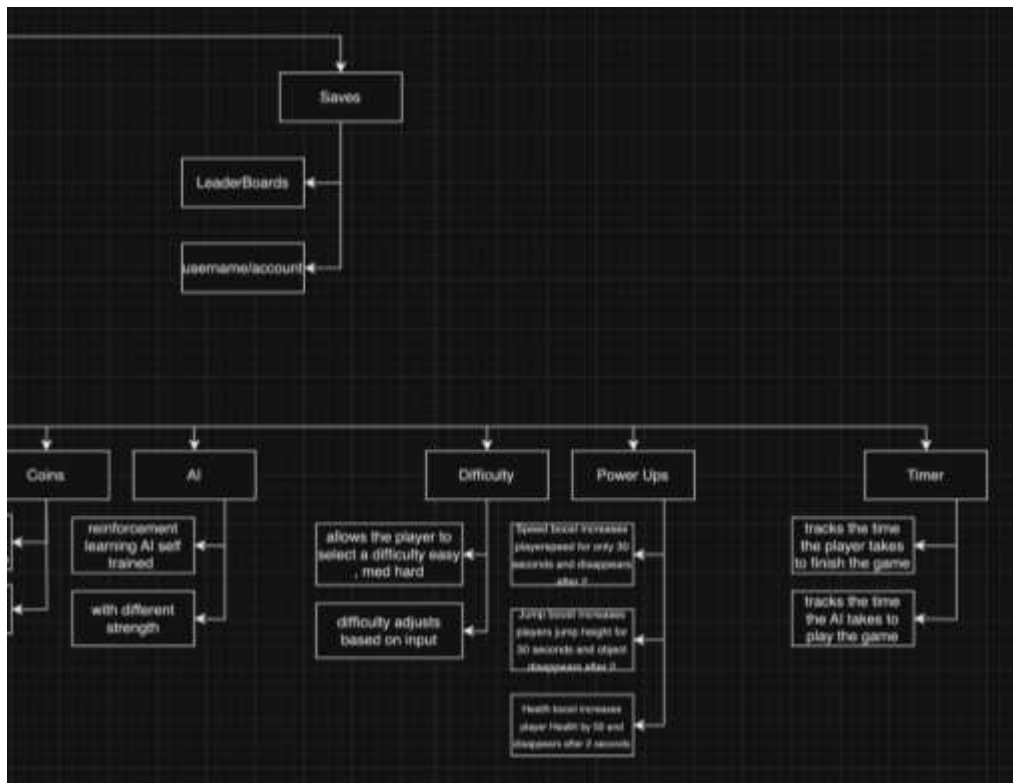
Justification - Complex file handling is a limitation as currently in C# I have only covered basic file handling. it would require much research, but I'd need to look at some examples of how people have done it before as this won't require any new coding knowledge.

## Design

After having researched my games along with my testers and after accumulating the feedback I have outlined my user requirements and have formed a systems diagram of how my program will be structured as the systems diagram allows me to tackle one small problem at a time. Shown below



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## Game Screens

Here I will outline and show what each game screens functions will be and where each element of the screen will be placed however the colour maybe vary in my actual design and whether buttons have sharp or rounded corners, but the basic principles will stay constant

### Sign in page

- This is the screen the user will first be greeted by where they can enter their username and password then click the sign in button to go to their account
- Then if the user doesn't have an account there's a button for the user to press and go to the register page

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A screenshot of a web form titled "Maze Escape". The form is centered on a light gray background. It features two input fields: "Username:" and "Password:", each followed by a white rectangular text box. Below these fields is a blue rectangular button with the word "Register" in white text.

- As you can see in my first draft, I wanted to keep it nice and simple as my product is aimed at a younger demographic as it would be simpler concept for them to understand.

### Register page

- Once the user has pressed the register button on the page they will be greeted with this page
- Where they can now enter their desired username and password
- After this they then click create account and it will be saved to the file holding account sign in information
- They also now have a return button shown clearly and large so it's easy to see in the top left-hand corner so after they create and account, they can then sign in to it on the sign in page

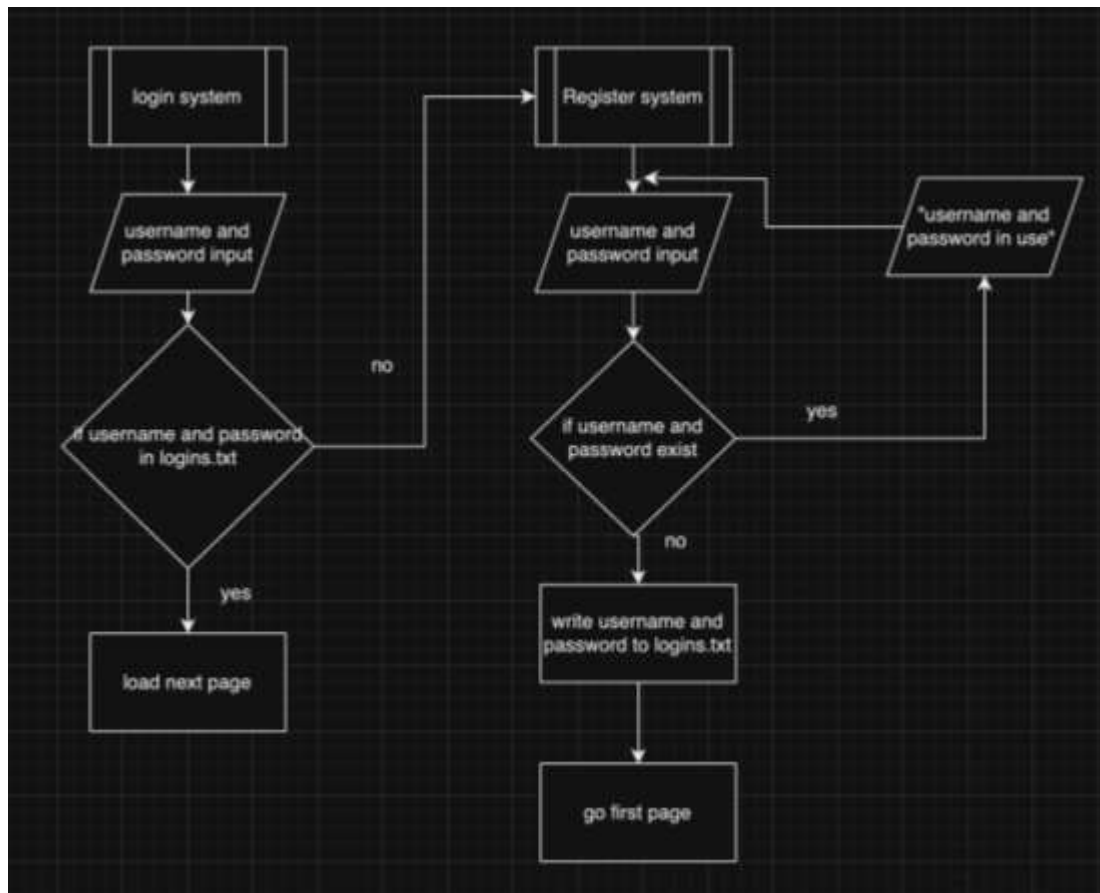
A screenshot of a web form titled "Register". The form is centered on a light gray background. In the top left corner, there is a blue rectangular button with the word "Return" in white text. The main title "Register" is centered at the top. Below the title are two input fields: "Username:" and "Password:", each followed by a white rectangular text box. At the bottom center of the form is a blue rectangular button with the text "Create Account" in white.

- Once again, I wanted to maintain the simplicity for the reason of the young demographic

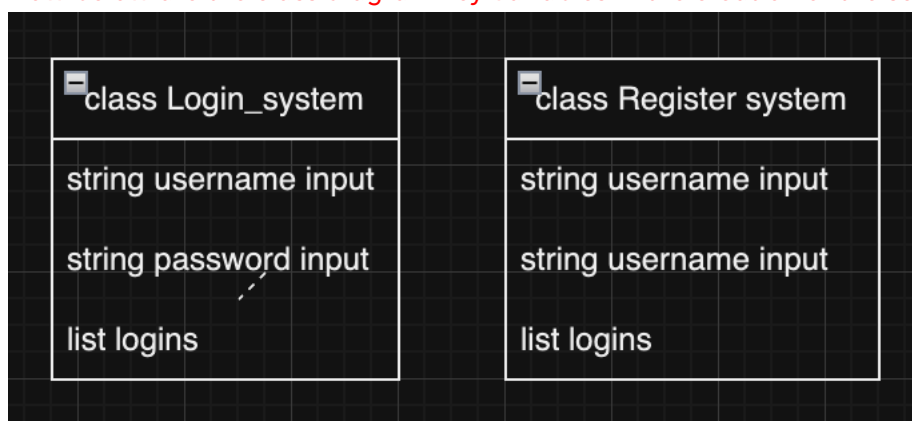
Candidate Number:3879

Candidate name: Oscar Winter

Below is the flow chart for both the login and register system in one this as they will work side by side



Now below are the class diagram key variables in the creation of the script



This outlines key variables that will be used in the respective classes how they are used can be seen in the flow chart above. The two input will be inputted using text boxes and the data structure used is a list and will be the text file being iterated through and added to the list. Validation will be used at this stage in the registration to check the username doesn't match any that are currently in use and that the password has 8 character and a capital letter.

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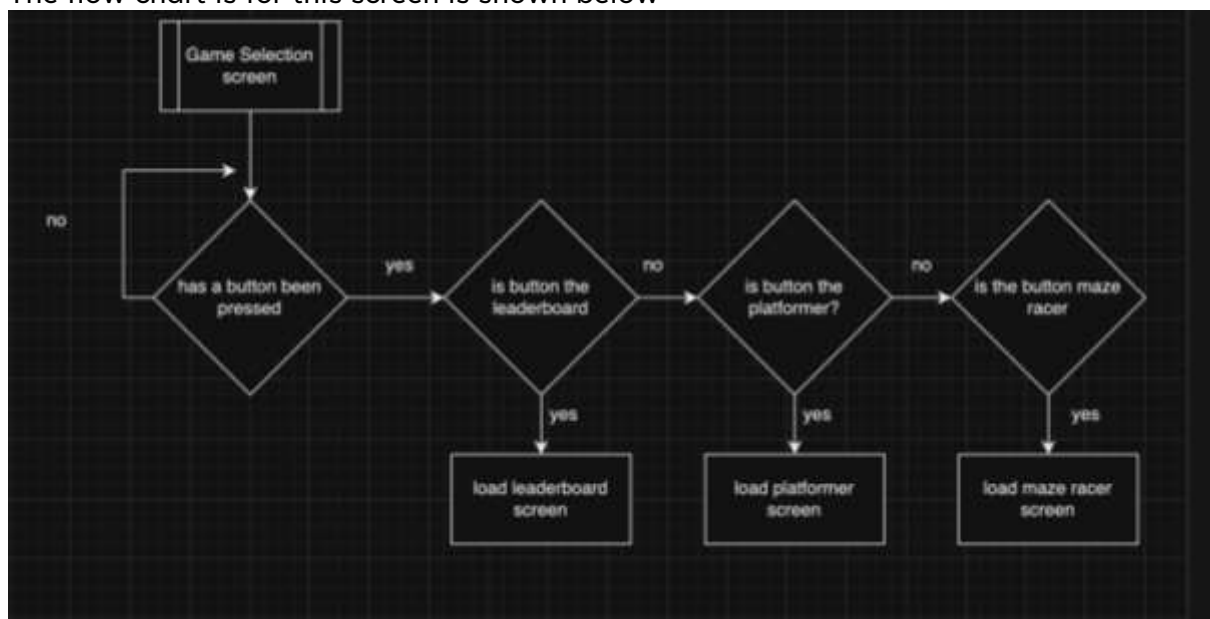
Candidate name: Oscar Winter

### Maze escape selection screen

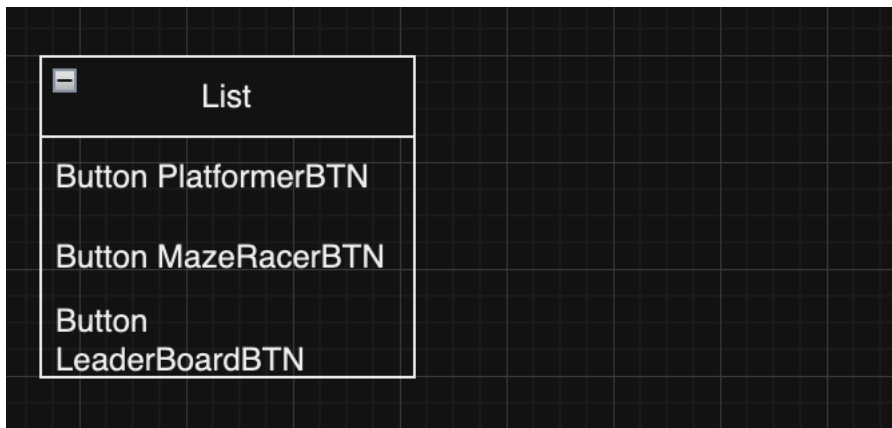
- Once the user has logged in all their game data will have been loaded
- Next, they are presented with the game selection screen where they are able to choose between either Platformer or maze racer
- And a third option to see the leader board for the fastest times on the hardest difficulty



The flow chart for this screen is shown below



Now below are the class diagram key variables in the creation of the script

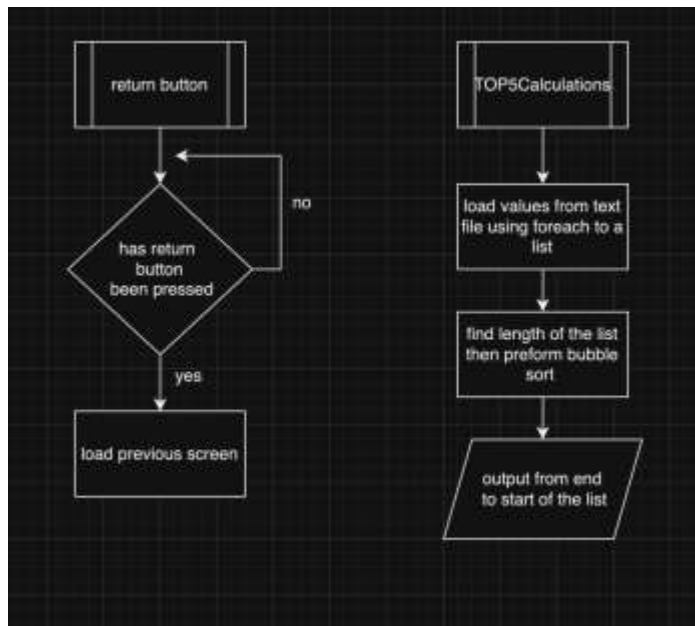


This outlines key variables that will be used in the respective classes how they are used can be seen in the flow chart above. Each key variable holds the state of the button. And is used to check when a button pressed if a certain button is pressed then a corresponding menu / game will be presented through the activation of a method.

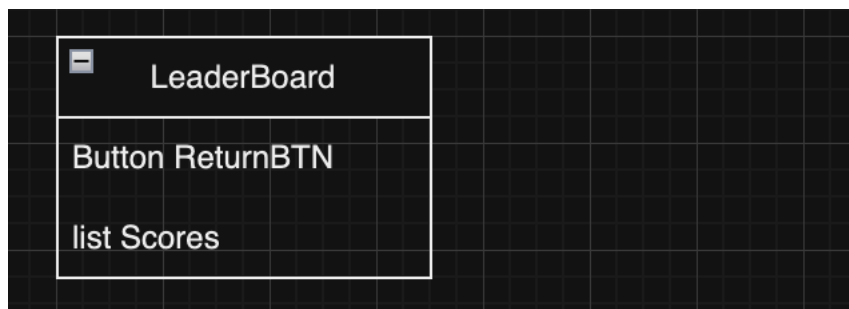
### Leader Board Screen

- Once on the leader board screen you'll see and easily visible return button in the top left
- As well as the top 5 highest scores achieved
- Like shown below





Now below are the class diagram key variables in the creation of the script

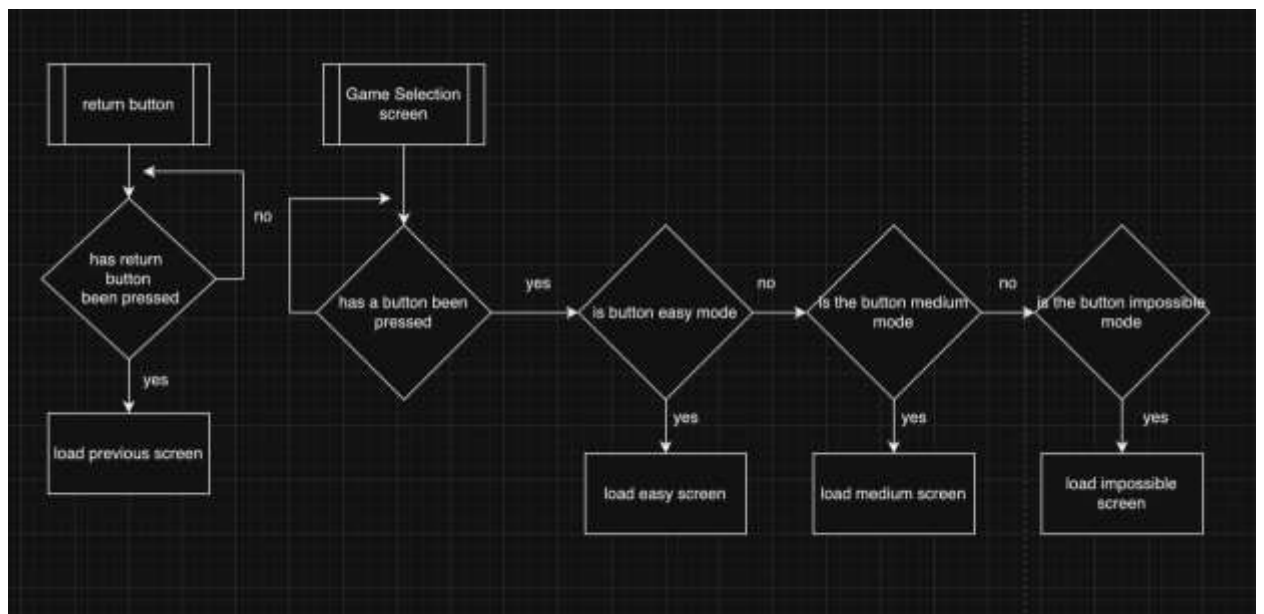


This outlines key variables that will be used in the respective classes how they are used can be seen in the flow chart above. Return button will be used the same as the previous buttons. And list scores will be created by iterating through the text file and adding to the list to then be used for further calculation to determine the top scores to be displayed on the menu using TMLPro.

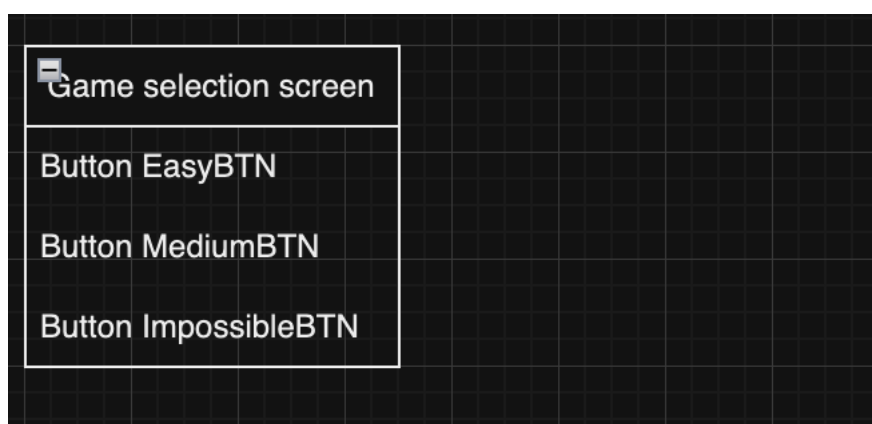
### Level selection screen/Difficulty

- Upon clicking Maze runner, you will be presented with the following screen
- Where the user has a back button in the top left
- Has the option to select one of the three game modes easy, medium, impossible
- After one is pressed it will take the user to the appropriate difficulty game mode
- When a difficulty is selected it will load a brain suitable to the difficulty selected which has been trained to a controlled strength
- This is because my audience will have a range of cognitive abilities and in order to make the game suitable for all audiences, I have the several game modes so the user can decide on their own ability

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Now below are the class diagram key variables in the creation of the script



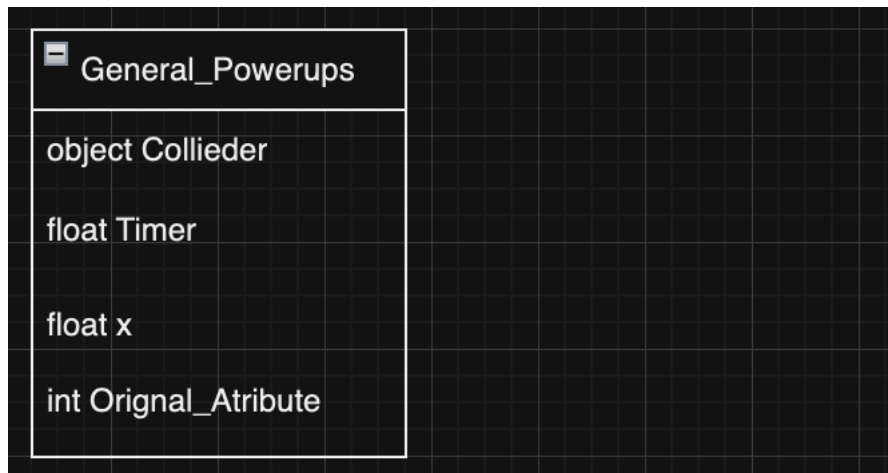
This outlines key variables that will be used in the respective classes how they are used can be seen in the flow chart above. Each key variable holds the state of the button. And is used to check when a button pressed this will then activate a matching method used to load the correct scene.

## Power ups

- The player will have access to three powerups of which are a speed boost a jump boost and an extra life. responding to the response about my game research where the testers said they didn't enjoy the lack of powerups in PacMan.
- The power up will only exist in scene 2 seconds after the collider has been triggered to stop the players from repeatedly collecting it
- The first two power ups jump boost and speed boost will last for a duration of 30 seconds this is to encourage the player to advance quickly through the level as one of my testers said they enjoyed the added pressure of being chased by the ghosts in PacMan
- After the 30 second time the players values will return to the original
- For the health power up after it collected it will also disappear in 2 seconds and will add a live to the players life as my testers said they liked in hollow knight the ability to regain health
- The general algorithm for these power ups is show in the flow chart below for easy visualisation of how I'll be coding it



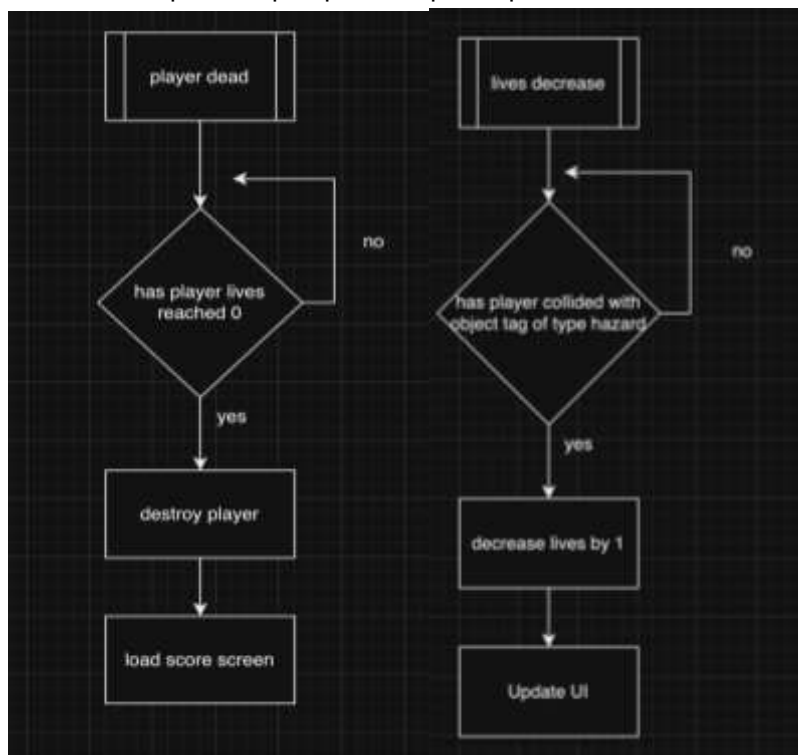
Now below are the class diagram key variables in the creation of the script



This outlines key variables that will be used in the respective classes how they are used can be seen in the flow chart above. Object collier can be used to hold the current collision state the float timer will be used as the main 30 second timer x will similarly be used for time before destruction and lastly the attribute to be changed so it can alter the playing experience.

### Player death and lives

- For the player to die their number of lives my reach 0
- And on players death they will be shown their score, so the player immediately knows how well they did.
- The player starts with 3 lives if the player collides with a entity which has the tag hazard there lives goes down by one and this is updated in the UI as the testers said its useful to see the number of lives left fo they know when to be risk or defensive
- If the health power up is picked up its updated in the lives variable and the UI



Now below are the class diagram key variables in the creation of the script



This outlines key variables that will be used in the respective classes how they are used can be seen in the flow chart above. Object collider can be used to hold the current collision state then int lives will be used to store the current lives to then go onto be outputted in the UI.

## Controls

- My controls for my platformer game will be using the A and D to move left and right respectively ill achieve this by moving the along the x axis by a given amount then when clicking the jump button, the players transform will be set to a high value along the y axis. Ive chose these controls as response from my testers on my research game
- Then for my maze runner game ill use WASD to move the player A and D moves along the x axis then W and S for the y axis using heuristics.



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- When navigating menus, I'll be using the default cursor as this is a widely adopted way of navigating menus making it easy for first time users especially as my demographic is younger

Now below are the class diagram key variables in the creation of the script



This outlines key variables that will be used in the respective classes how they are used can be seen in the flow chart above. Object collider can be used to hold the current collision state then the move input can be used to physically move the players transform.

## Enemies

- My enemies will follow my player around withing there are using the A\* pathfinding algorithm.
- I want my enemies to have the tag hazard so in previously shown algorithm if they collide with the player, they reduce the players lives
- CLASS PathfindingManager
- INSTANCE VARIABLE: sharedInstance
- 
- FUNCTION Initialize ()
- SET sharedInstance = this (Singleton pattern to store the instance)
- END FUNCTION
- 
- FUNCTION FindPath (startNode, goalNode)
- INITIALIZE openList as an empty list
- 
- FOR EACH node IN allSceneNodes (all nodes in the scene)
- SET node.costFromStart = infinity (default gScore)
- END FOR
- 
- SET startNode.costFromStart = 0 (initialize start node)
- SET startNode.estimatedCostToGoal = Distance between startNode and goalNode
- ADD startNode to openList
- 
- WHILE openList is not empty

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- SET lowestCostIndex = index of node with the lowest F score in openList
- 
- SET currentNode = openList[lowestCostIndex] (node with the lowest F score)
- REMOVE currentNode from openList
- 
- IF currentNode is the goalNode
- INITIALIZE pathList as an empty list
- ADD goalNode to pathList
- 
- WHILE currentNode is not the startNode
- SET currentNode = currentNode.previousNode (trace back the path)
- ADD currentNode to pathList
- END WHILE
- 
- REVERSE pathList to get the correct order from startNode to goalNode
- RETURN pathList
- END IF
- 
- FOR EACH neighborNode IN currentNode.connectedNodes
- SET tentativeCostFromStart = currentNode.costFromStart + Distance  
between currentNode and neighborNode
- 
- IF tentativeCostFromStart < neighborNode.costFromStart
- SET neighborNode.previousNode = currentNode
- SET neighborNode.costFromStart = tentativeCostFromStart
- SET neighborNode.estimatedCostToGoal = Distance between  
neighborNode and goalNode
- 
- IF neighborNode is not in openList
- ADD neighborNode to openList
- END IF
- END IF
- END FOR
- END WHILE
- 
- RETURN null (if no path is found)
- END FUNCTION
- 
- FUNCTION FindNearestNode(position)
- SET closestNode = null
- SET minimumDistance = infinity
- 
- FOR EACH node IN allSceneNodes (all nodes in the scene)
- SET distanceToNode = Distance between position and node's position

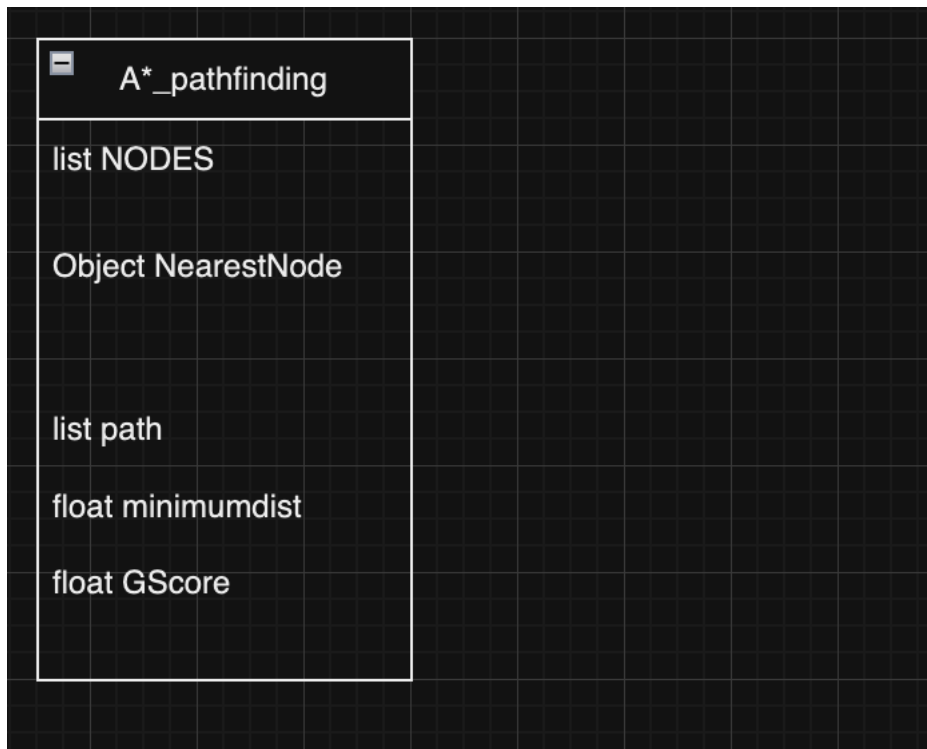
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- 
- IF distanceToNode < minimumDistance
- SET minimumDistance = distanceToNode
- SET closestNode = node
- END IF
- END FOR
- 
- RETURN closestNode
- END FUNCTION
- 
- FUNCTION FindFurthestNode(position)
- SET furthestNode = null
- SET maximumDistance = 0
- 
- FOR EACH node IN allSceneNodes (all nodes in the scene)
- SET distanceToNode = Distance between position and node's position
- 
- IF distanceToNode > maximumDistance
- SET maximumDistance = distanceToNode
- SET furthestNode = node
- END IF
- END FOR
- 
- RETURN furthestNode
- END FUNCTION
- 
- FUNCTION GetAllSceneNodes ()
- RETURN allSceneNodes (all the nodes in the scene)
- END FUNCTION
- END CLASS

Here I have chosen to plan my A\* algorithm script in pseudo code as if I were to use a flow chart like I have previously it would get very large and complicated to read and understand defeating the point of trying to simplify a problem into a more manageable form.

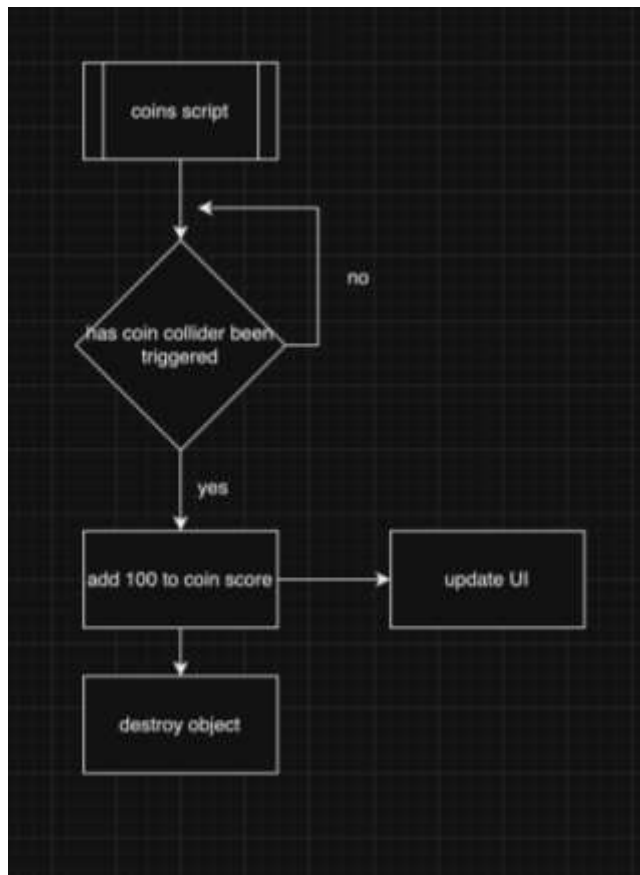
Now below are the class diagram key variables in the creation of the script



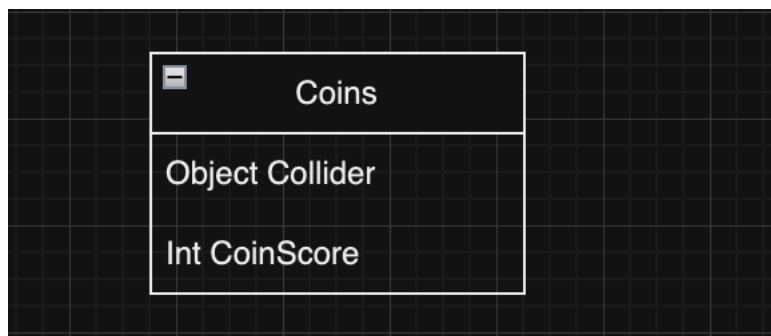
This outlines key variables that will be used in the respective classes how they are used can be seen in the flow chart above. List of nodes is a key variable as all the nodes need a standard data structure to fit into so normal operations can be performed. Object Nearest Node is needed as the algorithm needs to save the closest node as it iterates. List path is needed to store the final path through the graph. Float minimum distance is need so that all comparisons are fair as its set to infinity. Float GScore can be used to compare the viability of nodes to then find the nearest one.

## Coins

- All coins will be in scene as the start
- If the player collides with a coin the coin will be deleted from scene
- 100 will be added to the score
- And the Score UI will be updated as in PacMan my testers found it helpful to see their live score, so they know how they are doing compared to previous runs
- The overview of the algorithm is shown below



Now below are the class diagram key variables in the creation of the script

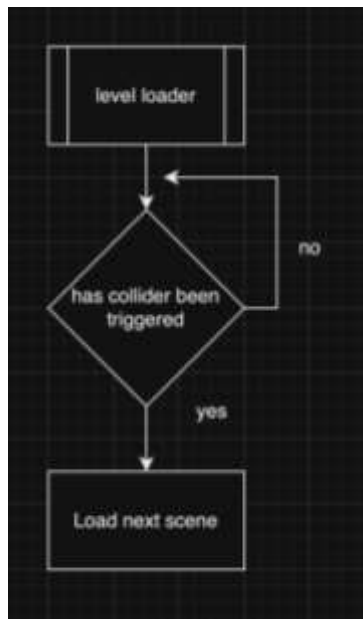


This outlines key variables that will be used in the respective classes how they are used can be seen in the flow chart above. Object collider can be used to hold the current collision state then int coin score can be used to hold the current coin score to be used for UI during playtime and after completion.

### Level loader

- The player after finishing a level must enter a box
- This then loads the next game scene once triggered
- This will be coloured so it's easy for players to see and not get lost
- The algorithm for this is below.

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Now below are the class diagram key variables in the creation of the script

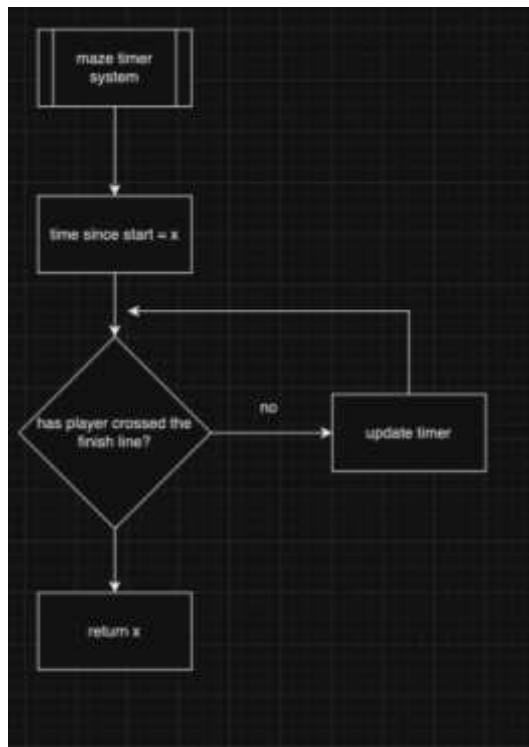


This outlines key variables that will be used in the respective classes how they are used can be seen in the flow chart above. Object collider can be used to hold the current collision state that can then be used to set off a method taking the player to the next level.

### Timer

- The timer needs to start at 0 and count upwards
- The timer will continue to count upwards until the player crosses the finish line
- Then this will be displayed and used in calculations of overall score

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Now below are the class diagram key variables in the creation of the script



This outlines key variables that will be used in the respective classes how they are used can be seen in the flow chart above. Object collider can be used to hold the current collision state then when it's changed the timer can stop. The TimeSinceStart can be used to incrementally increase time for recording this will be stored in a text file and used to determine the top scores.

## AI

- For my Maze AI this involved training my own AI using an API through command prompt I had to install packages including ML-agents and python then running scenes in unity trains the AI after training is given an average reward and other data stored in a text file this tells me how strong the AI is this helps me to choose brains of certain difficulty's so I can load it into the scene depending on the difficulty selected. There's a script required where I override functions to add and remove rewards for the AI as well as its transform so the AI can progressively learn based off the values the flow chart

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for this script is below, as well as an example of the result data and how I determine the strength

- A strength of approximately 10 for impossible (perfect score)
- A strength of approximately 6.5 for medium (fairly reliable)
- A strength of approximately 3.5 for easy (beginner level)



```
PushButtonAgent_Config.yaml
File Edit View
behaviors:
  PushButton:
    trainer_type: ppo
    hyperparameters:
      batch_size: 256
      buffer_size: 1024
      learning_rate: 3.0e-4
      beta: 5.0e-4
      epsilon: 0.2
      lambda: 0.99
      num_epoch: 3
      learning_rate_schedule: linear
      beta_schedule: constant
      epsilon_schedule: linear
    network_settings:
      normalize: false
      hidden_units: 128
      num_layers: 2
    reward_signals:
      extrinsic:
        gamma: 0.99
        strength: 1.0
    max_steps: 500000
    time_horizon: 64
    summary_freq: 10000
Ln 23, Col 22 100% Windows (CRLF) UTF-8
```

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- This also required me to learn how to create virtual environments in python so any error that the AI possible make wouldn't affect my local device a list of all commands I will use are show below

```
Microsoft Windows [Version 10.0.22621.1702]
(c) Microsoft Corporation. All rights reserved.

C:\GameDev\Unity\ML Agents>python
Python 3.9.13 (tags/v3.9.13:6de2ca5, May 17 2022, 16:36:42) [MSC v.1929 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>> exit()

C:\GameDev\Unity\ML Agents>py
Unable to create process using '"D:\Game Dev\Unity\Packages\Other\python.exe"'; The system cannot find the file specified
-

C:\GameDev\Unity\ML Agents>python -m venv venv

C:\GameDev\Unity\ML Agents>venv\scripts\activate
The system cannot find the path specified.

C:\GameDev\Unity\ML Agents>venv\scripts\activate
The system cannot find the path specified.

C:\GameDev\Unity\ML Agents>venv\Scripts\activate

(venv) C:\GameDev\Unity\ML Agents>python -m pip install --upgrade pip

Successfully installed MarkupSafe-2.1.2 Pillow-9.0.0 PySnp-2.4.0 Scikit-20.1.0 SQLAlchemy-2.0.0 Tqdm-4.64.0
23.5.7 charset-normalizer-3.1.0 cloudpickle-2.2.1 filelock-3.12.0 google-auth-2.18.1 google-auth-oauthlib-1.0.0 grpcio-
.54.2 gym-0.26.2 gym-notices-0.0.8 h5py-3.8.0 idna-3.4 importlib-metadata-6.6.0 markdown-3.4.3 mlagents-0.30.0 mlagents-
envs-0.30.0 numpy-1.21.2 oauthlib-3.2.2 pettingzoo-1.15.0 protobuf-4.23.1 pyasn1-0.5.0 pyasn1-modules-0.3.0 pypwin32-22
3 pypwin32-386 pyyaml-6.0 requests-2.30.0 requests-oauthlib-1.3.1 rsa-4.9.0 six-1.16.0 tensorboard-2.13.0 tensorboard-data-
server-0.7.0 urllib3-1.26.15 werkzeug-2.3.4 wheel-0.40.0 zipp-3.15.0

(venv) C:\GameDev\Unity\ML Agents>pip3 install torch torchvision torchaudio

Collecting pip
  Using cached pip-23.1.2-py3-none-any.whl (2.1 MB)
Installing collected packages: pip
  Attempting uninstall: pip
    Found existing installation: pip 22.0.4
    Uninstalling pip-22.0.4:
      Successfully uninstalled pip-22.0.4
  Successfully installed pip-23.1.2

(venv) C:\GameDev\Unity\ML Agents>ip install mlagents

Installing collected packages: mpmath, typing-extensions, sympy, networkx, Jinja2, torch, torchvision, torchaudio
Successfully installed Jinja2-3.1.2 mpmath-1.3.0 networkx-3.1 sympy-1.12 torch-2.0.1 torchaudio-2.0.2 torchvision-0.15.2
typing-extensions-4.5.0

(venv) C:\GameDev\Unity\ML Agents>pip install protobuf==3.20.3

Microsoft Windows [Version 10.0.22621.1702]
(c) Microsoft Corporation. All rights reserved.

C:\GameDev\Unity\ML Agents>venv\scripts\activate

(venv) C:\GameDev\Unity\ML Agents>mlagents-learn --run-id=Test1



Version information:
ml-agents: 0.30.0,
ml-agents-envs: 0.30.0,
Communicator API: 1.5.0,
PyTorch: 2.0.1+cpu
[INFO] Listening on port 5884. Start training by pressing the Play button in the Unity Editor.
```

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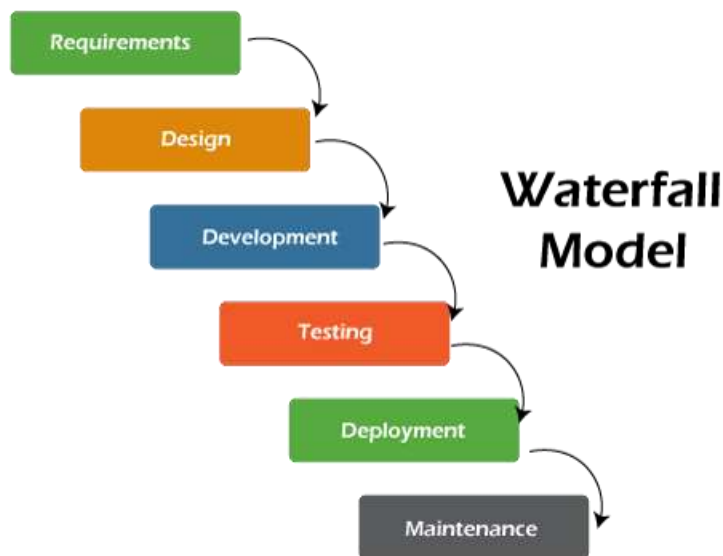
How do my algorithms link



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## Common Software development cycles and what I plan to use

The traditional waterfall model of software development is very well-known and well-defined. The waterfall model is based on a series of stages which are completed in sequence, from start to finish. Because this model is based on the stages being complete in sequence it makes it a poor choice of model for projects which include large amounts of client response as there is little to no opportunity for iteration. This is why I've decided I don't want to take this approach as I believe the best way to achieve the product my users will enjoy the most is using an iterative development cycle



The iterative development cycle I have decided to use is the Agile method. Agile project management is a process for managing a project that involves constant collaboration and working in iterations. It works off the basis that a project can be continuously improved upon throughout its lifecycle and adapt to changes quickly. The three big reasons I chose this iterative development cycle is the Customer and stakeholder alignment, rapid progresses and its ability to improve as time goes on. Hence why I previously stated I think it an ideal option



The agile works but gathering your ideas with your user basses ideas and (the same as which ive done in my analysis section) then using this to start the development and design process ( what I'm doing now ) then beginning the implementation phase which will be what I start next where you create a product then you have it tested and then you develop and improve it iteratively until your satisfied leaving the user with a high quality product they've been involved with.

My first iteration will be the game selection menus, and all the game play my second iteration will be the login and register screens then my third iteration will be my final product.

## Testing data

movement

|                           |                  |                                                                                              |
|---------------------------|------------------|----------------------------------------------------------------------------------------------|
| Maze solver (iteration 2) |                  |                                                                                              |
| Test data                 | Expected outcome | Justification                                                                                |
| W                         | valid            | W is the standard control for moving up in games making it easier for the user to understand |
| A                         | valid            | A is the standard control for moving up in games making it easier                            |

|                             |                  |                                                                                                            |
|-----------------------------|------------------|------------------------------------------------------------------------------------------------------------|
|                             |                  | for the user to understand                                                                                 |
| S                           | valid            | S is the standard control for moving up in games making it easier for the user to understand               |
| D                           | valid            | D is the standard control for moving up in games making it easier for the user to understand               |
| Left arrow                  | invalid          | Left arrow is invalid as it is not the control I'll be used as it was found to be too cramped in research  |
| Right arrow                 | invalid          | Right arrow is invalid as it is not the control I'll be used as it was found to be too cramped in research |
| Down arrow                  | invalid          | Down arrow is invalid as it is not the control I'll be used as it was found to be too cramped in research  |
| Up arrow                    | invalid          | Up arrow is invalid as it is not the control I'll be used as it was found to be too cramped in research    |
| 2D platformer (iteration 1) |                  |                                                                                                            |
| Test data                   | Expected outcome | Justification                                                                                              |
| A                           | valid            | A is the standard control for moving left in platformers making it easier for the user to understand       |
| D                           | valid            | D is the standard control for moving right in platformers making it                                        |

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|             |         |                                                                                                            |
|-------------|---------|------------------------------------------------------------------------------------------------------------|
|             |         | easier for the user to understand                                                                          |
| L           | valid   | L is what I'll be using as in my research games it was found the controls where to cramped.                |
| Space bar   | invalid | Space bar is invalid as this is not what my jump button will be                                            |
| Left arrow  | invalid | Left arrow is invalid as it is not the control I'll be used as it was found to be too cramped in research  |
| Right arrow | invalid | right arrow is invalid as it is not the control I'll be used as it was found to be too cramped in research |

#### Menus (iteration 3)

| Test input                | Expected outcome                     | Justification                                                                                                 |
|---------------------------|--------------------------------------|---------------------------------------------------------------------------------------------------------------|
| Left click button         | Triggers command connected to button | Left Click Button is the standard control for moving in all menus making it easier for the user to understand |
| Right click button        | Invalid                              | right arrow is invalid as it is not the control I'll be used as it would be confusing                         |
| left click return button  | Loads the previous scene             | left Click Button is the standard control for moving in all menus making it easier for the user to understand |
| right click return button | Invalid                              | right arrow is invalid as it is not the control I'll be used as it would be confusing                         |

These tests will be completed in conjunction with the creation of the parts of the game this will be to ensure that I have codes my iterations correctly to help mitigate potential errors later on.

#### Alpha testing criteria for my testers

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Menus (iteration 3)

| Action                                                  | explanation                                                                                                    | Expected outcome                                                      |
|---------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| login scene is loaded first                             | The first menu to be displayed to the user should be the login menu                                            | User is presented with the login screen                               |
| Player using input fields                               | When the player is trying to sign in or create a new account the user should be able to use the keyboard       | Letters appear in the input field                                     |
| The player clicks the return button                     | Its button so the user is able to go the previous page after finishing with their current page                 | The player is taken to the previous scene                             |
| The player is able to use their mouse to navigate menus | Most common interfaces use mouses to navigate so the player will be able to intuitively use the mouse controls | When the player moves the mouse, the cursor will move correspondingly |
| Clicking on register button                             | This should take the user to the register page so they can create an account                                   | The scene for the register page is loaded                             |
| Clicking on sign in button                              | If credentials match the user is taken to the. Game selection page                                             | Takes the player to the game selection page                           |
| Clicking on the platformer button                       | This takes the player to the first level of the platformer so start the levels                                 | Loads the scene for the first game level                              |
| Clicking on leader board button                         | Takes the player to the leader board where they can view the top scores                                        | Loads the scene for the leader board                                  |
| Clicking on the maze solver button                      | When clicking the maze solver button, it should take the user to the difficulty selection screen               | Loads the scene for difficulty selection                              |
| Clicking on easy difficulty button                      | The user clicks this button to load a particular difficulty of brain into the maze solving AI                  | Loads Scene for the easy difficulty                                   |
| Clicking on medium difficulty button                    | The user clicks this button to load a particular difficulty of brain into the maze solving AI                  | Loads Scene for the medium difficulty                                 |
| Clicking on impossible difficulty button                | The user clicks this button to load a particular                                                               | Loads Scene for the impossible difficulty                             |

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|                                                          |                                                                                          |                            |
|----------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|
|                                                          | difficulty of brain into the maze solving AI                                             |                            |
| In the leader board the top 5 scores should be displayed | This gives the player a motive to beat and lets them know how they compare to top scores | Top 5 scores are displayed |

#### Enemies and player (iteration 1)

| action                                      | explanation                                                                                                                             | Expected outcome                                                                          |
|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Enemies should engage                       | When the player is within range of the node network the enemy should begin to follow the path calculated for it                         | The enemy starts to follow the player around the node network                             |
| Character bumping into surfaces             | When traversing the game, the charter will collided with walls and floors this test is to make sure the collisions are all set properly | The player is unable to walk through/ fall through any surfaces                           |
| Character jumping (should only jump once) f | When player the game the player should only be able to. Jump once so they are required to play the game as intended                     | The player can only jump will on the floor if the player does try too, they will not move |
| Character jumping on a danger               | The game has objects under a group call danger if the player does collide with a danger, they should lose a life                        | The player loses a life                                                                   |

#### Timer, AI and UI (iteration 2)

| Action                                                 | Explanation                                                                                                                                                                                                      | Expected outcome |
|--------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| Timer should tick for the player while they are racing | the game needs to keep track of how long is taking the player to reach the finish lines so it can be returned at the end, so this test is to make sure the timer starts ticking after the player sets difficulty | The timer ticks  |
| Timer should tick for the AI while its racing          | the game needs to keep track of how long is taking the AI to reach the finish lines so it can be returned                                                                                                        | The timer ticks  |

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|                                                              |                                                                                                                                                          |                                                              |
|--------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
|                                                              | at the end, so this test is to make sure the timer starts ticking after the player sets difficulty                                                       |                                                              |
| Going over the finish line (timer should end) for the player | When the player reaches the finish line the timer should stop, and the end screen should be loaded and the final time returned                           | The timer end and end screen are shown, and time is returned |
| AI should be roughly of difficulty selected                  | The player selects the AI difficulty this is, so the player picks a difficulty suitable to their ability so it's important this is functioning correctly | The AI is of approximate difficulty selected                 |
| Going over the finish line (timer should end) for the AI     | When the player reaches the finish line the timer should stop, and the AI final time returned                                                            | The timer end and time is returned                           |
| UI should update the score in the maze game mode             | The player needs this to see their current statistics in game as it makes it easy for them to recognise if they are doing well                           | The UI should be updated on screen in the designated area    |
| UI should update the score in the platformer                 | The player needs this to see their current statistics in game as it makes it easy for them to recognise if they are doing well                           | The UI should be updated on screen in the designated area    |
| UI should update the timer in the maze game                  | The player needs this to see their current statistics in game as it makes it easy for them to recognise if they are doing well                           | The UI should be updated on screen in the designated area    |
| UI should display end score at end of maze game mode         | The player needs this to see their current statistics in game as it makes it easy for them to recognise if they are doing well                           | The UI should be updated on screen in the designated area    |
| UI should display end score at end of platformer             | The player needs this to see their current statistics in game as it makes it easy for them to recognise if they are doing well                           | The UI should be updated on screen in the designated area    |
| UI should update the coin score in the maze game mode        | The player needs this to see their current statistics in game as it makes it                                                                             | The UI should be updated on screen in the designated area    |

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|                                                           |                                                                                                                                |                                                           |
|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
|                                                           | easy for them to recognise if they are doing well                                                                              |                                                           |
| UI should update the coin score in the platformer         | The player needs this to see their current statistics in game as it makes it easy for them to recognise if they are doing well | The UI should be updated on screen in the designated area |
| UI should update player lives in the platformer game mode | The player needs this to see their current statistics in game as it makes it easy for them to recognise if they are doing well | The UI should be updated on screen in the designated area |

Powerups, death/lives, coins and level loader (iteration 1)

| Action                                                                        | Explanation                                                                                                                                | Expected outcome                       |
|-------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| On Collision with a speed power up                                            | The player during the game will encounter power ups after colliding with a power up it will have an effect on one of the player's stats    | The player's speed has increased       |
| On Collision with a jump power up                                             | The player during the game will encounter power ups after colliding with a power up it will have an effect on one of the player's stats    | The player's jump height has increased |
| On Collision with a health power up                                           | The player during the game will encounter power ups after colliding with a power up it will have an effect on one of the player's stats    | The player's health increases by 1     |
| On Collision with a power up it should stay in scene for only 2 seconds       | When the player is playing after colliding with the powerup to give the computer time to change the values it stays in scene for 2 seconds | Power up only stays for 2 seconds      |
| On Collision with a power up (jump and speed types) its effects will last 30s | When the player is playing after colliding with the powerup to give the computer time to change the values it stays in scene for 2 seconds | Power up effects only last 30 seconds  |

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|                                                                                                          |                                                                                                                       |                                                |
|----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| On collision with a health power up it should update the UI                                              | When the player is playing after colliding with the powerup the UI should go up by one to keep updated on their lives | Player health increase by 1 in the top right   |
| The players collision with a hazard decreasing player health                                             | The player has obstacles they need to navigate so it important that when they hit on, they lose health                | The player health is reduced by 1              |
| The player health reaching 0                                                                             | The player has lost all health                                                                                        | The game ends                                  |
| Player colliding with coin. 100 will be added to the coin score (for both platformer and maze game mode) | This will be saved to tally the players score at the end                                                              | 100 will be added to the coin score            |
| Player colliding with coin should cause the coin to be deleted from scene                                | This is so the player can continually collect the coins in the same place                                             | The coin will be deleted from scene            |
| When the player enters the level loader                                                                  | The player will be transported to the next level                                                                      | The player will be loading into the next scene |

I've had both testers 1 and 2 review this and there are their thoughts as I want to keep them as involved as possible

#### Tester 1

- I'm happy with all the scripts they are planned well
- I like the simplicity of the menus "I did inform tester 1 the design of the menus may change but I plan to keep it as close as possible to the original"

#### Tester 2

- I'm happy with all the criteria and like to see tester 1 and mine's ideas being used I look forward to seeing the final product

## Implementation

Now I have decomposed the iterations into separate tasks and have communicated and incorporated the testers ideas into the design of the product in now confident I can start implementing these ideas into the unity game engine

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## Iteration 1 (the platformer)

Throughout this iteration I will be developing the platformer section of the game this would include movement, coins, consumables(powerups) and my enemy scripts First I'll be implementing my player movement as many of the other tasks are dependent on the player being able to move and trigger colliders

### Movement

Below is my c# script I've written for player movement

```
using System.Collections;
using System.Collections.Generic;
using UnityEngine;
using UnityEngine.InputSystem;

public class Playerscript : MonoBehaviour
{
    [SerializeField] float RunSpeed = 6f;
    [SerializeField] public float JumpSpeed = 5f;
    //sets the variable runspeed and jumpspeed

    |
    Vector2 moveInput;

    Rigidbody2D myRigidbody;

    CapsuleCollider2D myBodyCollider2D;

    //this gives the compontes in scene a variable name to use in the script

    void Start()
    //this method gets the components of the ridgid
    //body and the capulase collied and loads them into the script for use
    {
        myRigidbody = GetComponent<Rigidbody2D>();
        myBodyCollider2D = GetComponent<CapsuleCollider2D>();
    }
    //this updates all the method in the method every frame
    void Update()
    {
        run();
    }

    void run()
    //this method takes the players vector2 value then has them move along the x axis
    {
        Vector2 playerVelocity = new Vector2(moveInput.x * RunSpeed, myRigidbody.velocity.y);
        myRigidbody.velocity = playerVelocity;
    }

    void OnJump(InputValue value)
    //this method checks the players on the floor then allows the player to jump if condistion is met
    {
        if (myBodyCollider2D.IsTouchingLayers(LayerMask.GetMask("Walls")))
        {
            myRigidbody.velocity += new Vector2(0f, JumpSpeed);
        }
    }
}
```

In this script I've used methods to help break down the problem into smaller tasks also by having my methods as separate methods to my update method this enables me to expand this script as I implement more methods to do with the player, I can just add them the update method. In terms of what I've done for the player movement is I've used unity's input system to when a designated key is pressed as I've loaded the input value into scene this then becomes a decimal value which I can have constantly update my player position in scene either left or right by updating the players vector 2 values.

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However for jump as this is a Boolean value when a key is pressed it returns a true or false (1/0) so I can't use it in scene like before so instead unity's input system triggers the methods condition to be met where I have then used selection in the form of an if statement to check the player is on the floor before jumping so the player cant repeatedly jump. After the condition is met, I then update the vector2 value along the y axis

Now tester 1 will test that the player movement script works against my testing criteria stated previously in the design section. Below is a click demonstrating the test

To find the video in file. it's in location project testing/iteration 1/Movement test

As you can see the player is able to move left and right as can be seen by the players position in respect to the cameras position which was successful. keys used where correct as I watch the testing of left and right arrow which were invalid and then the use of A for left which was valid and D for right which was valid all as expected. After this tester 1 then user space to try and jump this was invalid then they tried to use L to jump which was valid all as expected.

| action                                      | explanation                                                                                                                             | Was the outcome correct? |
|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| Character bumping into surfaces             | When traversing the game, the charter will collided with walls and floors this test is to make sure the collisions are all set properly | yes                      |
| Character jumping (should only jump once) f | When player the game the player should only be able to. Jump once so they are required to play the game as intended                     | yes                      |

| Test data   | Expected outcome | Outcome |
|-------------|------------------|---------|
| A           | valid            | valid   |
| D           | valid            | valid   |
| L           | valid            | valid   |
| Space bar   | invalid          | invalid |
| Left arrow  | invalid          | invalid |
| Right arrow | invalid          | invalid |

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## Coins

Below is my c# script I've written for coin pickup and object deletion

```
using System.Collections;
using System.Collections.Generic;
using UnityEngine;

public class CoinPickup : MonoBehaviour
{
    [SerializeField] int PointsForCoin = 100;

    int CollectedPoints;

    //initializes all variables needed for the method

    void OnTriggerEnter2D(Collider2D other)
    //this method checks that what the coin has collided with is the player
    //then calls a separate method from the gamesession manager then destroying the game object
    {
        if (other.tag == "Player")
        {
            FindObjectOfType<GameSessionManager>().AddToScore(PointsForCoin);

            int CollectedPoints += 100;
            Debug.Log(CollectedPoints);
            Destroy(gameObject);
        }
    }
}
```

First, I have set all variables I'll need for the script in this case just the points for the score and I've set this as a serialized field so I'm able to adjust in the game engine without changing the script. This helps keep the script stable as not having to enter the script means there's a lower chance of accidental editing of the script.

After testing this worked via the output from my debug.log below I could then start work on the Add to score method in the game session manager script/object



```
[15:38:23] 100
UnityEngine.Debug:Log (object)

[15:38:27] 100
UnityEngine.Debug:Log (object)
```

This was the output after collecting two coins which was as expected so then I started work on the game manager script and the AddToScore method.

Now below this is an image for the AddToScore method.

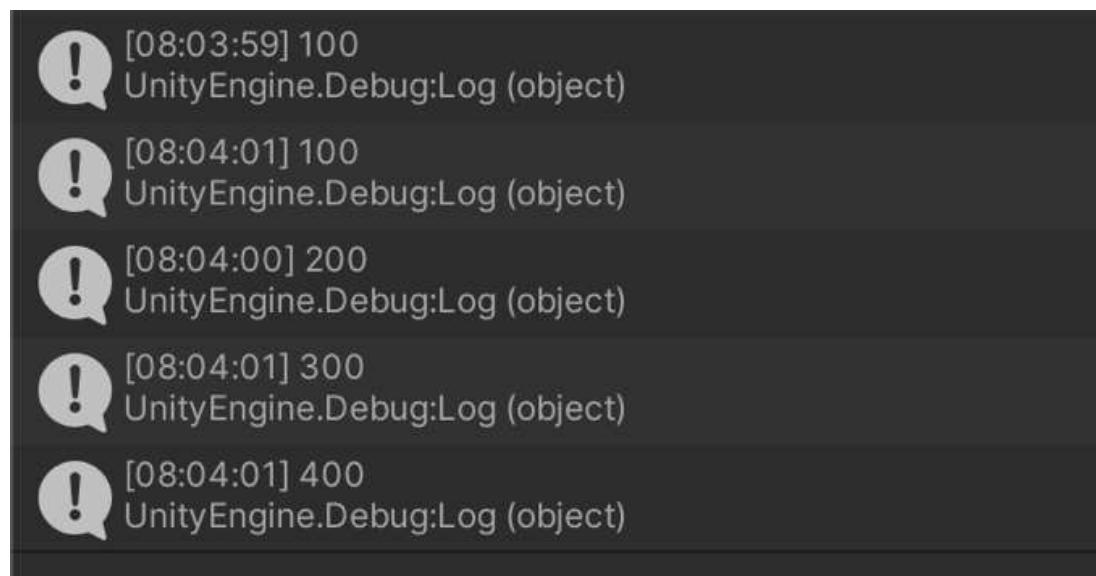
Candidate name: Oscar Winter

```
using UnityEngine;
using UnityEngine.SocialPlatforms.Impl;

public class GameSessionManager : MonoBehaviour
{
    int CoinScoreTotal = 0;

    //here i initialized the CoinScoreTotal variable
    public void AddToScore(int pointsToAdd)
    //this method makes a running total of the coin score
    //then debug.logs it to check outputs
    {
        CoinScoreTotal += pointsToAdd;
        Debug.Log(CoinScoreTotal);
    }
}
```

After this I then ran the program again and got this for my output after collecting 4 coins:



The screenshot shows the Unity console with five log entries, each preceded by a warning icon (a circle with an exclamation mark). The logs show the running total of coins collected over time:

- [08:03:59] 100  
UnityEngine.Debug:Log (object)
- [08:04:01] 100  
UnityEngine.Debug:Log (object)
- [08:04:00] 200  
UnityEngine.Debug:Log (object)
- [08:04:01] 300  
UnityEngine.Debug:Log (object)
- [08:04:01] 400  
UnityEngine.Debug:Log (object)

As seen in the image above this now outputs my running total rather than the amount of each coin as expected. Now that this is successful, I can start to implement a UI for the running total.

To achieve my user interface, I have used the unity Text Mesh Pro this allows me to add things like scores to the screen using a canvas so all I need to do is set a variable and link the Text to it then just update the text every time the methods called. I also need to import the TMP library so I can manipulate objects of this type.

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```
using System.Collections;
using System.Collections.Generic;
using UnityEngine;
using UnityEngine.SocialPlatforms.Impl;
using TMPro;

public class GameSessionManager : MonoBehaviour
{
    [SerializeField] TextMeshProUGUI Score;
    int CoinScoreTotal = 0;

    //here i initialized the CoinScoreTotal and score variable
    public void AddToScore(int pointsToAdd)
    //this method makes a running total of the coin score
    //then updates the UI it to check outputs
    {
        CoinScoreTotal += pointsToAdd;

        Score.text = CoinScoreTotal.ToString();
    }
}
```

This is the script after the stated changes. Now that coins are completed ill have tester 2 test this. The video showing this is below.

To find the video in file. it's in location project testing/iteration 1/Coins test

As seen in the video tester 2 collided with the coins and on collision the score in the top left went up and the object was destroyed as expected.

| Action                                                                                                   | explanation                                                               | Was outcome expected? |
|----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|-----------------------|
| Player colliding with coin. 100 will be added to the coin score (for both platformer and maze game mode) | This will be saved to tally the players score at the end                  | yes                   |
| Player colliding with coin should cause the coin to be deleted from scene                                | This is so the player can continually collect the coins in the same place | yes                   |

### Lives and death

Below where my first scripts for handles the player lives UI. Then ill implement death after.

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```
void Die()
{
    if (myBodyCollider2D.IsTouchingLayers(LayerMask.GetMask("Dangers")))
    {
        transform.position = new Vector2(-4.320046f, 1.201181f);

        isAlive = true;
        FindObjectOfType<GameSessionManager>().Death();
    }
}
```

This is my method for when the player collides with a danger it should set the players transform to the start position then call the method Death from my Game Session manager splitting up methods like this allows me to break the problem into small more manageable and easier to maintain methods. Script for Death method is below

```
public void Death()
{
    PlayerLives--;

    Lives.text = PlayerLives.ToString();
}
```

This then takes one off the players life then updates the UI. however, I was running into a logic issue while testing as show below when colliding with a danger is registered a lot more deaths than expected. I believe this was because the program was taking too long to reset the transform and kept registering contact every frame. To combat this, I am going to try deleting the game object first to see if that fixes the issue then resetting the game scene.



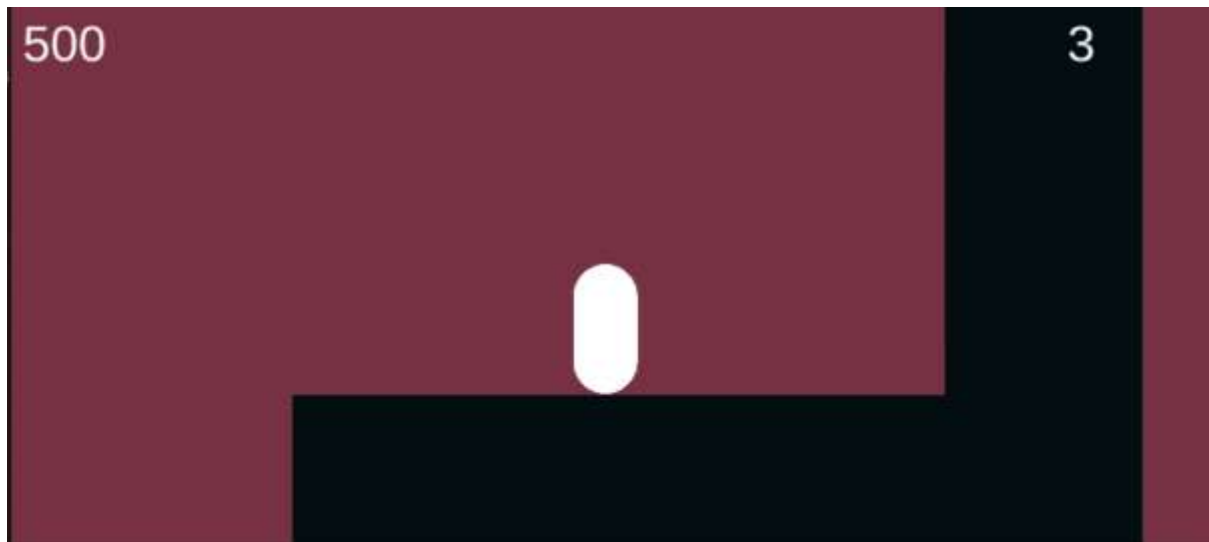
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To find the video in file. it's in location project testing/iteration 1/Hazards test 1 (logic error)

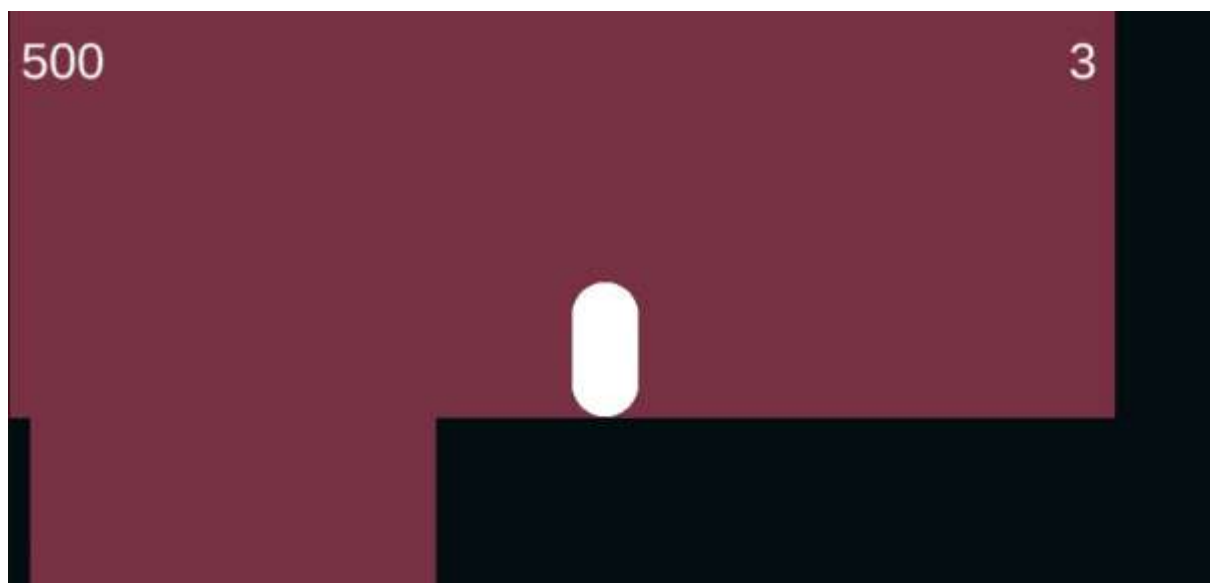
After changing the code to the solution suggested I got closer but after the players death it then causes the lives to rest but for a second after death you see the lives decrease only once.

A video showing this test is below (this test was conducted by me as this is not a complete solution and I'm still developing).



To find the video in file. it's in location project testing/iteration 1/Hazards test 2 (logic error)

Now after this test after some thinking I thought I need to make it so that there is a time delay between the player making contact with the hazard and then be transformed back to the starting position. And as die () is in the update function it registers several deaths. to fix this I thought of setting a timer so after a period of time almost equal to



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To find the video in file. it's in location project testing/iteration 1/Hazard test 3 (complete)

the time of contact between the player and the hazard so only one death is registered below is my finished code and Tester 2 testing this.

Here I watched tester 2 jump and as expected the lives only decreases by one and then resets the transform back to the start.

```
void Die()
//this method checks id the player has died then resets the players transform
//then calls the timers1 fuction
{
    if (myBodyCollider2D.IsTouchingLayers(LayerMask.GetMask("Dangers")))
    {
        transform.position = new Vector2(-4.320046f, 1.201181f);
        Hastouched1 = true;

        Timers1();
    }
}

void Timers1()
//this method checks if the players touched a danger then starts the timer
//then is calls the death function from the gamesession manager
{
    if (Hastouched1 == true)
    {
        elapsedTime1 += Time.deltaTime;
        if (elapsedTime1 >= 0.01388257f)
        {
            isAlive = true;

            Hastouched1 = false;

            FindObjectOfType<GameSessionManager>().Death();
            elapsedTime1 = 0f;
        }
    }
}

public void Death()
//this substracts one from theplayers live and updates the UI
{
    transform.position = new Vector2(-4.320046f, 1.201181f);
    PlayerLives--;
    Lives.text = PlayerLives.ToString();
}
```

For my final iteration of this code, I have used three methods the first two methods are in the playermovement script the first Die (). First uses and if statement as I need to check if the player is touching any dangers in the scene after this it them resets the

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transform to minimise the time the player is in contact with the dangers as this is in update I don't want it to continually decrease the players lives so I then set has touched to true and set a timer before it can be triggered again where I call my next method

called Timers1() where I've used nested if statement to first check that the player has touched the danger then to start the timer and the timer will continue in the update function till its reached 0.0139 seconds as I used debug.log to work out the average time of contact then set has touched to false, so it stops. then it resets the elapsed time to 0 so the player can die again on the next turn. after this is then call the death () method I had to do this through finding the object as it's in a separate script

For my death () method decrease the players lives by one then update the UI so the player can see their lives in real time.

| action                                                       | explanation                                                                                            | Was the outcome expected? |
|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|---------------------------|
| The players collision with a hazard decreasing player health | The player has obstacles they need to navigate so it important that when they hit on, they lose health | yes                       |
| The player health reaching 0                                 | The player has lost all health                                                                         | yes                       |

### Powerups

As there are three different powerups that require 3 different algorithms ill split this into two parts one for jump boost and speed boost and one for health.

### Speed boost

Below is my code for my speed boost object

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```
using System.Collections;
using System.Collections.Generic;
using UnityEngine;

public class Speedboost : MonoBehaviour
{
    CircleCollider2D myBodyCollider2D;
    // Start is called before the first frame update

    public bool Newtimer = false;

    public float elapsedTime = 0f;

    void Update()
    //this calls the timers method every frame
    {
        Timers();
    }

    void Timers()
    //this is used to start the timer and checks if the time reaches 2
    //then destroying the objects
    {
        if (Newtimer == true)
        {
            elapsedTime += Time.deltaTime;
            if (elapsedTime >= 2f)
            {
                elapsedTime = 0f;
                Newtimer = false;
                Destroy(gameObject);
            }
        }
    }

    private void OnCollisionEnter2D(Collision2D collision)
    //checks for collision and matches the condition for the timer
    {
        Newtimer = true;
    }
}
```

Here I have my first script that's attached to the powerups object itself and is used to destroy the object after 2 seconds as this gives time for my other script to start and also gives a better effect in the game.

I have done this using three separate functions update which calls the timers method every frame to update the timer then I have used nested if statements to first trigger that the player has collided with the power up this then starts the timer and keeps calling it till elapsed time is equal to 2 then resetting elapsed time and changing the condition to false so it doesn't continue to count infinitely then destroys the game object so the player can't use the same power up over and over

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The last function I used is the on collision enter to set the condition for Timer's function.

Below is my second script which is attached to the player movement script as the variables are required to change are already in this script.

```
private void OnCollisionEnter2D(Collision2D collision)
//This method checks for collision and the correct object
//then sets the condition for Timer to true and increase run speed
{
    if (myBodyCollider2D.IsTouchingLayers(LayerMask.GetMask("Speedboost")))
    {
        Hastouched = true;
        runspeed = 12f;
    }
}
```

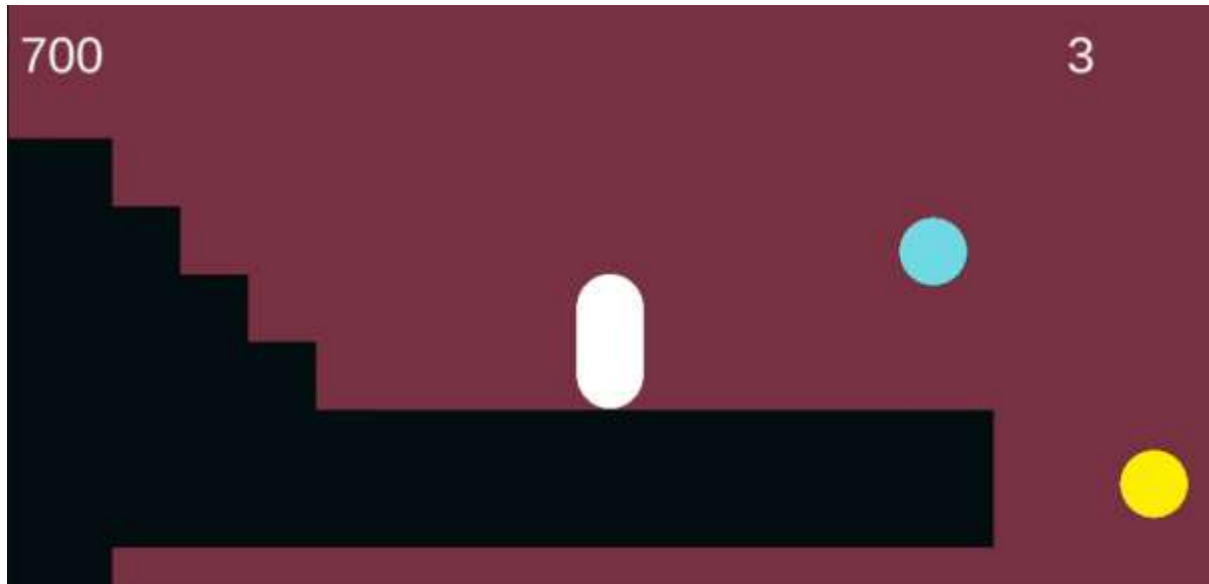
This is the first method I used which is on collision enter where first an if statement to check the object it collided with is on the layer Speed Boost. Sets has touched (the condition for the next method to be true). And increased the run speed to 12.

```
void Timers()
//this method checks the condition previous is true
//starts the timer and after 30s then resets all variable to previous values
{
    if (Hastouched == true)
    {
        elapsedTime += Time.deltaTime;
        if (elapsedTime >= 30f)
        {
            runspeed = 7f;
            elapsedTime = 0f;
            Hastouched = false;
            // Change the run speed and stop further updates
        }
    }
}
```

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This is another method from the same script that is in update so after the previous method triggers the if condition to be true it then uses nested ifs as this is update elapsed time until its equal to 30 then returns all variable to previous values.

After this I then had tester 1 test it the table and video corresponding are below



To find the video in file. it's in location project testing/iteration 1/PowerUps 1

Below are the testing tables I have also included my jump boost power up as they use almost exact scripts with the only difference being some variable names and the variable changed on collision. The script for this is below the testing table as well as the video demonstrating it works.

| action                            | explanation                                                                                                                              | Expected result                      | What the result expected? |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|---------------------------|
| On Collison with a speed power up | The play during the game will encounter power ups after the colliding with a power up it will have an effect on one of the players stats | The player's speed has increased     | yes                       |
| On Collison with a jump power up  | The play during the game will encounter power ups after the colliding with a power up it will have an effect on                          | The players jump hight has increased | yes                       |

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|                                                                              |                                                                                                                                                      |                                       |     |
|------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|-----|
|                                                                              | one of the players stats                                                                                                                             |                                       |     |
| On Collison with a power up it should stay in scene for only 2 seconds       | When the player is player is playing after colliding with the powerup to give the computer time to change the values it stays in scene for 2 seconds | Power up only stays for 2 seconds     | yes |
| On Collison with a power up (jump and speed types) its effects will last 30s | When the player is player is playing after colliding with the powerup to give the computer time to change the values it stays in scene for 2 seconds | Power up effects only last 30 seconds | yes |

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Scripts for jump boost

```
using System.Collections;
using System.Collections.Generic;
using UnityEngine;

public class JumpBoost : MonoBehaviour
{
    CircleCollider2D myBodyCollider2D;
    // Start is called before the first frame update

    public bool Newtimer = false;

    public float elapsedTime = 0f;

    void Update()
    //this calls the timers method every frame
    {
        Timers();
    }
    void Timers()
    //this is used to start the timer and checks if the time reaches 2
    //then destroying the objects
    {
        if (Newtimer == true)
        {
            elapsedTime += Time.deltaTime;
            if (elapsedTime >= 2f)
            {
                elapsedTime = 0f;

                Newtimer = false;

                Destroy(gameObject);
            }
        }
    }

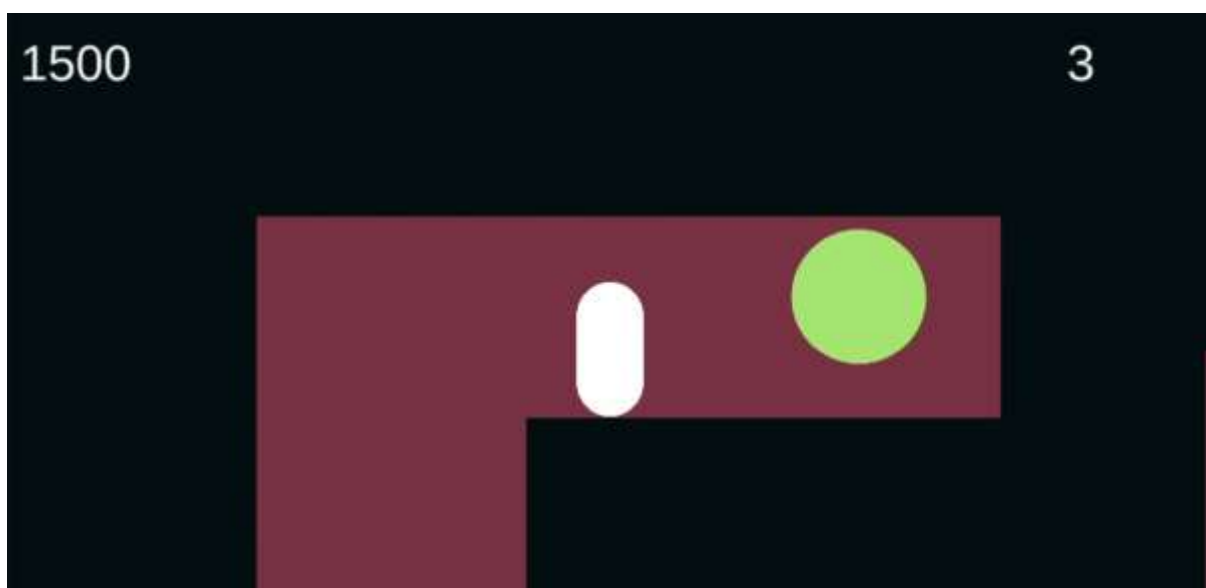
    private void OnTriggerEnter2D(Collider2D collision)
    //checks for collision and matches the condition for the timer
    {
        Newtimer = true;
    }
}
```

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```
void timers2()
{
    if (Hastouched2 == true)
    {
        elapsedTime2 += Time.deltaTime;
        if (elapsedTime2 >= 30f)
        {
            jumpspeed = 15f;
            elapsedTime2 = 0f;
            Hastouched2 = false;
            // Change the run speed and stop further updates
        }
    }
}
```

```
private void OnTriggerEnter2D(Collider2D collision)
{
    if (myBodyCollider2D.IsTouchingLayers(LayerMask.GetMask("Jumpboost")))
    {
        Hastouched2 = true;
        //first checks to see if the player has collided with a hazard or enimes
        jumpspeed = 30f;
        // Check if 30 seconds have passed
    }
}
```



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To find the video in file. it's in location project testing/iteration 1/PowerUps 2

Next is my scripts for my health power up

Below is my first script attached to the object

```
using System.Collections;
using System.Collections.Generic;
using UnityEngine;

public class HealthBoost : MonoBehaviour
{
    CircleCollider2D myBodyCollider2D;
    // Start is called before the first frame update

    public bool Newtimer = false;

    public float elapsedTime = 0f;

    void Update()
    //this calls the timers method every frame
    {
        Timers();
    }

    void Timers()
    //this is used to start the timer and checks if the time reaches 2
    //then destroying the objects
    {
        if (Newtimer == true)
        {
            elapsedTime += Time.deltaTime;
            if (elapsedTime >= 2f)
            {
                elapsedTime = 0f;
                Newtimer = false;
                Destroy(gameObject);
            }
        }
    }

    private void OnTriggerEnter2D(Collider2D collision)
    //checks for collision and matches the condition for the timer
    //calls the add life function
    {
        Newtimer = true;
        FindObjectOfType<GameSessionManager>().AddLife();
    }
}
```

Here I have my first script that's attached to the powerups object itself and is used to destroy the object after 2 seconds as this gives time for my other script to start and also gives a better effect in the game.

I have done this using three separate functions update which calls the timers method every frame to update the timer then I have used nested if statements to first trigger that the player has collided with the power up this then starts the timer and keeps calling it till elapsed time is equal to 2 then resetting elapsed time and changing the condition to false so it doesn't continue to count infinitely then destroys the game object so the player can't use the same power up over and over

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The last function I used is the on collision enter to set the condition for Timer's function.

Then calls the Add Life method from the game session manager script

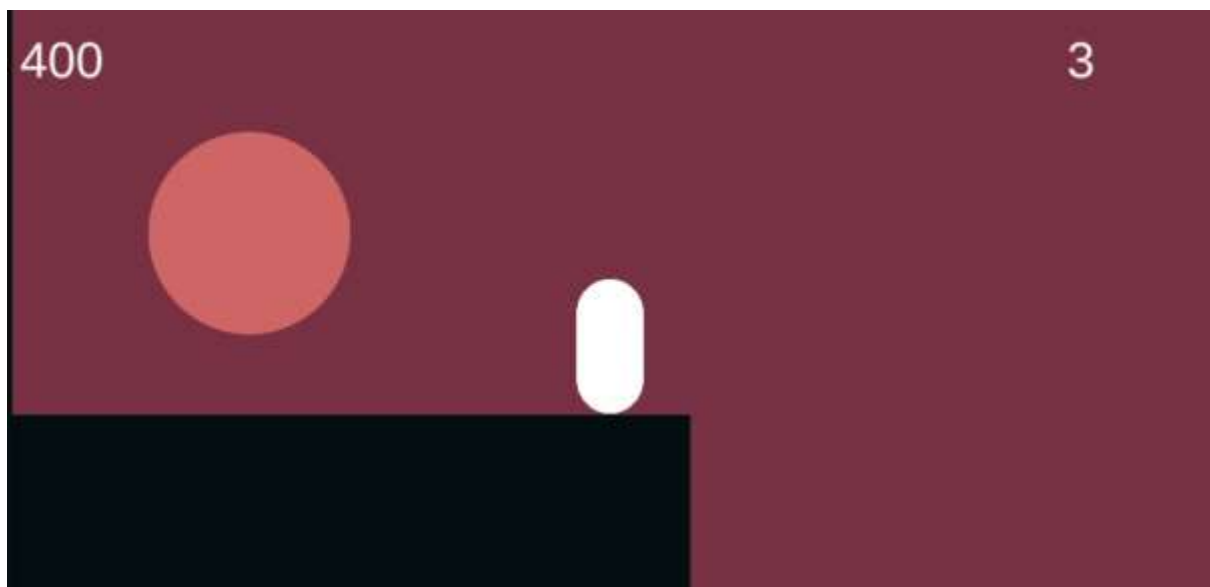
Below is my second script which is attached to the game session manager script as the variables are required to change are already in this script.

```
public void AddLife()  
//adds one to the players life  
//then updates the UI  
{  
    PlayerLives++;  
    Lives.text = PlayerLives.ToString();  
}
```

This is the method where it increases the players health by one and then updates the players UI

There's no need for another method like the other too as there no limit to how long this is in effect for.

Below is a video of this in action and the testing tables carried out by tester 2.



To find the video in file. it's in location project testing/iteration 1/PowerUps 3

| action | explanation | Expected result | Was the result expected? |
|--------|-------------|-----------------|--------------------------|
|--------|-------------|-----------------|--------------------------|

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|                                                                         |                                                                                                                                            |                                              |     |
|-------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|-----|
| On Collision with a health power up                                     | The play during the game will encounter power ups after the colliding with a power up it will have an effect on one of the players stats   | The players health increases by 1            | yes |
| On Collision with a power up it should stay in scene for only 2 seconds | When the player is playing after colliding with the powerup to give the computer time to change the values it stays in scene for 2 seconds | Power up only stays for 2 seconds            | yes |
| On collision with a health power up it should update the UI             | When the player is playing after colliding with the powerup the UI should go up by one to keep updated on their lives                      | Player health increase by 1 in the top right | yes |

Level loader

Below is my script for my NextLevel loader

```
using System.Collections;
using System.Collections.Generic;
using UnityEngine;
using UnityEngine.SceneManagement;

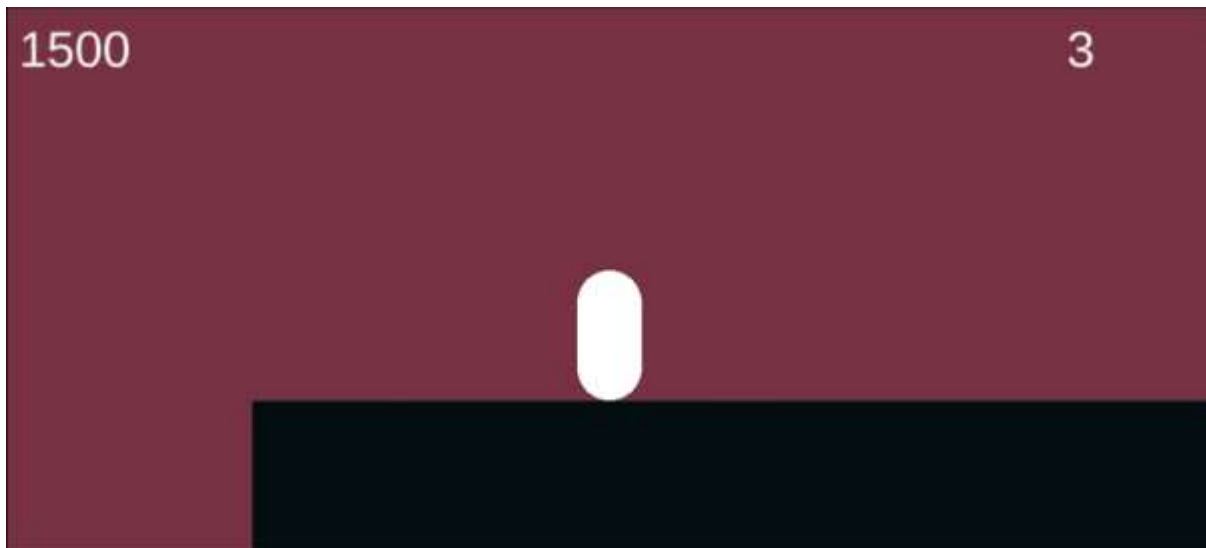
public class NextScene : MonoBehaviour
{
    private void OnTriggerEnter2D(Collider2D other)
    //this method finds the current scene then loads the next scene in order
    {
        int currentSceneIndex = SceneManager.GetActiveScene().buildIndex;

        SceneManager.LoadScene(currentSceneIndex + 1);
    }
}
```

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in this script I've used a single method which is triggered when the player enters the NextScene objects collider where first it finds the integer value of the current active scene then adds one to it and loads the new scene. As I have used static in previous variables my values stay persistent through scenes.



To find the video in file. it's in location project testing/iteration 1/Level Loader test

Below is the test carried out by tester 2 and the testing table filled out by tester 2

| action                                  | explanation                                      | Expected result                                | Was the result expected? |
|-----------------------------------------|--------------------------------------------------|------------------------------------------------|--------------------------|
| When the player enters the level loader | The player will be transported to the next level | The player will be loading into the next scene | yes                      |

## Enemies

Initially I made my A\* script below is my first iteration of the script. Below is the first iteration of the generate path script.

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```
public List<Node> GeneratePath(Node start, Node end)
{
    List<Node> openSet = new List<Node>();
    List<Node> allNodes = new List<Node>(FindObjectsOfType<Node>());
    List<Node> previousNodes = new List<Node>(new Node[allNodes.Count]);

    foreach (Node n in allNodes)
    {
        n.gScore = float.MaxValue;
    }

    start.gScore = 0;
    start.hScore = Vector2.Distance(start.transform.position, end.transform.position);
    openSet.Add(start);

    while (openSet.Count > 0)
    {
        int lowestF = default;

        for (int i = 1; i < openSet.Count; i++)
        {
            if (openSet[i].FScore() < openSet[lowestF].FScore())
            {
                lowestF = i;
            }
        }

        Node currentNode = openSet[lowestF];
        openSet.Remove(currentNode);

        if (currentNode == end)
        {
            List<Node> path = new List<Node>();
            path.Add(end);

            Node currentPathNode = currentNode;

            while (previousNodes[allNodes.IndexOf(currentPathNode)] != null)
            {
                currentPathNode = previousNodes[allNodes.IndexOf(currentPathNode)];
                path.Add(currentPathNode);
            }

            return path;
        }

        foreach (Node connectedNode in currentNode.connections)
        {
            float tentativeGScore = currentNode.gScore + Vector2.Distance(currentNode.transform.position, connectedNode.transform.position);

            if (tentativeGScore < connectedNode.gScore)
            {
                connectedNode.gScore = tentativeGScore;
                connectedNode.hScore = Vector2.Distance(connectedNode.transform.position, end.transform.position);

                previousNodes[allNodes.IndexOf(connectedNode)] = currentNode;

                if (!openSet.Contains(connectedNode))
                {
                    openSet.Add(connectedNode);
                }
            }
        }
    }

    return null;
}
```

Below the first iteration of the Nearest Node script

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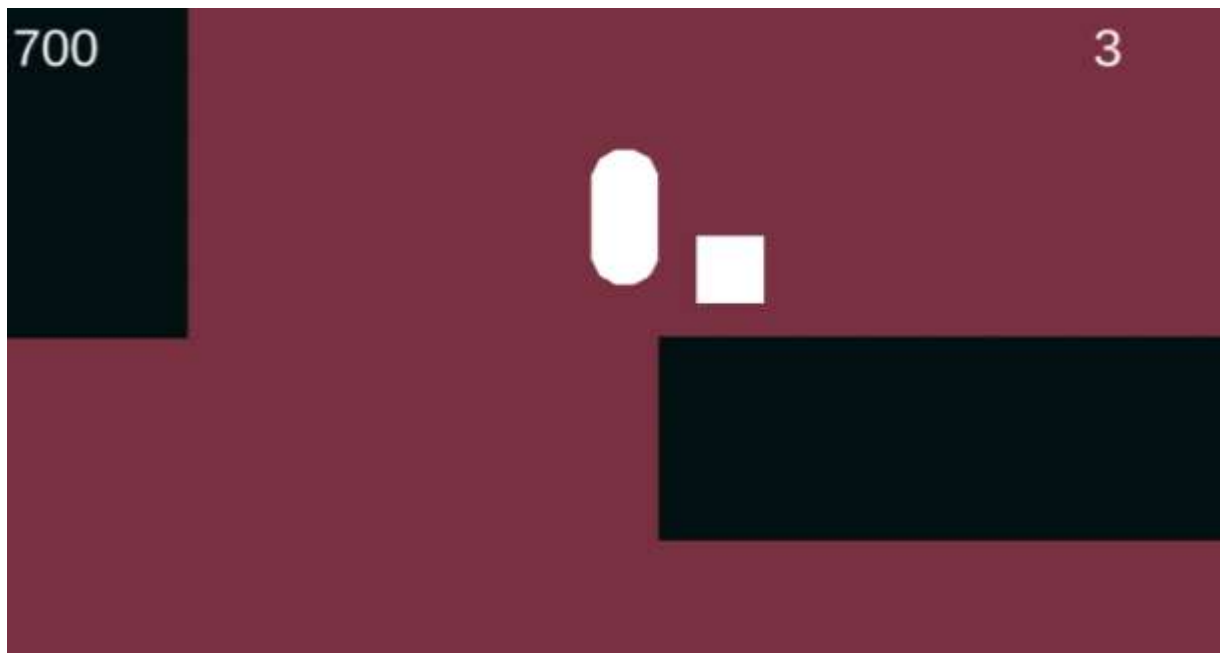
```
public Node NearestNode(Vector2 position)
{
    Node foundnode = null;
    float maxDistance = 0;

    foreach (Node node in NodesInScene())
    {
        float currentDistance = Vector2.Distance(position, node.transform.position);
        if (currentDistance > maxDistance)
        {
            maxDistance = currentDistance;
            foundnode = node;
        }
    }

    return foundnode;
}
```

First iteration of the NPC\_Controller script stays constant throughout.

After running this script, the game was found to crash. And I have to force quit. After reviewing the script, I was unable to find my error. So, I started doing research into potential coding issues that can crash unity. Below is a screen shot of this test ive used a screen shot as the game froze there is not enough content to justify a video



This is the information I found.

Damaged or missing game files can cause the game to malfunction. Bugs in the game code; Programming error, memory leaks, or other issues in the game code. After reading this I put the crashing down to there being an issue in the games code. This is because as I can run the game however as soon as the A\* manger script has to be run due to me moving away from the closest node it crashes.

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This means the error must reside in the generate path method or nearest node method.

I then did research into the operation in code that can cause unity to crash and of all the one I found only two where applicable to me. Operations using infinity/default float value and division by 0.

After reviewing both scripts with this now in mind I found after looking for the node with the lowest f score instead of setting it to the node found I set it too infinity.

This then meant that both arguments would be infinite so it would look like  $\infty < \infty$  most likely causing the crash. After change this to LowestF=1; this now made for iteration 2 of my scripts. Bellow can be found the 2 iterations of the script

```
public List<Node> GeneratePath(Node start, Node end)
{
    List<Node> openSet = new List<Node>();
    List<Node> allNodes = new List<Node>(FindObjectsOfType<Node>());
    List<Node> previousNodes = new List<Node>(new Node[allNodes.Count]);

    foreach (Node n in allNodes)
    {
        n.gScore = float.MaxValue;
    }

    start.gScore = 0;
    start.hScore = Vector2.Distance(start.transform.position, end.transform.position);
    openSet.Add(start);

    while (openSet.Count > 0)
    {
        int lowestF = default;

        for (int i = 1; i < openSet.Count; i++)
        {
            if (openSet[i].fScore() < openSet[lowestF].fScore())
            {
                lowestF = i;
            }
        }

        Node currentNode = openSet[lowestF];
        openSet.Remove(currentNode);

        if (currentNode == end)
        {
            List<Node> path = new List<Node>();
            path.Add(end);

            Node currentPathNode = currentNode;

            while (previousNodes[allNodes.IndexOf(currentPathNode)] != null)
            {
                currentPathNode = previousNodes[allNodes.IndexOf(currentPathNode)];
                path.Add(currentPathNode);
            }

            return path;
        }

        foreach (Node connectedNode in currentNode.connections)
        {
            float tentativeGScore = currentNode.gScore + Vector2.Distance(currentNode.transform.position, connectedNode.transform.position);

            if (tentativeGScore < connectedNode.gScore)
            {
                connectedNode.gScore = tentativeGScore;
                connectedNode.hScore = Vector2.Distance(connectedNode.transform.position, end.transform.position);

                previousNodes[allNodes.IndexOf(connectedNode)] = currentNode;

                if (!openSet.Contains(connectedNode))
                {
                    openSet.Add(connectedNode);
                }
            }
        }
    }

    return null;
}
```

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```
public Node NearestNode(Vector2 position)
{
    Node foundnode = null;
    float maxDistance = 0;

    foreach (Node node in NodesInScene())
    {
        float currentDistance = Vector2.Distance(position, node.transform.position);
        if (currentDistance > maxDistance)
        {
            maxDistance = currentDistance;
            foundnode = node;
        }
    }

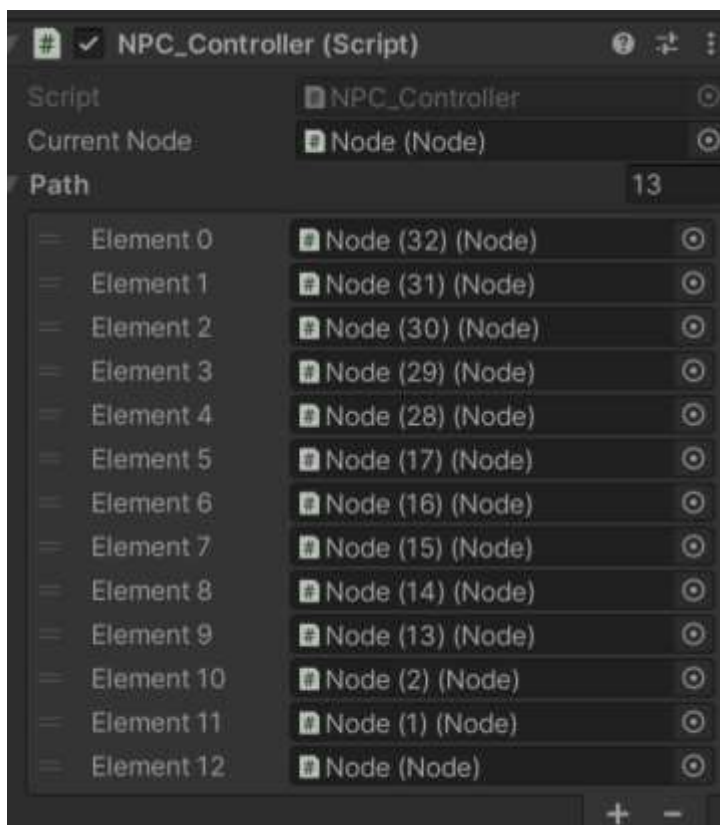
    return foundnode;
}
```

After running iteration 2 of the script as per the test below. It can be seen while the script running now the AI moves away from the player. as well as bouncing back and forth between nodes.

To find the video in file. it's in location project testing/iteration 1/A\* iteration2

As the enemy is moving further away this means the issue is either in the nearest node script or in the finding of the node with the lowest F score.

After looking in the inspector while the scripts running. This is what's shown.



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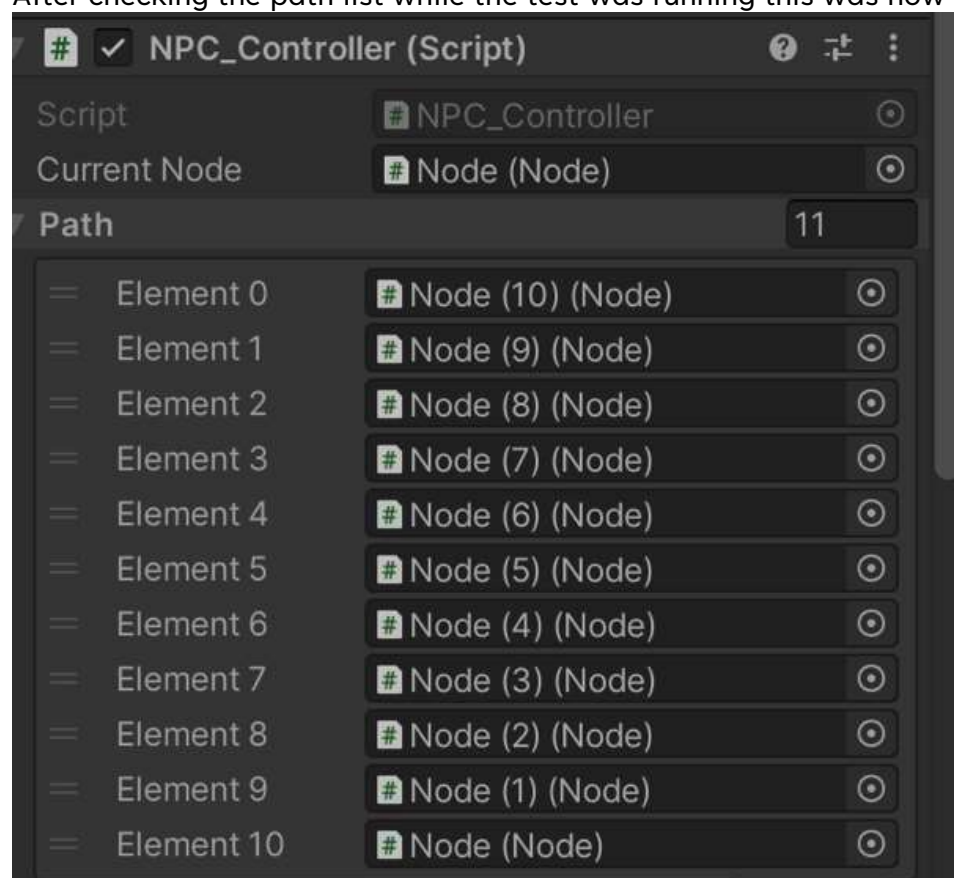
Candidate name: Oscar Winter

This shows it finding the furthest node from the player instead of the nearest confirming the issue is in the nearest node function. To fix this I needed to change the maxdistance to equal infinity. and flip the less than or equal too this now means instead of current distance increasing with each iteration of the loop now it will decrease until the closest node is returned. The script corresponding to the fixes can be found in the 4<sup>th</sup> iteration of the script.

After these fixes, the player did stop moving away from the player however now bounces between the paths. Below is the test showing this.

To find the video in file. it's in location project testing/iteration 1/A\* iteration3

After checking the path list while the test was running this was now seen



The player was placed closest to node 10 this means the script would be moving away from the player in the exact opposite direction to intended after reviewing the script while tracking the correct path and adding each element to the path list I never reversed it as the method I was using required backtracking.

To fix this after the loop I added path. Reverse; using the ins build reverse function. This concludes iteration 3 of the working A\* algorithm script which will be explained below along with tests.

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As this is one of my larger more complex scripts, I have decided to split this up into describing each method while explaining what the code does below is the first method I'll be talking about

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```

public List<Node> GeneratePath(Node start, Node end)
{
    List<Node> openSet = new List<Node>();
    List<Node> allNodes = new List<Node>(FindObjectsOfType<Node>()); // Store all nodes in a list
    List<Node> previousNodes = new List<Node>(new Node[allNodes.Count]); // Tracks the previous node for each node

    // Initialize all nodes
    foreach (Node n in allNodes)
    {
        // this says for every node which is in scene it calculates the cost from the start
        {
            n.gScore = float.MaxValue;
            // sets to infinity
        }

        start.gScore = 0;
        // indicates the starting node
        start.hScore = Vector2.Distance(start.transform.position, end.transform.position);
        // this finds an estimated cost from the start to end node
        openSet.Add(start);
        // this now adds the start node previously set to the list

        while (openSet.Count > 0)
        {
            // this makes it so while the list isn't empty
            {
                int lowestF = default;

                for (int i = 1; i < openSet.Count; i++)
                {
                    // then this finds the node with the lowest f score and sets it to lowest f
                    {
                        if (openSet[i].fScore() < openSet[lowestF].fScore())
                        {
                            lowestF = i;
                        }
                    }

                    Node currentNode = openSet[lowestF];
                    openSet.Remove(currentNode);
                    // here it then removes that node from the list

                    if (currentNode == end)
                    {
                        // checks if the current node is the end node
                        {
                            List<Node> path = new List<Node>();
                            // makes a new empty list for the path

                            path.Add(end);
                            // adds the goal node to the path list

                            Node currentPathNode = currentNode;

                            // Work backwards using the previousNodes list
                            while (previousNodes[allNodes.IndexOf(currentPathNode)] != null)
                            {
                                // now checks to see if we get to the start node as we work backwards
                                {
                                    currentPathNode = previousNodes[allNodes.IndexOf(currentPathNode)];
                                    // goes to the previous node sets it to the current node
                                    path.Add(currentPathNode);
                                    // then adds it to the path (now it's in reverse so we want to un-reverse)
                                }

                                path.Reverse();
                                // we do this by reversing the reversed path
                                return path;
                            }
                        }

                        foreach (Node connectedNode in currentNode.connections)
                        {
                            float tentativeGScore = currentNode.gScore + Vector2.Distance(currentNode.transform.position, connectedNode.transform.position);
                            // this finds the distance from start and adds the distance from the neighboring nodes

                            if (tentativeGScore < connectedNode.gScore)
                            {
                                // compares the values
                                {
                                    connectedNode.gScore = tentativeGScore;
                                    connectedNode.hScore = Vector2.Distance(connectedNode.transform.position, end.transform.position);

                                    // Update the previous node
                                    previousNodes[allNodes.IndexOf(connectedNode)] = currentNode;

                                    if (!openSet.Contains(connectedNode))
                                    {
                                        openSet.Add(connectedNode);
                                    }
                                }
                            }
                        }
                    }
                }

                return null;
            }
        }
    }
    // returns null if no path is found
}

```

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first in my generate path function my first data structure is setup. A list called openSet is used to hold nodes that are still to be explored I decided to use a list as it's a dynamic data structure meaning I can change its contents at runtime and is much easier to use compared to an array. (my list is used to store my items of a graph). Previous node list is also made to keep track of the previous node

Next, I set all nodes GScore to infinity or default this tells the program that no suitable path has been found yet. This then allows me to later ensure a fair starting point for the program

After this I then need to set the root node to a value the system can recognise to it can be used as the starting node, I do this but setting the root nodes GScore to 0 as there's no cost to move to itself. Next, I need to find a heuristic estimate from the start node to the goal node. Lastly the starting node is added to openSet so it's the first node to be assessed. This now means the node graph is ready to be to search.

Now I use a while loop to loop through open set until no nodes are left to be searched (the list the empty). I have a second loop that then goes through all other nodes finding the one with the lowest Fscore and setting that to my current node as it's the most promising path. This node is then removed from open set so it's not re assessed.

Next, I use an if statement checking to see if the current node is actually the goal node / end node if yes then a new empty list is created called path. I decided to use a list as it's a dynamic data structure meaning I can change its contents at runtime and is much easier to use compared to an array this will be storing the order of nodes to visit indicating the path the enemy should take from the start node to the goal node. Nodes are then added starting from the current node then by using my previous node list to find the previous item which is then systematically added using a while loop until the start node it the current node.

After this the path is now in the path list however this is in reverse order as we started from the end node working backwards to the start node. To get the path facing the correct direction I can use the in-built method reverse for lists to have the path facing the correct direction.

Next, I have a for each loop that is used to compare connections to see if a better path becomes available. I've done this by first calculating an approximate GScore by adding current GScore to the distance between the current node and the connected node then if the tentative score is lower than the current value. the node is updated as before using previous node as this means the node is closer so the previous node it then set to the current node. As the distance would have to be negative for this to happen. And updating the other variables so the new current node can go through the same procedure

Then the current node is checked to see if it's in openSet if it's not its added for more exportation so the path can change as the player moves.

Lastly if no path is found null is returned to show this

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Next, I have created another method as this needs to be broken into smaller more manageable sections. This method is called Nearest node this is used to find a good starting point for the generate path method above the code I made for this is below.

```
public Node NearestNode(Vector2 position)
{
    Node foundnode = null;
    float minDistance = float.MaxValue;

    foreach(Node node in NodesInScene())
        //loops through all nodes in scene till the node with the smallest vector 2 vaule is found
        {
            float currentDistance = Vector2.Distance(position, node.transform.position);
            if(currentDistance < minDistance)
            {
                minDistance = currentDistance;
                foundnode = node;
            }
        }

    return foundnode;
    //returns nearest node
}
```

The script first passes through the vector 2 position, so the script knows its location in scene. after this there's a foreach loop that goes through all nodes in scene from the nodes in scene method that I'll explain later where it first finds current distance by taking the position of the object to the nodes transform then these are compared until the closest node is found then the found node is set the node and found node is returned for later use.

Lastly for the A\* manager script I have a node in scene method that just goes through scene objects and returns all objects that are of type node.

```
public Node[] NodesInScene()
{
    return FindObjectsOfType<Node>();
    //returns all nodes in scene
}
```

As this is all in one script and not attached to any objects if I run the scene nothing will happen so I need to make a script that can all these methods from the external script when needed hence why it was important for me to make all methods public. The second script I used is called Non player character controller script as this will be the script attached to my enemy's.

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```
public class NPC_Controller : MonoBehaviour
{
    public Node currentNode;
    public List<Node> path;

    public PlayerMovement player;
    private float speed = 1;

    private void Start()
    {
    }

    private void Update()
    {
        if (path.Count == 0)
        {
            path = AStarManager.Instance.GeneratePath(currentNode, AStarManager.Instance.NearestNode(player.transform.position));
        }
        CreatePath();
    }

    void CreatePath()
    {
        if (path.Count > 0)
        {
            int x = 0;
            transform.position = Vector3.MoveTowards(transform.position, new Vector3(path[x].transform.position.x, path[x].transform.position.y, -2), speed * Time.deltaTime);

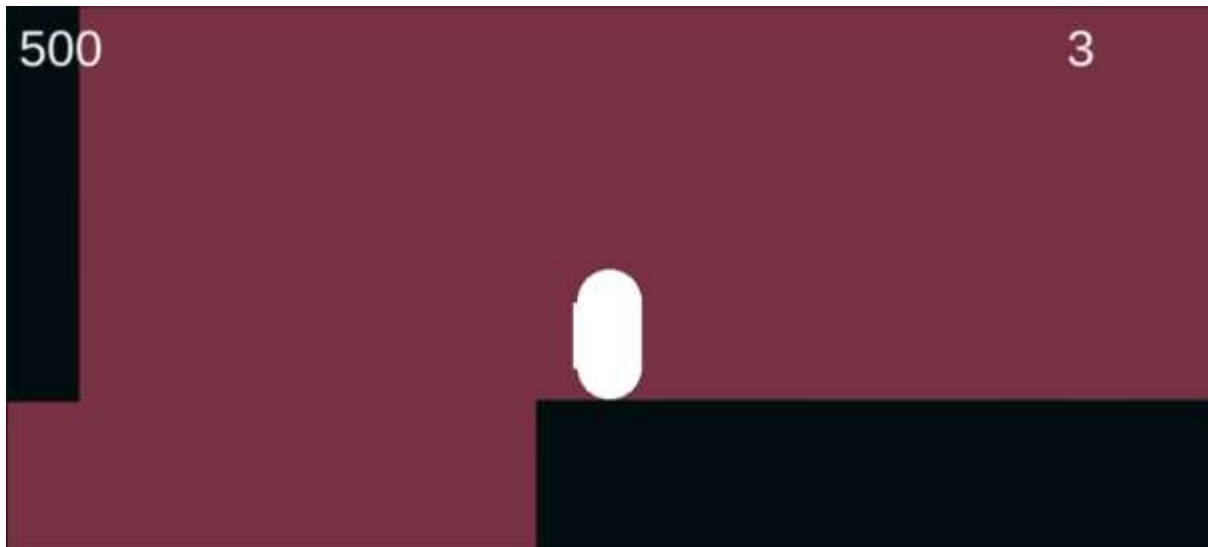
            if (Vector2.Distance(transform.position, path[x].transform.position) < 0.1f)
            {
                currentNode = path[x];
                path.RemoveAt(x);
            }
        }
    }
}
```

First in the update function as this is the only thing that will go in update rather than make a method its more efficient just to write directly in update. First in update I check to see if the path is empty using an if statement as if the list is empty, we need to use the singleton instance of the A\* manager to find the nearest node to the players position using nearestnode () then generatepath () to find a path between the npcs current node and the players nearest node. Then the create path method is called explained next.

The create path method is necessary as it handles actually setting the transform between nodes in scene of the enemy. I have done this by first checking there are nodes in path still to travel too as if there's no more nodes to travel to the npcs at the goal. However, if there are I set the npcs transform equal to the vector 3 then use move towards so its incremental for smooth movement I set this vector 3 value to the transform of the node in path that next to travel to and I use speed as time. delt time so speed is independent of frame rate. Then I used another if statement so that if the enemy's distance from the node it less than 0.1f it then moves to the next node to allow for smoother movement, so it doesn't move in a grid like pattern.

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Below I have my first tests to see if everything is working as intended before adding the loss of lives below is the video of the test, I conducted checking it was moving between nodes correctly.



To find the video in file. it's in location project testing/iteration 1/A\* test 1

Here I saw the script worked as intended so I can add the damage however the movement between nodes it too clear so ill fix this by increasing my if statement in the npc controller to move to the next. Node when you are almost there to a higher float value.

I have now made those amendments as the die method is already on the player script, I just needed to add the enemy to layer hazards and then add a collider set to is trigger. Below are the updates I made to the npc script where I increased the distance to move to the next node from 0.1f to 0.5f and multiplied the speed by 2 this gave me smooth and faster pace that I was going for. Below is my test conducted by tester 1.

```
using System.Collections;
using System.Collections.Generic;
using UnityEngine;

public class NPC_Controller : MonoBehaviour
{
    public Node currentNode;
    public List<Node> path;

    public PlayerMovement player;
    private float speed = 1;

    private void Start()
    {
    }

    private void Update()
    {
        if (path.Count == 0)
        {
            path = AIManager.Instance.GeneratePath(currentNode, AIManager.Instance.NearestNode(player.transform.position));
            CreatePath();
        }

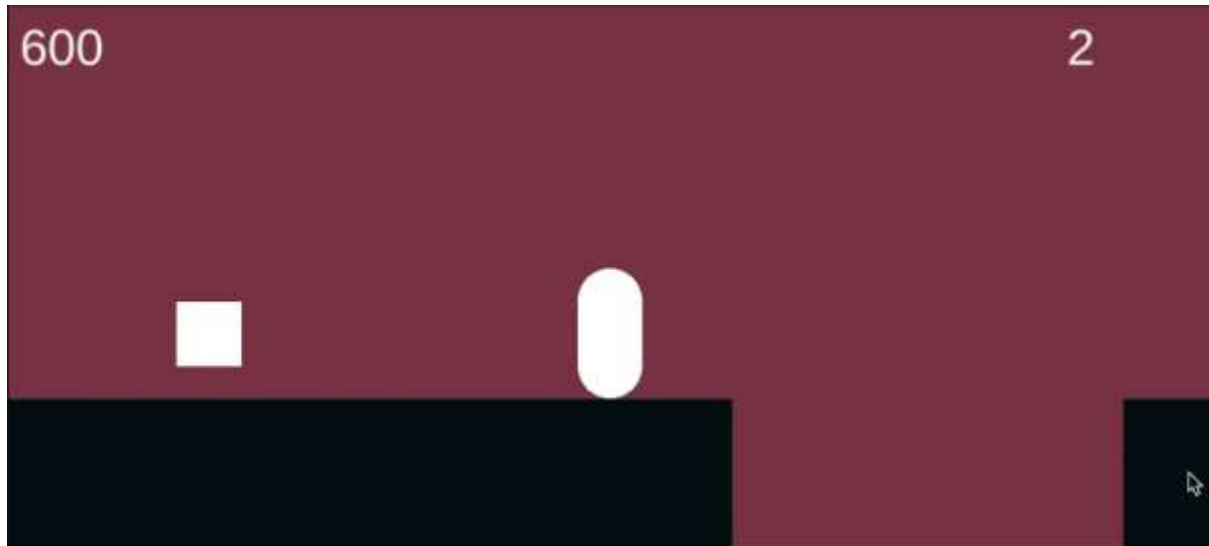
        void CreatePath()
        {
            if (path.Count > 0)
            {
                int x = 0;
                transform.position = Vector2.MoveTowards(transform.position, new Vector2(path[x].transform.position.x, path[x].transform.position.y, -2), speed * Time.deltaTime);

                if (Vector2.Distance(transform.position, path[x].transform.position) > 0.5f)
                {
                    currentNode = path[x];
                    path.RemoveAt(x);
                }
            }
        }
    }
}
```

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To find the video in file. it's in location project testing/iteration 1/A\* test 2



| action                        | explanation                                                                                                      | Expected outcome                                              | Was the result expected? |
|-------------------------------|------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|--------------------------|
| Enemies should engage         | When the player is within range of the node network the enemy should begin to follow the path calculated for it  | The enemy starts to follow the player around the node network | yes                      |
| Character jumping on a danger | The game has objects under a group call danger if the player does collide with a danger, they should lose a life | The player loses a life                                       | yes                      |

### Review of this iteration

In this iteration I have now completed the platformer game mode that consist of the player, enemies, powerups, coins and the level loader of which 2 of the powerups speed boost and jump boost and coin will be useful prefabs for my second iteration helping to speed up the implementation of the second iteration. My first iteration also has the 3 levels as specified. the ending scene menus previously spoken about will be

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added in my third iteration all about the menus of the game. This now concludes my first iteration where thoughts and input from each of my testers is below.

#### Tester 1

So far, I am happy with the platformer game modes development I like how the controls aren't cramped like it was in PacMan and the gameplay feels smooth with all features discussed working.

#### Tester 2

While playing the final product I was happy to see the game come together with its problem solving and challenges that it presents I'm especially fond of the time limit of the powerups and how it gives an extra sense of urgency, and I look forward to seeing what else the other parts of the game have to offer. as of now I'm happy with all the things implemented.

#### Iteration 2(Maze Racer)

Now as iteration 1 have been complete parts of code previously used can now be reused for things like power ups and coins, I can do this by using prefabs and slightly changing variable names to meet the circumstance. in this iteration will focus on the maze game mode which has an adjustable AI difficulty that the player has to race against.

#### Movement

Below is my script for the players movement

Candidate name: Oscar Winter

```
using System.Collections;
using System.Collections.Generic;
using UnityEngine;

public class Playermove : MonoBehaviour
{
    [SerializeField] float RotateSpeed = 60f;
    [SerializeField] float movespeed = 4f;

    void Start()
    {
    }

    // Update is called once per frame
    void Update()
    {
        float HSpeedfloat = Input.GetAxis("Horizontal") * RotateSpeed * Time.deltaTime;
        float Speedfloat = Input.GetAxis("Vertical") * movespeed * Time.deltaTime;
        //takes the standerent WASD inputs and converts them to vlaues

        transform.Translate(HSpeedfloat, 0, 0);
        transform.Translate(0, Speedfloat, 0);
        //sets players transform to the value previously determined
    }
}
```

First this in this player movement script I decided to not use unity's input systems as this won't give me the correct motion, I'm after so instead I've use unity's input. Get axis to use WASD and convert them into float amounts that I multiple by time. delta time to make them frame time independent where I can then change the player transform by translation as seen above.

Below are my testers testing table carried out by tester 1 and the video can be found at location project testing/iteration 2/Movement test

| action                          | explanation                                                                                                                             | Expected outcome                                                | Was the result expected? |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|--------------------------|
| Character bumping into surfaces | When traversing the game, the charter will collided with walls and floors this test is to make sure the collisions are all set properly | The player is unable to walk through/ fall through any surfaces | yes                      |

| Test data   | Expected outcome | Outcome |
|-------------|------------------|---------|
| W           | valid            | valid   |
| A           | valid            | valid   |
| S           | valid            | valid   |
| D           | valid            | valid   |
| Left arrow  | invalid          | invalid |
| Right arrow | invalid          | invalid |
| Down arrow  | invalid          | invalid |
| Up arrow    | invalid          | invalid |

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## Powerup

Next, I have my powerups for the maze racer where I will be reusing the first script that's attached to the object itself.

```
using System.Collections;
using System.Collections.Generic;
using UnityEngine;

public class Speedboost : MonoBehaviour
{
    CircleCollider2D myBodyCollider2D;
    // Start is called before the first frame update
    public bool Newtimer = false;
    public float elapsedTime = 0f;

    void Update()
    //this calls the timers method every frame
    {
        Timers();
    }

    void Timers()
    //this is used to start the timer and checks if the time reaches 2
    //then destroying the objects
    {
        if (Newtimer == true)
        {
            elapsedTime += Time.deltaTime;
            if (elapsedTime == 2f)
            {
                elapsedTime = 0f;
                Newtimer = false;
                Destroy(gameObject);
            }
        }
    }

    private void OnCollisionEnter2D(Collision2D collision)
    //checks for collision and matches the condition for the timer
    {
        Newtimer = true;
    }
}
```

This here is the same as my previous description as the script hasn't changed. Here I have my first script that's attached to the powerups object itself and is used to destroy the object after 2 seconds as this gives time for my other script to start and also gives a better effect in the game.

I have done this using three separate functions update which calls the timers method every frame to update the timer then I have used nested if statements to first trigger that the player has collided with the power up this then starts the timer and keeps calling it till elapsed time is equal to 2 then resetting elapsed time and changing the condition to false so it doesn't continue to count infinitely then destroys the game object so the player can't use the same power up over and over

Now on contact the powerup will disappear but as is the game engine doesn't know to change any variables. so next I have a method that attaches to the player movement script as this is what already has the Variable I require to change. Below are those methods. Using similar previous code.

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```
void Timers()
//this method checks the condition previous is true
//starts the timer and after 30s then resets all variable to previous values
{
    if (Hastouched == true)
    {
        elapsedTime += Time.deltaTime;
        if (elapsedTime >= 30f)
        {
            movespeed = 4f;

            RotateSpeed = 4f;

            elapsedTime = 0f;

            Hastouched = false;
            // Change the run speed and stop further updates
        }
    }
}

private void OnCollisionEnter2D(Collision2D collision)
//This method checks for collision and the correct object
//then sets the condition for Timer to true and increase run speed
{
    if (myBodyCollider2D.IsTouchingLayers(LayerMask.GetMask("Speedboost")))
    {
        Hastouched = true;

        movespeed = 10f;

        RotateSpeed = 10f;
    }
}
```

This is the first method I used which is on collision enter where first an if statement to check the object it collided with is on the layer Speed Boost. Sets has touched (the condition for the next method to be true). And increased the move speed to 10f and the Rotate speed to 10f as well so its proportional otherwise one axis will be moving faster than the other.

Now the condition for the Timers method has been it will now be triggered This is another method from the same script that is in update so after the previous method triggers the if condition to be true it then uses nested ifs as this is in update elapsed time will increase until its equal to 30 equivalent of 30 seconds then returns all variable to previous values. Move speed to 4f and rotate speed to 4f.

Then I have tester 2 test this the testing table is below, and the video of this test can be found at location

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project testing/iteration 2/SpeedPowerUp test

| action                                                                       | explanation                                                                                                                                          | Expected result                       | What the result expected? |
|------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|---------------------------|
| On Collison with a speed power up                                            | The play during the game will encounter power ups after the colliding with a power up it will have an effect on one of the players stats             | The player's speed has increased      | yes                       |
| On Collison with a power up it should stay in scene for only 2 seconds       | When the player is player is playing after colliding with the powerup to give the computer time to change the values it stays in scene for 2 seconds | Power up only stays for 2 seconds     | yes                       |
| On Collison with a power up (jump and speed types) its effects will last 30s | When the player is player is playing after colliding with the powerup to give the computer time to change the values it stays in scene for 2 seconds | Power up effects only last 30 seconds | yes                       |

Timers for Player and infrastructure for AI

Now I have the powerups and movement working next, I want to work on the timers for the game to return the value of the time it takes for either the player or the AI to finish the run. The reason my I can only lay out the infrastructure of my AI is that it hasn't been trained yet so I can't implement it however I will come back to this after implementing the AI. Below is the script I have made for the player and Ais finish times.

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```
using System.Collections;
using System.Collections.Generic;
using TMPro;
using UnityEngine;
using UnityEngine.SocialPlatforms.Impl;

public class FinishLine : MonoBehaviour
{
    [SerializeField] TextMeshProUGUI PlayerTimeUI;
    [SerializeField] TextMeshProUGUI AITimeUI;

    public bool PlayerComplete = false;
    public bool AIComplete = false;

    public float PlayerTimeScore = 0;
    public float AITimeScore = 0;

    new Collider2D myBodyCollider2D;
    //initialises all my variables

    // Start is called before the first frame update
    void Start()
    {
        ..
    }

    // Update is called once per frame
    void Update()
    {
        PlayerTime();
        AITime();
    }

    void PlayerTime()
    //checks the player hasn't completed yet then continues the timer and updating the UI
    {
        if (PlayerComplete != true)
        {
            PlayerTimeScore += Time.deltaTime;

            PlayerTimeUI.text = PlayerTimeScore.ToString();
        }
        else
        {
            Debug.Log(PlayerTimeScore);
        }
    }

    void AITime()
    //checks the AI hasn't completed yet then continues the timer and updating the UI
    {
        if (AIComplete != true)
        {
            AITimeScore += Time.deltaTime;

            AITimeUI.text = AITimeScore.ToString();
        }
        else
        {
            Debug.Log(AITimeScore);
        }
    }

    private void OnTriggerEnter2D(Collider2D collision)
    //once collided with conditions to true
    {
        PlayerComplete = true;
        AIComplete = true;
    }
}
```

First in this script I initialize all the variable I will need for the player and AIs timers. First method I have is the Player Time method where first I use an if statement that acts like a while loop as it's called in update. Where until the condition has been met (Player complete is true) is will add the time to the method and updates the UI. And for the AITime method this is the exact same. Then I have on trigger enter so when the player or AI reaches the finish line/has collision with the object it sets the conditions to true.

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Below is tester 2s testing table and the video corresponding is in location project testing/iteration 2/Timers test1

| Action                                                       | Explanation                                                                                                                                                                                                      | Expected outcome                                             | What the result expected? |
|--------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|---------------------------|
| Timer should tick for the player while they are racing       | the game needs to keep track of how long is taking the player to reach the finish lines so it can be returned at the end, so this test is to make sure the timer starts ticking after the player sets difficulty | The timer ticks                                              | yes                       |
| Timer should tick for the AI while its racing                | the game needs to keep track of how long is taking the AI to reach the finish lines so it can be returned at the end, so this test is to make sure the timer starts ticking after the player sets difficulty     | The timer ticks                                              | yes                       |
| Going over the finish line (timer should end) for the player | When the player reaches the finish line the timer should stop, and the end screen should be loaded and the final time returned                                                                                   | The timer end and end screen are shown, and time is returned | yes                       |
| UI should update the timer in the maze game                  | The player needs this to see their current statistics in game as it makes it easy for them to recognise if they are doing well                                                                                   | The UI should be updated on screen in the designated area    | yes                       |

Reinforcement AI

Next, I've decided to do my enemy AI as this links nicely into the infrastructure I set out for the timers previously while the structure of the algorithm is still fresh in my memory. In this section I will first train an AI using the ML agent's attachment in the unity registry after training the AI on my map for different periods of time for different

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strengths which I can check in the text file of the training results. Then I will load the brain into an object which will then allow it to move independently in game.

First, I download the unity package in the unity registry. After this I then need to use some command line / command line python. First, I need to make a virtual file environment, so the python scripts don't affect my system. This is done using this line of python.

```
Python -m venv venv
```

After this my virtual file environment has been created. Next, I need to open this venv I'll do this in command line by typing.

```
Venv/scripts/activate
```

Then I need to make sure all libraries, so I use the following commands to confirm this.

```
Python -m install --upgrade pip
```

```
Pip install mlagents
```

```
Pip3 install torch torchvision torchaudio
```

After this ml agents is now ready for use in the unity engine. Next, I made this script overriding the ml agent methods to attach to the object. During training. This can be found below

```
using Unity.MLAgents.Actuators;
using Unity.MLAgents.Sensors;
using Unity.MLAgents;
using UnityEngine;

// Unity Script (1 agent reference / 0 references)
public class MoveToTargetAgent : Agent
{
    [SerializeField] private Transform target;

    //imports the targets transform as the AI can use it as reference

    // references
    private void OnCollisionEnter2D(Collision2D collision)
    {
        //checks collisions if tag is recognised a reward is given then ends the episode and adds the observation
        {
            if (collision.tag == "FinishLine")
            {
                AddReward(+75);
                EndEpisode();
            }
            else if (collision.tag == "Wall")
            {
                AddReward(-35);
                EndEpisode();
            }
        }
    }

    // references
    public override void CollectObservations(VectorSensor Result)
    {
        //overrides observation method so at the end of the episode
        //it takes the ts transform as data to be fed to the neural network
        Result.AddObservation((Vector2)transform.position);
        Result.AddObservation((Vector2)target.position);
    }

    // references
    public override void OnActionReceived(ActionBuffers Movement)
    {
        //after data is evaluated in the neural network then the agents local position is adjusted accordingly
        transform.localPosition += new Vector3(Movement.ContinuousActions[0], Movement.ContinuousActions[1]) * Time.deltaTime;
    }
}
```

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In this script first I need to attach the transform element of the target (as is called in the APIs notes) to a variable so I can feed that data into the sensors later after each episode for evaluation by the neural network.

Next, I made a OnCollision enter method this is, so the Agent knows when it collides with particular objects, I did this by checking the tags of the object it collided with and then depending on the object either give a positive reward for reaching the target or a negative reward for hitting the walls. Then ending the episode.

Then I had to change the subclass type to agent this is so I can access the methods in the Mlagents API I need to override so I can send data and observation in and out of my project.

The first script I overwrote is the Collect observations script I did this one first as this is the order in which the events happen so in this method, I added two observations so I added to results the agents transform, and the targets transform which will be fed to the neural network

The next method I overwrote was the on action received this was so after previous values had been evaluated then it would change the agent's local position accordingly. Where I use its vector values and add on the Movement to each respective continues action 0 and 1 for x and y then multiplying by delta time so it's not frame time depended.

After these events repeat many times over the AI should learn. For my test to be successful I should see the agent moving in random directions this means the scripts working.

This is not in my test data as its only for training the AI and not actually I script active in the game so I will have I test I carried out as evidence the script is functional but there is no testing table to go with it.

So next I type `mlagents-learn --run-id=" insert name"`

This a command I installed earlier in the ml agent's packages, and this starts the training and after executing looks like this. So, when the scene is started in unity the training automatically starts then when I end the scene is automatically saves this is so I can then use the brain later.

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```
C:\Windows\System32\cmd.exe X + -
C:\Users\oscar\Mazevainiaiproto\venv\lib\site-packages\torch\__init__.py:749: UserWarning: torch.set_default_tensor_type() is deprecated as of PyTorch 2.1, please use torch.set_default_dtype() and torch.set_default_device() as alternatives. (Triggered internally at C:\actions-runner\work\pytorch\pytorch\builder\windows\pytorch\torch\csrc\tensor\python_tensor.cpp:433.)
  _C._set_default_tensor_type(t)

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```

Candidate name: Oscar Winter

```
default_settings: null
behaviors:
  My Behavior:
    trainer_type: ppo
    hyperparameters:
      batch_size: 1024
      buffer_size: 10240
      learning_rate: 0.0003
      beta: 0.005
      epsilon: 0.2
      lamdb: 0.95
      num_epoch: 3
      shared_critic: false
      learning_rate_schedule: linear
      beta_schedule: linear
      epsilon_schedule: linear
    network_settings:
      normalize: false
      hidden_units: 128
      num_layers: 2
      vis_encode_type: simple
      memory: null
      goal_conditioning_type: hyper
      deterministic: false
    reward_signals:
      extrinsic:
        gamma: 0.99
        strength: 1.0
      network_settings:
        normalize: false
        hidden_units: 128
        num_layers: 2
        vis_encode_type: simple
        memory: null
        goal_conditioning_type: hyper
        deterministic: false
    init_path: null
    keep_checkpoints: 5
    checkpoint_interval: 500000
    max_steps: 500000
    time_horizon: 64
    summary_freq: 50000
```

I then repeated this 3 time for 3 strengths 0.3, 0.7 and approximately 1 as you can see strength can be viewed above then I just have to copy this folder into my project assets and attach it to my object (Agent).

After this you can find my three final tests below carries out by tester 1 for the first two then tester 2 for the last one as well as the testing tables.

project testing/iteration 2/EasyAI

project testing/iteration 2/MediumAI

project testing/iteration 2/ImpossibleAI

| Action                                      | explanation                                                 | outcome                                      | What outcome expected? |
|---------------------------------------------|-------------------------------------------------------------|----------------------------------------------|------------------------|
| AI should be roughly of difficulty selected | The player selects the AI difficulty this is, so the player | The AI is of approximate difficulty selected | yes                    |

Candidate Number:3879

Candidate name: Oscar Winter

|                                                          |                                                                                               |                                    |     |
|----------------------------------------------------------|-----------------------------------------------------------------------------------------------|------------------------------------|-----|
|                                                          | picks a difficulty suitable to their ability so it's important this is functioning correctly  |                                    |     |
| Going over the finish line (timer should end) for the AI | When the player reaches the finish line the timer should stop, and the AI final time returned | The timer end and time is returned | yes |

Coins/full score

Below is my c# script I've written for coin pickup and object deletion re used from iteration 1.

```
using System.Collections;
using System.Collections.Generic;
using UnityEngine;

public class CoinPickup : MonoBehaviour
{
    [SerializeField] int PointsForCoin = 100;

    int CollectedPoints;

    //initializes all variables needed for the method

    void OnTriggerEnter2D(Collider2D other)
    //this method checks that what the coin has collided with is the player
    //then calls a separate method from the gamesession manager then destroying the game object
    {
        if (other.tag == "Player")
        {
            FindObjectOfType<GameSessionManager>().AddToScore(PointsForCoin);

            int CollectedPoints += 100;
            Debug.Log(CollectedPoints);
            Destroy(gameObject);
        }
    }
}
```

First, I have set all variables I'll need for the script in this case just the points for the score and I've set this as a serialized field so I'm able to adjust in the game engine without changing the script. This helps keep the script stable as not having to enter the script means there's a lower chance of accidental editing of the script.

To achieve my user interface, I have used the Unity Text Mesh Pro. This allows me to add things like scores to the screen using a canvas so all I need to do is set a variable and link the Text to it then just update the text every time the methods are called. I also need to import the TMP library so I can manipulate objects of this type.

Candidate Number: 3879

Candidate name: Oscar Winter

```
using System.Collections;
using System.Collections.Generic;
using UnityEngine;
using UnityEngine.SocialPlatforms.Impl;
using TMPro;

public class GameSessionManager : MonoBehaviour
{
    [SerializeField] TextMeshProUGUI Score;
    int CoinScoreTotal = 0;

    //here i initialized the CoinScoreTotal and score variable
    public void AddToScore(int pointsToAdd)
    //this method makes a running total of the coin score
    //then updates the UI it to check outputs
    {
        CoinScoreTotal += pointsToAdd;

        Score.text = CoinScoreTotal.ToString();
    }
}
```

This is the script after the stated changes. Now that coins are completed ill have tester 1 test this as tester 2 has already completed a coin test. The video showing this is below.

To find the video in file. it's in location project testing/iteration 2/Coins test

As seen in the video tester 1 collided with the coins and on collision the score in the top left went up and the object was destroyed as expected.

| Action                                                                                                   | explanation                                                                                                                     | Was outcome expected? |
|----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| Player colliding with coin. 100 will be added to the coin score (for both platformer and maze game mode) | This will be saved to tally the players score at the end                                                                        | yes                   |
| Player colliding with coin should cause the coin to be deleted from scene                                | This is so the player can continually collect the coins in the same place                                                       | yes                   |
| UI should update the coin score in the maze game mode                                                    | The player needs this to see their currents statistics in game as it makes it easy for them to recognise if they are doing well | yes                   |

Now for the full score to calculate this I will take my two performance parameters to make an overall performance score I plan to do this by taking my current coin score then dividing it by my current time then multiplying by 1000 to get the overall score.

Candidate Number:3879

Candidate name: Oscar Winter

Here is the script I made to calculate the overall score and update it. and my testing evidence and table below.

```
using System.Collections;
using System.Collections.Generic;
using UnityEngine;
using UnityEngine.SocialPlatforms.Impl;
using TMPro;

public class TotalScoreCreate : MonoBehaviour
{
    [SerializeField] TextMeshProUGUI totalScore;

    public FinishLine finishLine;
    public GameManager sessionManager;

    //loads the other script from which variables i need to use are
    // Start is called before the first frame update
    void Start()
    {
    }

    // Update is called once per frame
    void Update()
    {
        //allows for real time calculating of the total score from all the scripts and updates the UI
        float TotalScore = (GameManager.CoinScoreTotal / finishLine.PlayerTimeScore) * 1000;
        totalScore.text = (((int)TotalScore)).ToString();
    }
}
```

To find the video in file. it's in location project testing/iteration 2/TotalScore test.

| Action                                           | Explanation                                                                                                                    | Expected outcome                                          | What the result expected? |
|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|---------------------------|
| UI should update the score in the maze game mode | The player needs this to see their current statistics in game as it makes it easy for them to recognise if they are doing well | The UI should be updated on screen in the designated area | yes                       |

## Conclusion

In this iteration I have now completed the Maze racer game mode which consisted of the powerups, reinforcement learning AI, Coins/total score, player movement and the finish lines tracking time. the ending scene menus previously spoken about will be added in my third and next iteration all about the menus of the game. This now concludes my first iteration where thoughts and input from each of my testers is below.

Tester 1

Candidate Number:3879

Candidate name: Oscar Winter

So far, I am happy with the Maze game modes development I like how the timers help to encourage faster timing, and the gameplay feels smooth with all features discussed working. I await the game loop being complete in the final iteration bringing the game full circle

Tester 2

All features are as discussed, and the development is going smoothly I enjoy the challenge of trying to remember routes and problem solving. To try and find a most efficient path with is especially helpful trying to beat the impossible AI. After talking to tester 1 I am also looking forward to the menus to help complete the full game loop

Iteration 3 (menus)

Registration / Login system

First, I started with creating my registration page I've done this using file handling for permanent data storage the screen shot of all methods of the script is below.

```
using System;
using System.Collections;
using System.Collections.Generic;
using System.IO;
using UnityEngine;
using UnityEngine.UI;
using UnityEngine.SceneManagement;
using System.Linq;
using TMPro;

public class R : MonoBehaviour
{
    public InputField UserINP;
    public InputField PasINP;
    public Button RegBTN;
    public Button GTLBTN;

    public TextMeshProUGUI Warnings;

    //creating variables to be used

    ArrayList Creds;

    // Start is called before the first frame update
    void Start()
    {
        RegBTN.onClick.AddListener(WriteCredentials);
        GTLBTN.onClick.AddListener(GTLP);

        //sets so it checks for the state of the button
        //and if its true calls the method in brackets

        FileExist();
        //calls file exists method
    }
}
```

In this first screen shot I first import and library's that are required notable ones being Unity Engie. UI so while I'm coding it understands keyword such as button or input field next, I created my class called R standing for registration then I initialised all variables I'll be using throughout the script and declared my array list the reason I chose to use an array list is so standard operations can be performed. And then I begin my start method which is a method that is called once before the first frame. Then then I call my button variables and say once they are triggered / clicked it will call the method in the parameter I applied this technique to both my registration button and my go to login button. sets so it checks for the state of the button and if it's true calls the method in brackets. Then lastly, I call my file exist method that will be described later on.

```
}  
  
void GTLP()  
//loads loginpage from my scenes when called  
{  
    SceneManager.LoadScene("LoginPage");  
}
```

Next is a short method that is used to load the login page using the scene manager where it takes a string parameter of the scene name and is in the button listener previously stated so when the login button is pressed it loads the login page scene.

```
void FileExist()  
//this method checks if the file path is already on the devices local storage  
{  
    if (File.Exists(Application.dataPath + "/Cred.txt"))  
    {  
        Creds = new ArrayList(File.ReadAllLines(Application.dataPath + "/Cred.txt"));  
        //adds all data from the text file into the arraylist so it can be searched  
        //more efficiently  
    }  
    else  
    {  
        File.WriteAllText(Application.dataPath + "/Cred.txt", "");  
        //creates a file if not already  
    }  
}
```

My third method is my fileexist method this uses an if statement to check to see if the file is stored on the local device if the Boolean value of this is found to be true it then adds all the data from the text file into my array list as the array list is much easier to search through compared to straight off the text file as well as being more efficient. However, if this is found to be false then a new file is created with the name /Cred.txt.

Candidate name: Oscar Winter

```
void WriteCredentials()
//method used to write the players data to the text file
{
    bool Exists = false;
    //initializes new boolean variable

    Creds = new ArrayList(File.ReadAllLines(Application.dataPath + "/Cred.txt"));

    //adds all data from the text file into the arraylist so it can be searched
    //more efficiently
    string Validation = PasINP.text;

    //making a temp variable for validation later on
    foreach (var i in Creds)
    //loops through the array list
    {
        if (i.ToString().Contains(UserINP.text))
            //if the string is in the file already sets exists to true
            {
                Exists = true;
            }
    }
    if (Exists)
        //warning text is changed to show usernames taken
        {
            Warnings.text = "UserName is taken";
        }
    else
        //if username isnt taken
        {
            if (Validation.Length < 8)
                //validation checks if password is long enough
                {
                    Warnings.text = "password needs to be at least 8 characters";
                }
            else
                //if passwords long enough
                {
                    bool containsInt = Validation.Any(char.IsDigit);
                    //as a boolean value to check if the input has a number

                    if (containsInt)
                        //if true it then adds details to a file
                        {
                            Creds.Add(UserINP.text + ":" + PasINP.text);
                            File.WriteAllLines(Application.dataPath + "/Cred.txt", (String[])Creds.ToArray(typeof(string)));
                            Warnings.text = "Account Registered";
                        }
                    else
                        //otherwise produces a warning
                        {
                            Warnings.text = "password needs to contain a number";
                        }
                }
        }
}
```

In this new method write Credentials I first create my initial Boolean value to be used later for checking the username entered isn't already in use. Next, I add everything to my array list so I can use standard operations on it I will be using a foreach loop. Then I set a temporary variable to check my validation. Now I use the foreach loop to iterate through all items of my list using .contains to return a Boolean value into the if statement where if the username already exists it sets the condition for my next if statement where a warning is outputted stating the username is already in use. Otherwise, if the username is not in use. it then uses my previously set temporary variable to check its length then if this isn't correct another warning is outputted stating the password has to be at least 8 characters. However if the password has a length longer than 8 it is then checked for a digit I've done this by setting a new Boolean variable and then taking the users input and then .any(char.isdigit) this takes each char in the string then uses is digit to check whether any of the values return true ( a digit was found) then depending of the output uses file handling to add the users inputs for username and password with a colon between to the creds array list then

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Candidate name: Oscar Winter

opening the file stating the data path then writing the data as that previously added to point in the array list and outputting the message account registered. Otherwise, it will produce a warning saying the password needs to contain a number.

Below are the testing tables filled by tester 2 and the video of the test can be found at file location.

project testing/iteration 3/Registration test

| Action                                                  | explanation                                                                                                    | Expected outcome                                                      | Was outcome expected |
|---------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|----------------------|
| registration scene is loaded first                      | The first menu to be displayed to the user should be the registration menu                                     | User is presented with the registration screen                        | yes                  |
| Player using input fields                               | When the player is trying to sign in or create a new account the user should be able to use the keyboard       | Letters appear in the input field                                     | yes                  |
| The player is able to use their mouse to navigate menus | Most common interfaces use mouses to navigate so the player will be able to intuitively use the mouse controls | When the player moves the mouse, the cursor will move correspondingly | yes                  |
| Clicking on register button                             | This should register the user to the register page so they can create an account                               | The player is registered                                              | yes                  |
| Clicking on login in button                             | Player should be presented with the login scene                                                                | Login scene is loaded                                                 | yes                  |

| Test input         | Expected outcome                     | Outcome |
|--------------------|--------------------------------------|---------|
| Right click button | Triggers command connected to button | valid   |
| Left click button  | Invalid                              | invalid |

Next, I need to move onto the login system the script I wrote for this is below split into the different methods I've used many of which have been adapted from the registration system.

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```
}  
void login()  
{  
    bool isExists = false;  
  
    //initializes boolean variable  
  
    Creds = new ArrayList(File.ReadAllLines(Application.dataPath + "/Creds.txt"));  
  
    //adds to the creds array list so standard operations can be performed  
  
    foreach (var i in Creds)  
    //loops through all variable/ lines in the file  
    {  
        string line = i.ToString();  
  
        if (i.ToString().Substring(0, i.ToString().IndexOf(":"))==UserINP.text &&  
            i.ToString().Substring(i.ToString().IndexOf(":") + 1)==(PasINP.text))  
            //uses and if statement to essentially parse each side of the colons  
            //where they are individually evaluated for a match  
            {  
                isExists = true;  
                //sets is exists to true as the user exists  
                break;  
            }  
    }  
  
    if (isExists)  
    //takes user to login page  
    {  
        loginpage();  
    }  
}
```

In this first screen shot I first import and library's that are required notable ones being Unity Engine. UI so while I'm coding it understands keyword such as button or input field next, I created my class called login then I initialised all variables I'll be using throughout the script and declared my array list the reason I chose to use an array list for the same reasons as my registration script. And then I begin my start method which is a method that is called once before the first frame. Then then I call my button variables and say once they are triggered / clicked it will call the method in the parameter I applied this technique to both all buttons. sets so it checks for the state of the button and if it's true calls the method in brackets. Then lastly check the file is on the local machine.

Candidate name: Oscar Winter

```
void login()
{
    bool isExists = false;

    //initializes boolean variable

    Creds = new ArrayList(File.ReadAllLines(Application.dataPath + "/Creds.txt"));

    //adds to the creds array list so standard operations can be performed

    foreach (var i in Creds)
    //loops through all variable/ lines in the file
    {
        string line = i.ToString();

        if (i.ToString().Substring(0, i.ToString().IndexOf(":")).Equals(UserINP.text) &&
            i.ToString().Substring(i.ToString().IndexOf(":") + 1).Equals(PasINP.text))
            //uses and if statement to essentially parse each side of the colons
            //where they are individually evaluated for a match
            {
                isExists = true;
                //sets is exists to true as the user exists
                break;
            }
    }

    if (isExists)
    //takes user to login page
    {
        loginpage();
    }
}
```

Next, I have my login method which is attached to my regBTN. so, when regBTN is pressed these methods triggered. First, I initialize the Boolean variable isExists to be used to determine whether the user is already registered. Next, I use the same technique I've used previously where I add all the lines of the file storing credentials into an array list so I can perform standard operations like loops as I've used before and now, I'll be using string formatting to parse each variable I inside Creds at the colon so they can be evaluated separately. first, I use the foreach loop to go through all variable in Creds. then using an if statement I find the left of the colon and compare it to the user inputs this is possible as they are both string values. then also if the right side of the semi colon is equal to the players inputted password that are then on the list and is Exists is set to true and the first indent if statement is valid and the login page method is called. else nothing happens.

```
void moveToRegister()
//takes player back to register on press
{
    SceneManager.LoadScene("Login_registration");
}

void loginpage()
//takes user to login page
{
    SceneManager.LoadScene("GameModeSelection");
}
```

Then my last two methods are used so when called they load the designated scene.

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After attaching this to my appropriate object I let tester two test it before is the location you can find the test. This test was not successful when clicking the button after entering correct details the player is not taken the game mode selection scene. This meant either my button listener wasn't working or is Exists never became true so there's an issue with the logic.

This test can be found at location project testing/iteration 3/login (failed)

So first I wanted to troubleshoot the easier of the two.

I did this by initially setting is exists to true so the script then when running the programme, the button worked as intended. So, this meant it was a logic error the evidence of this test can be found at file location project testing/iteration 3/login button troubleshooting

After reading the code several times I realised my mistake was I mistook my file path and a variable and had it looking for Creds.txt when it should have been looked for Cred.txt so I changed the file path to Cred.txt. after this tester 1 tester this the video evidence and the testing table can be found below.

project testing/iteration 3/login

project testing/iteration 3/login return button

| Action                                                  | explanation                                                                                                    | Expected outcome                                                      | Was the outcome expected? |
|---------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|---------------------------|
| Player using input fields                               | When the player is trying to sign in or create a new account the user should be able to use the keyboard       | Letters appear in the input field                                     | yes                       |
| The player clicks the return button                     | Its button so the user is able to go the previous page after finishing with their current page                 | The player is taken to the previous scene                             | yes                       |
| The player is able to use their mouse to navigate menus | Most common interfaces use mouses to navigate so the player will be able to intuitively use the mouse controls | When the player moves the mouse, the cursor will move correspondingly | yes                       |

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|                            |                                                                    |                                             |     |
|----------------------------|--------------------------------------------------------------------|---------------------------------------------|-----|
| Clicking on sign in button | If credentials match the user is taken to the. Game selection page | Takes the player to the game selection page | yes |
|----------------------------|--------------------------------------------------------------------|---------------------------------------------|-----|

| Test input                | Expected outcome                     | Outcome |
|---------------------------|--------------------------------------|---------|
| Right click button        | Triggers command connected to button | valid   |
| Left click button         | Invalid                              | invalid |
| Right click return button | Loads the previous scene             | valid   |
| Left click return button  | Invalid                              | invalid |

Game mode selection menu

The script used for the game selection menus can be found below.

Candidate name: Oscar Winter

```
using System;
using System.Collections;
using System.Collections.Generic;
using System.IO;
using UnityEngine;
using UnityEngine.UI;
using UnityEngine.SceneManagement;

public class ModeSelection : MonoBehaviour
{
    public Button MazeBTN;
    public Button PlatformerBTN;
    public Button LeaderBoardBTN;

    //initializing variables to attach to objects

    void Start()
    {
        MazeBTN.onClick.AddListener(MazeRacer);

        PlatformerBTN.onClick.AddListener(Platformer);
        LeaderBoardBTN.onClick.AddListener(LeaderBoard);
        //adds a listener so when the buttons pressed
        //the method is called
    }

    void LeaderBoard()
    //loads leaderboard scene
    {
        SceneManager.LoadScene("LeaderBoard");
    }

    void MazeRacer()
    //loads the difficulty selection scene
    {
        SceneManager.LoadScene("DifficultySelection");
    }

    void Platformer()
    //loads the platformer scene
    {
        SceneManager.LoadScene("Platformer");
    }
}
```

In this script I first initialize all variables I will need in the script. Their variables are public so they show in the inspector, and I can attach the objects to the variable (the buttons). Then in the start method, which is called before the first frame, I added a listener so when the buttons are left-clicked by the user, it calls the method attached. My next method, Leader Board, loads the Leader Board scene. Next, the Maze racer method loads the difficulty selection scene. Lastly, the platformer method loads the platformer scene.

I had a tester test this. The testing tables are below, and the location of the evidence of the tests can be found.

Candidate Number: 3879

| Action                                                  | explanation                                                                                                    | Expected outcome                                                      | Was the outcome expected? |
|---------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|---------------------------|
| The player is able to use their mouse to navigate menus | Most common interfaces use mouses to navigate so the player will be able to intuitively use the mouse controls | When the player moves the mouse, the cursor will move correspondingly | yes                       |
| Clicking on the platformer button                       | This takes the player to the first level of the platformer so start the levels                                 | Loads the scene for the first game level                              | yes                       |
| Clicking on leader board button                         | Takes the player to the leader board where they can view the top scores                                        | Loads the scene for the leader board                                  | yes                       |
| Clicking on the maze solver button                      | When clicking the maze solver button, it should take the user to the difficulty selection screen               | Loads the scene for difficulty selection                              | yes                       |

project testing/iteration 3/game mode selection menu (LeaderBoard button)

project testing/iteration 3/game mode selection menu (Platformer button)

project testing/iteration 3/game mode selection menu (maze racer button)

| Test input         | Expected outcome                     | Outcome |
|--------------------|--------------------------------------|---------|
| Right click button | Triggers command connected to button | valid   |
| Left click button  | Invalid                              | invalid |

## Ending screens

Next, I'll be coding the end screens to each the platformer and the Maze racer game modes I've decided to implement them in the same section as they use very similar code and will only seem repetitive if I gave them each a separate section. Below is each of the scripts I wrote for the end scenes and what the scripts do.

```
using System;
using System.Collections;
using System.Collections.Generic;
using System.IO;
using UnityEngine;
using UnityEngine.UI;
using UnityEngine.SceneManagement;
using TMPro;
using Unity.VisualScripting;

public class EndMaze : MonoBehaviour
{
    public Button HOME;

    public TextMeshProUGUI YOURTIME;

    public TextMeshProUGUI AITIME;

    public TextMeshProUGUI YOURSCORE;

    //initializing variables to attach to objects

    void Start()
    {
        HOME.onClick.AddListener(Home);
        //then i add a listener
        //so when the buttons pressed it calls the Home method
    }
    void Home()
    {
        SceneManager.LoadScene("GameModeSelection");
        //loads GameMode selection scene
    }
    // Update is called once per frame
    void Update()
    {
        YOURSCORE.text = (TotalScoreCreate.TotalScore1).ToString();
        YOURTIME.text = (FinishLine.PlayerTimeScore).ToString();
        AITIME.text = (AIFinishLine.AITimeScore).ToString();
        //set text equal to desired variable every frame
    }
}
```

First, I create the variables I will need to attach my objects to in the inspector. then in the start method I add a listener so when clicked the home method is called. Then the void home method loads the game mode selection scene as this is the first screen the user is presented with after login in.

Next in the update method which is called every frame I set the text in each of the TMPro variables to the correlating variables. As the variables I need are in other scripts I first reference their class after this the IED then loads the variables from that class into this one, so I then specify the variable I want and change it to a string as it'll either be an int or a float.

Candidate name: Oscar Winter

Now I had tester 2 test this for me the video of the test is below as well as the testing table.

project testing/iteration 3/MazeEnd test

note: in the video it may look like your score never changed however its value in that scene is initialized at 999 so when it displays 0 as tester 2 ignored all coins its value was updated to 0

| Action                                                  | explanation                                                                                                            | Expected outcome                                                      | Was the outcome expected? |
|---------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|---------------------------|
| UI should display end score at end of maze game mode    | The player needs this to see their statistics in game as it makes it easy for them to recognise if they are doing well | The UI should be updated on screen in the designated area             | yes                       |
| The player clicks the return button                     | Its button so the user is able to go the previous page after finishing with their current page                         | The player is taken to the previous scene                             | yes                       |
| The player is able to use their mouse to navigate menus | Most common interfaces use mouses to navigate so the player will be able to intuitively use the mouse controls         | When the player moves the mouse, the cursor will move correspondingly | yes                       |

| Test input         | Expected outcome                     | Outcome |
|--------------------|--------------------------------------|---------|
| Right click button | Triggers command connected to button | valid   |
| Left click button  | Invalid                              | invalid |

Next for the platformer end I'll be using very similar coding techniques as they largely have to do the same job.

Candidate name: Oscar Winter

```
using System;
using System.Collections;
using System.Collections.Generic;
using System.IO;
using UnityEngine;
using UnityEngine.UI;
using UnityEngine.SceneManagement;
using TMPro;
using Unity.VisualScripting;

public class PlatformerEnd : MonoBehaviour
{
    public Button HOME;

    public TextMeshProUGUI CoinScore;

    //initializing variables to attach to objects

    // Start is called before the first frame update
    void Start()
    {
        HOME.onClick.AddListener(Home);
        //then i add a listener
        //so when the button is pressed it calls the Home method
    }
    void Home()
    {
        SceneManager.LoadScene("GameModeSelection");
        //loads GameMode selection scene
    }

    // Update is called once per frame
    void Update()
    {
        CoinScore.text = (GameSessionManager.CoinScoreTotal).ToString();
        //set text equal to desired variable every frame
    }
}
```

As seen this script uses identical structure to the previous script the only changes are that there's only one metric to display at the end of the game and the metric its displaying is unique to that game mode so therefore is a different variable.

The testing for this was carried out by tester 1 the video can be found at location shown below as well as the testing tables.

project testing/iteration 3/PlatformerEnd test

| Action                                               | explanation                                                                                                            | Expected outcome                                          | Was the outcome expected? |
|------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|---------------------------|
| UI should display end score at end of maze game mode | The player needs this to see their statistics in game as it makes it easy for them to recognise if they are doing well | The UI should be updated on screen in the designated area | yes                       |

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|                                                         |                                                                                                                |                                                                       |     |
|---------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|-----|
| The player clicks the return button                     | Its button so the user is able to go the previous page after finishing with their current page                 | The player is taken to the previous scene                             | yes |
| The player is able to use their mouse to navigate menus | Most common interfaces use mouses to navigate so the player will be able to intuitively use the mouse controls | When the player moves the mouse, the cursor will move correspondingly | yes |

| Test input         | Expected outcome                     | Outcome |
|--------------------|--------------------------------------|---------|
| Right click button | Triggers command connected to button | valid   |
| Left click button  | Invalid                              | invalid |

#### Difficulty Selection menu

Now for the difficulty selection menu this will use similarly concepts to the buttons used previously where each button will load a separate scene that has each respective brain loaded in it.

Below is the used script.

Candidate name: Oscar Winter

```
using System;
using System.Collections;
using System.Collections.Generic;
using System.IO;
using UnityEngine;
using UnityEngine.UI;
using UnityEngine.SceneManagement;

public class LevelSelection : MonoBehaviour
{
    public Button EasyBTN;
    public Button HardBTN;
    public Button ImpossibleBTN;
    //initializing variables attached to button objects

    void Start()
    //called before the first frame
    {
        EasyBTN.onClick.AddListener(Easy);
        HardBTN.onClick.AddListener(Medium);
        ImpossibleBTN.onClick.AddListener(Impossible);
        //then i add a listener to each difficulty type button
        //so when the buttons pressed it calls the respective method
    }

    void Impossible()
    {
        SceneManager.LoadScene("Maze");
        //loads Impossible game mode scene
    }

    void Easy()
    {
        SceneManager.LoadScene("MazeEasy");
        //loads Easy game mode scene
    }

    void Medium()
    {
        SceneManager.LoadScene("MazeMedium");
        //loads Medium game mode scene
    }
}
```

First, I create the variables I will need to attach my objects to in the inspector (the buttons). then in the start method I add a listener so when clicked the method in the brackets is called. Then I have an impossible method that calls the maze scene that has the impossible brain loaded. Then similarly the Easy and Medium methods call the maze scene that has the easy brain loaded and the medium brain.

After this I then had tester 1 test this for all three buttons below are the locations you can find the videos. And the testing table.

project testing/iteration 3/EasyButtonTest

project testing/iteration 3/MediumButtonTest

project testing/iteration 3/ImpossibleButtonTest

| Action                                   | explanation                | Expected outcome                 | Was the outcome expected? |
|------------------------------------------|----------------------------|----------------------------------|---------------------------|
| The player is able to use their mouse to | Most common interfaces use | When the player moves the mouse, | yes                       |

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Candidate name: Oscar Winter

|                                          |                                                                                               |                                           |     |
|------------------------------------------|-----------------------------------------------------------------------------------------------|-------------------------------------------|-----|
| navigate menus (for all three)           | mouses to navigate so the player will be able to intuitively use the mouse controls           | the cursor will move correspondingly      |     |
| Clicking on easy difficulty button       | The user clicks this button to load a particular difficulty of brain into the maze solving AI | Loads Scene for the easy difficulty       | yes |
| Clicking on medium difficulty button     | The user clicks this button to load a particular difficulty of brain into the maze solving AI | Loads Scene for the medium difficulty     | yes |
| Clicking on impossible difficulty button | The user clicks this button to load a particular difficulty of brain into the maze solving AI | Loads Scene for the impossible difficulty | yes |

| Test input         | Expected outcome                     | Outcome |
|--------------------|--------------------------------------|---------|
| Right click button | Triggers command connected to button | valid   |
| Left click button  | Invalid                              | invalid |

## Leaderboard

Below is by script for my leader board it uses many techniques previously used throughout this iteration as well as some new ones

Candidate name: Oscar Winter

```
public class NewsScript343 : MonoBehaviour
{
    public TextMeshProUGUI CoinScore;
    public TextMeshProUGUI CoinScore1;
    public TextMeshProUGUI CoinScore2;
    public TextMeshProUGUI CoinScore3;
    public TextMeshProUGUI CoinScore4;

    public TextMeshProUGUI name;
    public TextMeshProUGUI name1;
    public TextMeshProUGUI name2;
    public TextMeshProUGUI name3;
    public TextMeshProUGUI name4;

    public Button EnterInfo;

    public InputField UserINP;
    public InputField TimeINP;

    ArrayList LeaderBoardvars;

    private void Start()
    {
        EnterInfo.onClick.AddListener(WriteCredentials);
        FileExist();
    }

    private void Update()
    {
        UpdatingLeaderBoard();
    }

    void FileExist()
    //this method checks if the file path is already on the devices local storage
    {
        if (File.Exists(Application.dataPath + "/HIScores.txt"))
        {
            LeaderBoardvars = new ArrayList(File.ReadAllLines(Application.dataPath + "/HIScores.txt"));
            //adds all data from the text file into the arraylist so it can be searched
            //more efficiently
        }
        else
        {
            File.WriteAllText(Application.dataPath + "/HIScores.txt", "");
            //creates a file if not already
        }
    }

    void WriteCredentials()
    //method used to write the players data to the text file
    {
        LeaderBoardvars = new ArrayList(File.ReadAllLines(Application.dataPath + "/HIScores.txt"));
        {
            //adds the info of the player
            UserINP.text + ":" + TimeINP.text
        };

        File.WriteAllLines(Application.dataPath + "/HIScores.txt", (String[])LeaderBoardvars.ToArray(typeof(string)));
    }
}
```

First, I will explain the first part of the script. Here I initialize the five placement position scores as well as the 5 corresponding names the button for the player to load there info and the two input fields for time and name and lastly the array list as the data from the file will be written to this so I can perform normal operations on the data such as a foreach loop or parse the data.

First, I have the start method where I call the file exist method I'll talk about shortly. And I add a listener to the button so when the player clicks the button it calls the write credentials method.

Next, I have the update method this is called once every frame. hence why I have the updating leader board method inside as this method requires updating every frame just in case multiple data entries are made it'll constantly be checking.

Now for the File exists method which is called during start this is because start is called before the first frame and it's important to confirm the file exists before trying

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to use it as if the file does not exist nothing will be stored or changed. So, in this method first it searches the device for a matching file path and returns a true or false depending on whether the search was successful if it was. The array list previously mentioned has all the contents of the file written to it. however, if this is not true the file path is created.

The last method for this screen shot is the write credentials method in this method I add the players into to the array list which is then written to the file so that for my later method that's in update has the new information ready in the file ready to be written back to the array list

```
void UpdatingLeaderBoard()
{
    LeaderBoardvars = new ArrayList(File.ReadAllLines(Application.dataPath + "/HIScores.txt"));

    //adds to the creds array list so standard operations can be performed

    var sortedList = LeaderBoardvars.Cast<string>().OrderBy(item => float.Parse(item));

    CoinScore.text = 9999.ToString();
    CoinScore4.text = 9999.ToString();
    CoinScore3.text = 9999.ToString();
    CoinScore2.text = 9999.ToString();
    CoinScore1.text = 9999.ToString();

    foreach (var i in sortedList)
    //loops through all variable/ lines in the file
    {
        float Current = float.Parse(i.ToString().Substring(i.ToString().IndexOf(":") + 1));
        float Next = float.Parse(TimeINP.text);

        if (Current < Next)
        //uses and if statement to essentially parse each side of the colons
        //where there individually evaluated for a match
        {
            CoinScore.text = Next.ToString();
            name.text = UserINP.text;

            CoinScore4.text = CoinScore3.text;
            name4.text = name3.text;

            CoinScore3.text = CoinScore2.text;
            name3.text = name2.text;

            CoinScore2.text = CoinScore1.text;
            name2.text = name1.text;

            CoinScore1.text = Current.ToString();
            name1.text = name.text;
            //sets is exists to true as the user exists
            break;
        }
    }
}
```

Here is my last method updating leader board this method first adds all the information from the file path holding all the scores into the array list then using order by and then smallest to biggest by => the array it is ordered and held in a new variable. Then I set initial values of the scores so when it compares later it will always be greater.

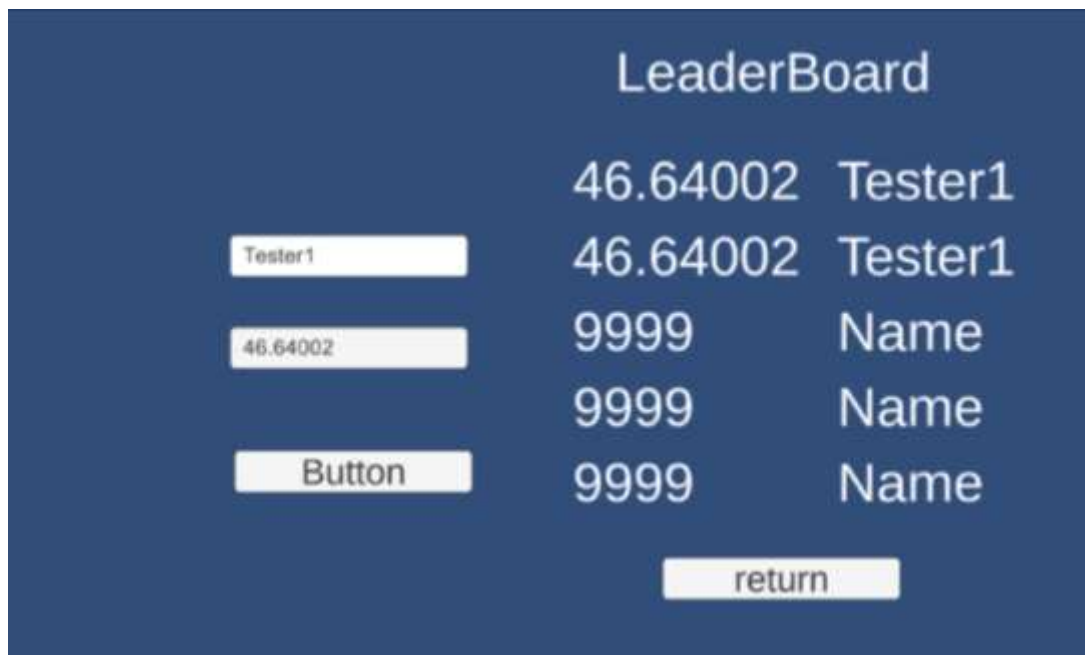
Then I use a foreach loop this iterates through all the variables in sorted list first finds the part of the variable to the right of the colon as this is how it was stored in the file.

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And then uses float.Parse to change it to a float value so it can be compared in the if statement. Then it checks if the current greatest item is greater than the entered score if it is then the positions are updated accordingly.

However, the first test was not successful. I have decided to use photo evidence for my tests here instead of videos as time taken to collect player data from playing the maze racer would take too long and produce very long videos. Below are the screen shots after a single entry



As seen above the entry was entered twice. After reviewing the code, the only place any of the TMPro variables are changed are in the Updating Leader Board method. After narrowing it down I realised that the data from the player is already in the file so when the array list that stores the files information is ordered the first item in the list would have already been tester1 entry so its compared and then the next place is also updated as its not greater than (as there the same value).so to fix this I need to change it so there's no comparison and instead use a loop to in the foreach . so, for each of the top 5 variables (top 5 scores) are each assigned to the score. Below is the solution

```
private void Start()
{
    EnterInfo.onClick.AddListener(WriteCredentials);
    FileExist();

    CoinScore.text = 9999.ToString();
    CoinScore4.text = 9999.ToString();
    CoinScore3.text = 9999.ToString();
    CoinScore2.text = 9999.ToString();
    CoinScore1.text = 9999.ToString();
}
```

Candidate name: Oscar Winter

```
void UpdatingLeaderBoard()
{
    LeaderBoardvars = new ArrayList(File.ReadAllLines(Application.dataPath + "/HiScores.txt"));

    //adds to the creds array list so standard operations can be performed

    var sortedList = LeaderBoardvars.Cast<string>().OrderBy(item => float.Parse(item));

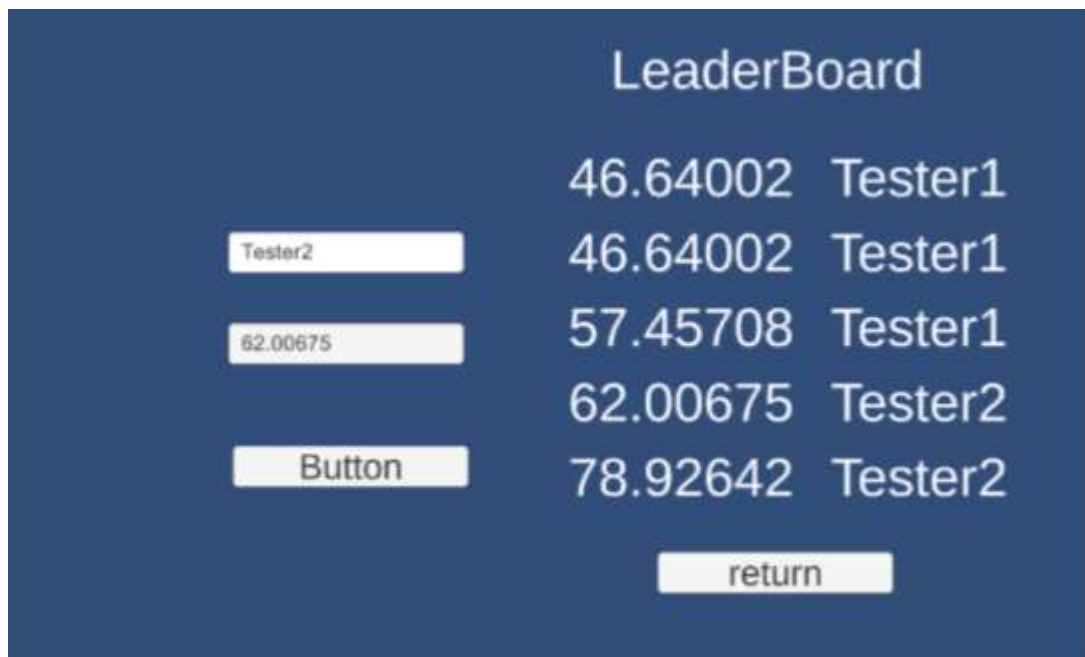
    foreach (var i in sortedList)
    //loops through all variable/ lines in the file
    {
        int x = 0;

        while (x != 5)
        {
            float Current = float.Parse(i.ToString().Substring(i.ToString().IndexOf(":") + 1));

            if (x == 0)
            {
                CoinScore.text = Current.ToString();
                x++;
            }
            else if (x == 1)
            {
                CoinScore1.text = Current.ToString();
                x++;
            }
            else if (x == 2)
            {
                CoinScore2.text = Current.ToString();
                x++;
            }
            else if (x == 3)
            {
                CoinScore3.text = Current.ToString();
                x++;
            }
            else if (x == 4)
            {
                CoinScore4.text = Current.ToString();
                x++;
            }
        }
    }
}
```

Here I have moved the variable initialization to start so it doesn't flicker between 9999 and the hi scores. And I also removed the current player scores parse and instead have while loop to iterate 5 time for the top five scores where there is separate if statements depending on the how many loops it done so I update the score for the player to see then add one to x, so it moves to the next score down. Below is the test tester 1 and tester 2 preformed entering there 6 data points. The evidence of this is below as well as the testing table.

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| Action                                                   | explanation                                                                                                    | Expected outcome                                                      | Was the outcome expected? |
|----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|---------------------------|
| Player using input fields                                | When the player is trying to sign in or create a new account the user should be able to use the keyboard       | Letters appear in the input field                                     | yes                       |
| The player clicks the return button                      | Its button so the user is able to go the previous page after finishing with their current page                 | The player is taken to the previous scene                             | yes                       |
| The player is able to use their mouse to navigate menus  | Most common interfaces use mouses to navigate so the player will be able to intuitively use the mouse controls | When the player moves the mouse, the cursor will move correspondingly | yes                       |
| In the leader board the top 5 scores should be displayed | This gives the player a motive to beat and lets them know how they compare to top scores                       | Top 5 scores are displayed                                            | yes                       |

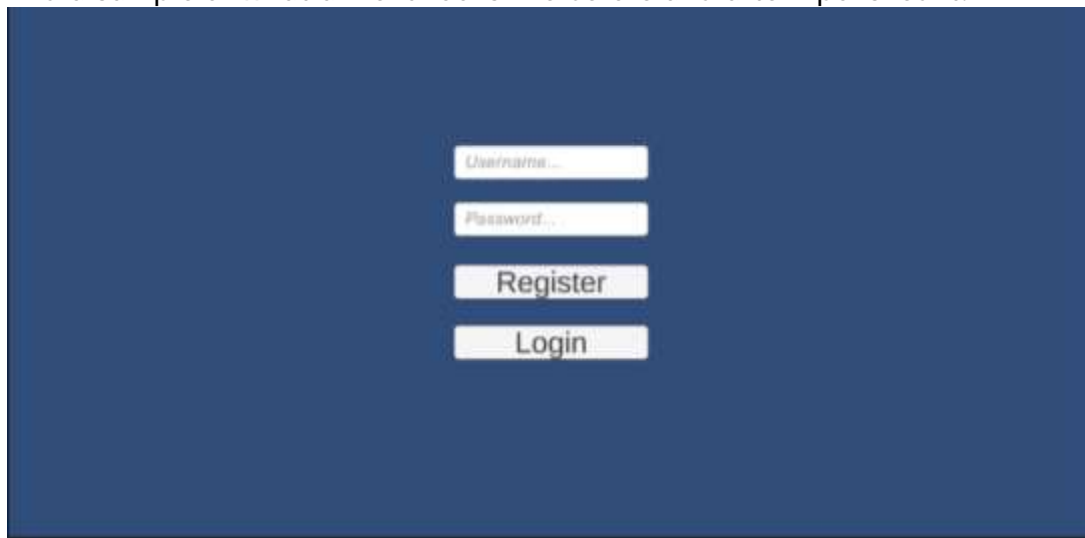
| Test input | Expected outcome | Outcome |
|------------|------------------|---------|
|------------|------------------|---------|

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|                           |                                      |         |
|---------------------------|--------------------------------------|---------|
| Right click button        | Triggers command connected to button | valid   |
| Left click button         | Invalid                              | invalid |
| Right click return button | Loads the previous scene             | valid   |
| Left click return button  | Invalid                              | invalid |

Polish

Game developers often refer to making a complete project more refined and tidier as polish so I will be using this term to mean this. So now I want to polish my menus I am going to do this by making all my text bold and placing a image being it to give it more atmosphere. This all requires no code so no testing is needed but below you will find a sample of what a menu looks like before and after I polished it.

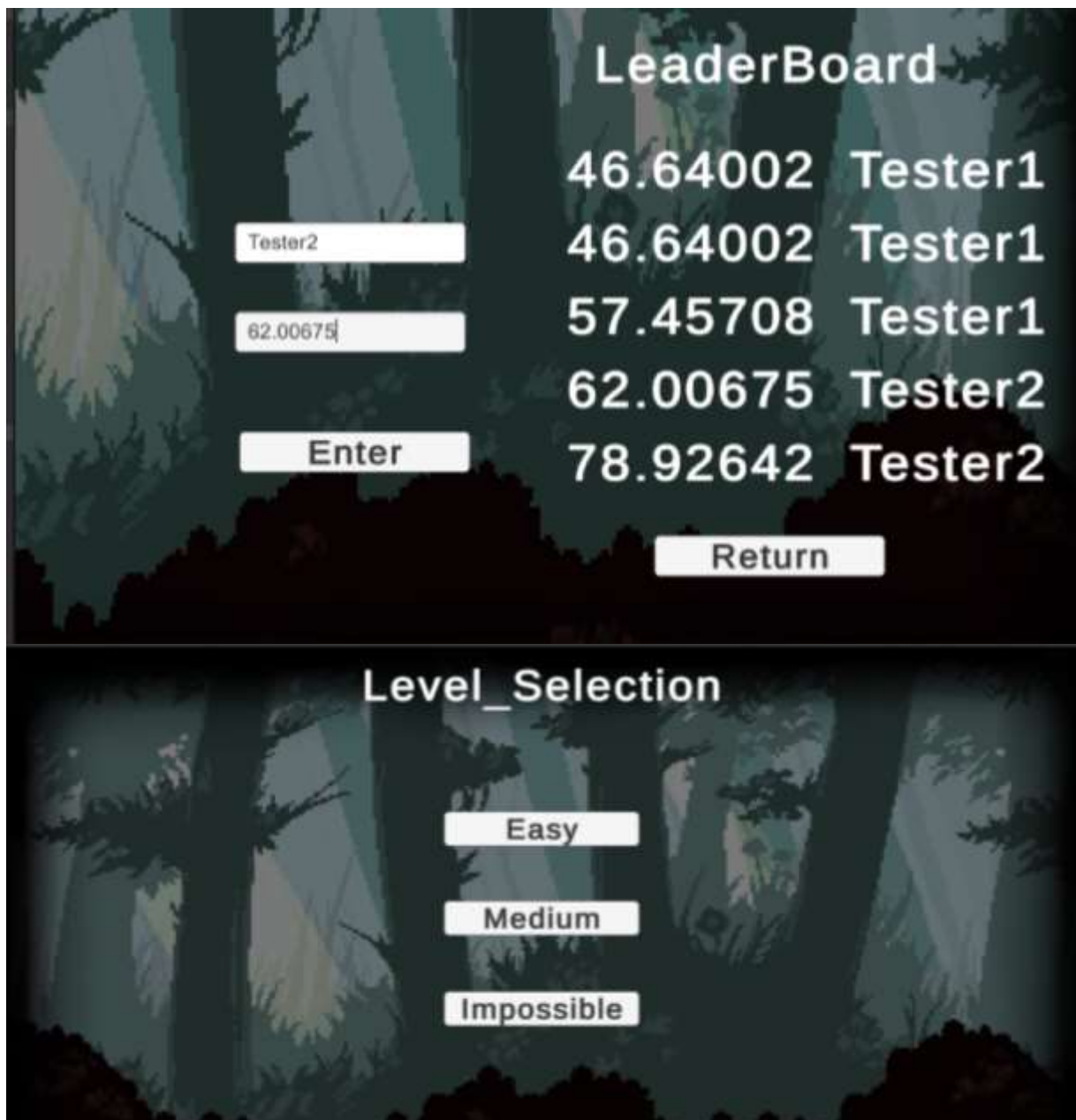


I repeated this process for all menus using the same image all will be show Below.

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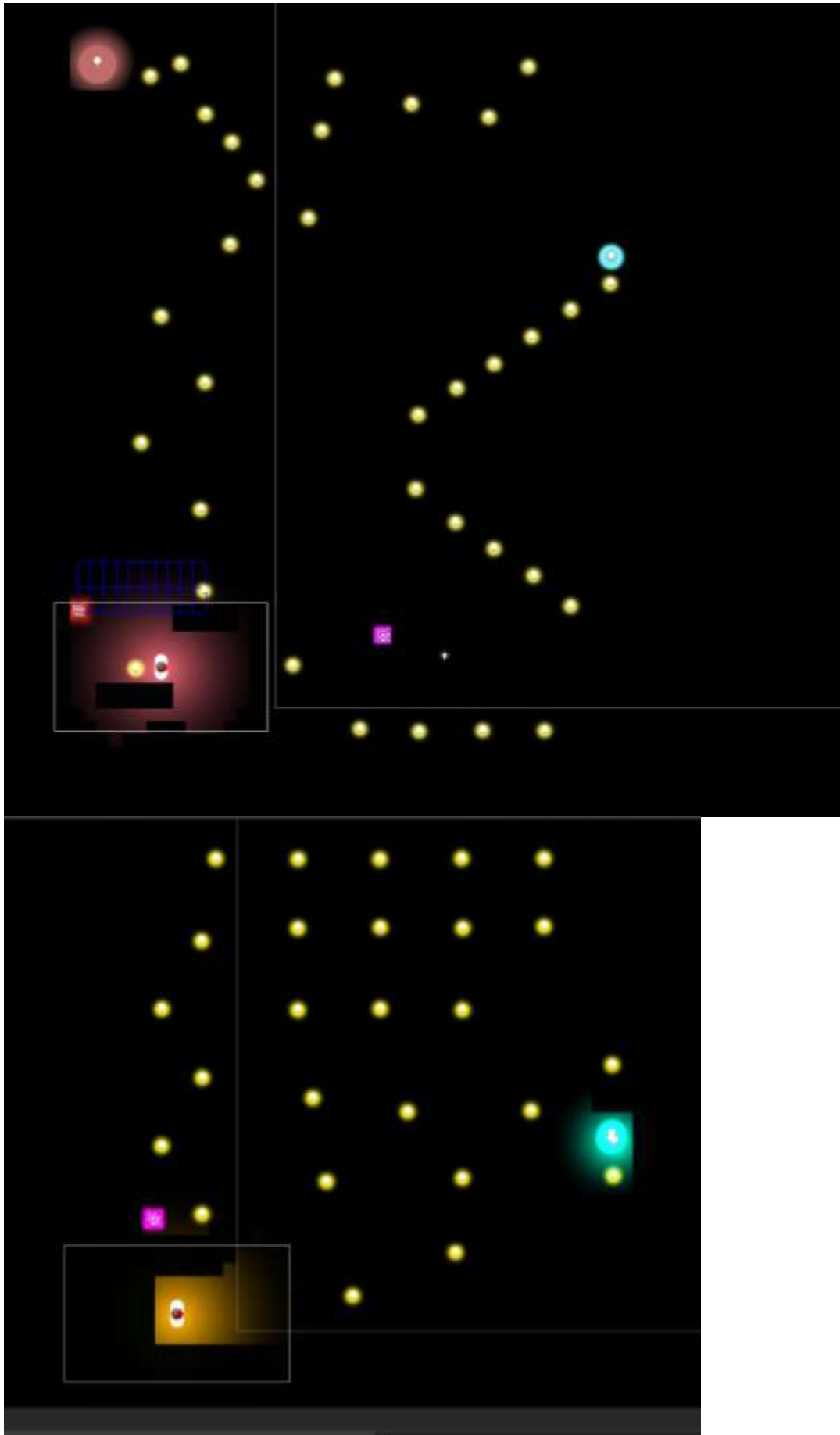
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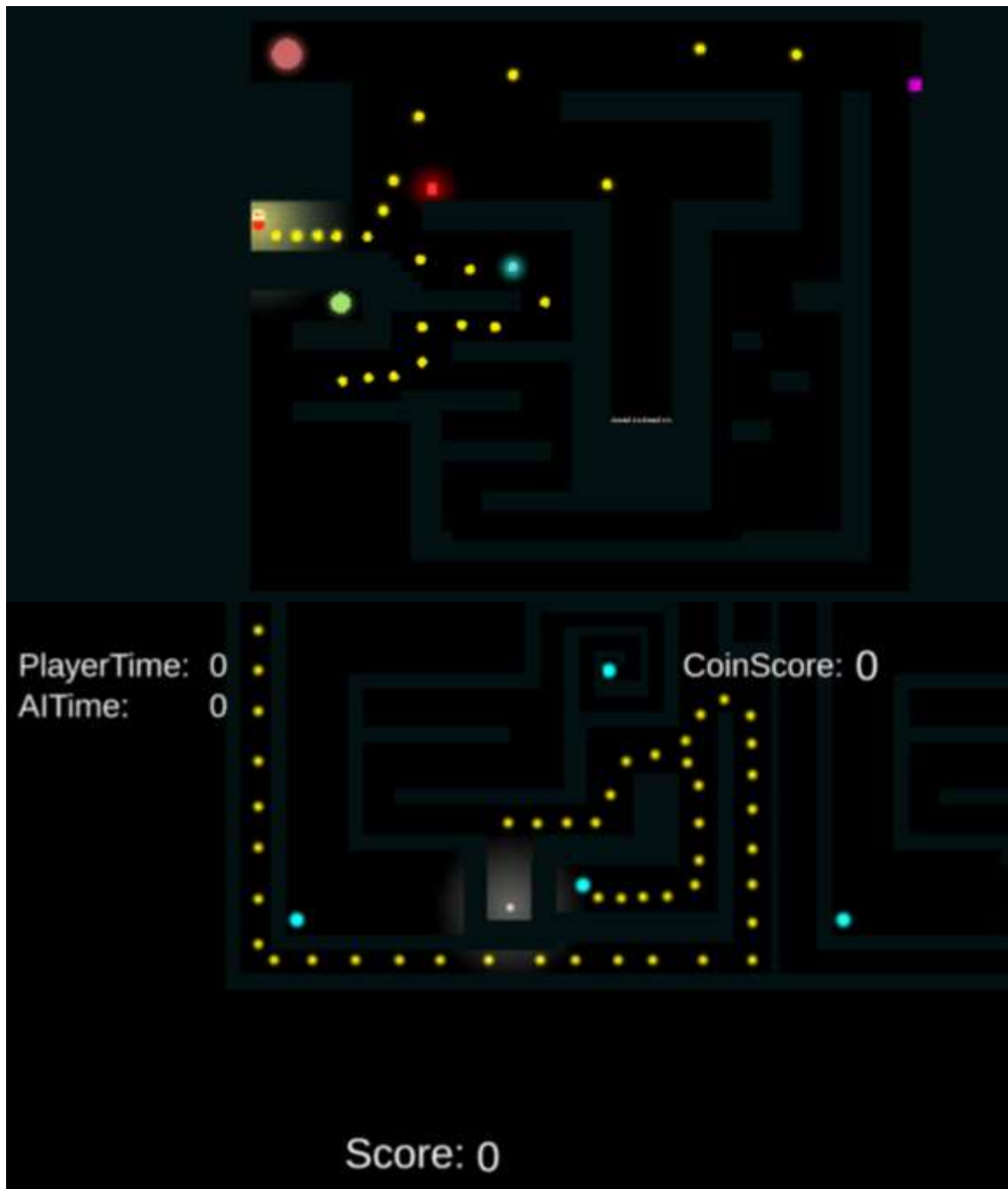
Now I have polished the menus I want to move onto polishing the game graphics I want to keep the original plain shapes but now I want to add all the coins in the correct places rather than just in places for testing. As well as this adds lighting to all important objects in each of the two game modes. Below you can find the pictures of the after the polishing results.

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Here I have temporally zoomed the camera out to give I larger view of each of the maps they may look dark now but the players light moves with them so the area around the player is always lit in addition to this I have given powerups a larger glow than the others this is to make them more noticeable similarly I did this with the end of level objects too. And I made the enemy glow red to indicate to the player the enemy is a danger.

Conclusion

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In this third and final iteration I have completed all menus, end screens, login system, polished the game and the Leaderboard system. all of this has now completed the game loop and brought all the elements of my previous iterations together. So, they can work as one large final product. Next the tester will give their thoughts on this iteration concluding my implementation.

Tester 1

I am very happy with this final product it seems well rounded and complete and complies to all previously discussed criteria. I enjoyed testing this iteration especially as the full game loop made it feel complete. I especially like the backdrop to the menus and the new lighting it created atmosphere for the game which it was missing before.

Tester 2

After this iteration I think the game has come a long way the menus tie the game together nicely as it can be played in a loop. in addition to this I enjoy the Leaderboard system as it brings a competitive element to the game. All features for iteration 3 have been implemented and are working as discussed

## Evaluation

In the evaluation section I will be analysing many factors of my project alongside my testers their factors include showing what test criteria I have met specified in the analysis, somewhat met and not met using post development tests how theses could be addressed how issues with partially or not met criteria could be resolved. and considering maintenance issues and the limitations of my solutions. And what I would like to add given more time.

### Post Development Evidence and meeting criteria

Below are the locations where you can find a full post development test (and entire playthrough of the game).

project testing/PostTestAnnotated

next I have the same table of Achievable success criteria from the analysis section with extra columns indicating whether the criteria were met, not met or partially met

| Achievable success | justification | Met, not met, partially met | How it's been met | Time stamp of evidence |
|--------------------|---------------|-----------------------------|-------------------|------------------------|
|--------------------|---------------|-----------------------------|-------------------|------------------------|

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|                                                                                                                                       |                                                                                                                                                                                                                     |     |                                                                                                                        |             |
|---------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|------------------------------------------------------------------------------------------------------------------------|-------------|
| The player must be able to log in using the login screen                                                                              | A log in is required for the player to be able to have their score up on the leader board as a username is required for display                                                                                     | Met | Player can be seen logging into the game using a username and password                                                 | 00:37       |
| The player must be able to use their mouse to navigate the UI menus of which being 3 either leader board or one of the two game modes | As a mouse is common and sensible input that everyone has to then access each of the menus                                                                                                                          | Met | Player can be seen at various points into post-test using their mouse to navigate                                      | 00:03       |
| The platformer game mode must have intelligent enemies                                                                                | As stated on page 9 the testers didn't like how challenging the game was to beat but I don't want to. Ake it too easy as there needs to be some drive behind the game so I'll make the enemies somewhat intelligent | Met | Platformer has enemies designed to follow the player intelligently using the A* algorithm where it adapts in real time | 01:23       |
| The controls in the platformer will use A and D to move left and right and I to jump                                                  | This is common for most basic platformers, so I'll be using a similar approach                                                                                                                                      | Met | Player can be seen moving in the platformer section left and right and jumping                                         | 00:59-03:00 |
| There will be at least 3 power ups in the platformer                                                                                  | As tester one said they enjoyed however they didn't like the lack of power ups therefore I'll be using 3 to give                                                                                                    | Met | All three powerups can be seen in use time stamps for each to the right                                                |             |

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|                                                                                  |                                                                                                                                                                |               |                                                                                                                               |                         |
|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-------------------------------------------------------------------------------------------------------------------------------|-------------------------|
|                                                                                  | tester one this added diversity                                                                                                                                |               |                                                                                                                               |                         |
| The player will be able to collect coins as they progress through the platformer | This is so the player can get scores which they can aim to beat each time they play                                                                            | Met           | Player can be seen collecting coins and disappearing                                                                          | 00:58                   |
| The platformer must have 3 levels minimum                                        | Depending on the time constraints ill at least build 3 levels to the game as I think this is a good number to start with                                       | Met           | Player can be seen going through each level at the time stamps to the right                                                   | 00:57<br>01:49<br>02:25 |
| The platformer must have a live UI updating score and lives                      | This is as both tester 1 and tester 2 said they liked seeing the live UI as it helped them keep track of their current lives and score                         | Met           | This can be seen as the lives and coin score changes from the start time stamp to the end time stamp show to the right        | 00:57-03:00             |
| The game must have an easy learning curve                                        | Games that have a high learning curve are often unenjoyable to play so I'll have my game have an easy learning curve to make it appeal to the younger ages     | Partially met | The game is easy and simple to understand but I believe there are improvements that can be made I will be address this later. | N/A                     |
| The player must have 3 lives                                                     | Testers 1 and 2 had no complaints or comments on the number of lives so I'm going to use 3 lives as it this seems to be a good indicator this is a good number | Met           | The players lives are initialized to 3 this can be seen when first entering a platformer at the timestamp to the right        | 00:58                   |
| The leader board must                                                            | I asked tester 2 what they would                                                                                                                               | Partially met | My leader board is only partially                                                                                             | N/A                     |

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|                                                                                         |                                                                                                                                                                                                                                                                                      |               |                                                                                                                                 |                                  |
|-----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|---------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| have the top 10 scores along with the username of the player                            | like the leader board to look like and they requested it to have the top 10 peoples scores along with their username                                                                                                                                                                 |               | complete as I only display the top 5 ill will address why this is what I can do to fix this later                               |                                  |
| There must be return buttons on every page                                              | The player needs a way to get out of menus or even quit the game if they need to therefore require a back button to do so                                                                                                                                                            | met           | Return buttons can be seen on the pages at the time stamps on the right                                                         | 00:32<br>03:01<br>07:04<br>07:27 |
| The player and the AI must have the ability to race one another at the same time        | This is so the player can see in real time weather they are beating the AI or not                                                                                                                                                                                                    | Met           | AI timer can be seen ticking at the same time as the player and also seen finishing the race. Two-time stamps this to the right | 03:10-6:53                       |
| The player must get at least three power ups / debuffs where the AI gets a maximum of 2 | As tester one said they enjoyed however they didn't like the lack of power ups therefore I'll be using 3 to give tester one this added diversity but the reason why the AI gets less powerups is to keep the game balanced as the AI smarter than the user (depending on difficulty) | Partially met | I will address this and how it's been partially met later and what I could do in future iterations                              | N/A                              |
| There must be a UI system to display the players time and score in real time            | This is as both tester 1 and tester 2 said they liked seeing the live UI. This would also allow                                                                                                                                                                                      | Met           | Real time updates to the players time as well as score can be seen in the top left and bottom centre                            | 03:10                            |

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|                                                                            |                                                                                                                                                                                                                          |         |                                                                                                                                               |                |
|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------|----------------|
|                                                                            | the player in this game mode to see if there on track to beating any of the top 10 scores                                                                                                                                |         | respectively. time stamps for this can be found to the right                                                                                  |                |
| There must be a visual indicator the player has gotten a power up / debuff | As my target audience is younger, they may not have noticed they have gotten a powerup / debuff the visual indicator makes it clear and easy for them to see                                                             | Not Met | I will address this and how it's been not met later and what I could do in future iterations                                                  | N/A            |
| The AI must have 3 levels of difficulty (noob, intermediate, expert)       | This is so that my game caters to all ages of my target stakeholders to different levels of ability                                                                                                                      | Met     | Difficulty section can be seen at the time stamp specified                                                                                    | 03:06          |
| The players game data must save                                            | I spoke with tester 2 and they said they wanted a game save so they can enter the same account and improve on their own personal scores                                                                                  | Met     | Leader board can be seen to save data (first time stamp). Login page can be seen to save usernames and passwords when logging. (time stamp 2) | 07:21<br>00:37 |
| Controlling the player in maze racing game mode                            | This will be done as specified before using the WASD to relate to the x and y positions of the player as tester 2 said they didn't like how cramped the arrow keys are as well as the unnatural position on the keyboard | Met     | Player can be seen moving using the standard WASD layout this can be seen the time stamp to the right.                                        | 03:10          |

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|                                            |                                                                                                                                               |     |                                                                                                                                             |             |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|-----|---------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| My game must be suitable for ages under 18 | This is because this is my specified age range for my stakeholders and my goal is to have my game be appropriate and appeal to this age range | Met | Met as all objects are age appropriate this can be seen then throughout the whole test as well as using recognisable layout for ease of use | 00:00-08:04 |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|-----|---------------------------------------------------------------------------------------------------------------------------------------------|-------------|

### Partially met, not met and met criteria

In this section I will discuss what limitations caused these criteria to only be partially met or not met. as well as how these criteria can be amended/addressed in further iterations.

- The game must have an easy learning curve
- The leader board must have the top 10 scores along with the username of the player
- The player must get at least three power ups / debuffs where the AI gets a maximum of 2
- There must be a visual indicator the player has gotten a power up / debuff
- The player must be able to log in using the login screen
- The player must be able to use their mouse to navigate the UI menus of which being 3 either leader board or one of the two game modes
- The platformer must have a live UI updating score and lives
- The platformer game mode must have intelligent enemies
- The controls in the platformer will use A and D to move left and right and I to jump
- There will be at least 3 power ups in the platformer
- The player will be able to collect coins as they progress through the platformer
- The platformer must have 3 levels minimum
- The player must have 3 lives
- There must be return buttons on every page
- The player and the AI must have the ability to race one another at the same time
- There must be a UI system to display the players time and score in real time
- The AI must have 3 levels of difficulty (noob, intermediate, expert)
- The players game data must save
- Controlling the player in maze racing game mode
- My game must be suitable for ages under 18

### Partially met criteria

- The game must have an easy learning curve
- The leader board must have the top 10 scores along with the username of the player

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- The player must get at least three power ups / debuffs where the AI gets a maximum of 2

“The game must have an easy learning curve”

This was the first of the criteria that was only partially met. This is because as I have taken extra care to use simple objects as well as globally standard controls such as WASD after consulting tester 1 and tester 2 presented them with the question. They said it does have an easy structure to follow however the feedback from both suggested a tutorial could prove useful to the younger audiences.

The reason this wasn't met is due to time constraints given extra time. To address this, I would in a further iteration add maps and a game mode option of tutorial displaying to the user how particular mechanics of the game operate. Thus, giving the game, a simpler learning curve. a potential way I would now go about implementing these features is first for the map is to have a second camera overlooking the whole map then creating a button where I create a method so on the press of the button the methods triggered that displays the whole view of the map. How I would then implement the tutorial game mode will be using the text mesh pro where I will create a video demonstrating a specific mechanic. With text explaining this where the player will be dropped into a scene/area which requires the use of the mechanic for completion thus forcing the user to learn how the mechanics work. This improves the learning curve as there's an easy and simplified way to understand the controls. Little additional code will be required.

“The leader board must have the top 10 scores along with the username of the player”

This criterion was only partially met. This is because as while I have a functioning leader board it is only designed to display the top 5 scores opposed to the top 10. This was because of the structure and design of my algorithm using several if, Elif statement makes it impractical to use this on a scale for top 10 scores as the algorithm is not suitable and would become inefficient and a problem for maintenance this will be further addressed in the section talking about maintenance.

To address this partially met criteria I will in further iterations I will change the structure of the leader board to instead use for loops to determine the position in the leader board by comparing the value to be entered the current value then adding one to move onto the next value until the value it found to be smaller than this means the position is found once the positions found it then needs to replace that value but setting it equal to it. And using a temp variable to shuffle all the items greater than it in the arraylist using a for loop after this it can then be transfer the text file data into and array list so it can be indexed then added back to the text file. This will make the algorithm more suitable in size due to the reduced if statements.

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“The player must get at least three power ups / debuffs where the AI gets a maximum of 2”

This criterion was only partially met. I have implemented a speed powerup. However it's not the 3 required for the player and I implemented none for the AI this is due to my research limitations as both my research games “hollow knight” and “PacMan” having limited powerups I didn't have any inspiration for potential powerups on a 2D maze racer. I did however reach 3 powerups in the platformer section speed, jump and health.

To address this partially met criteria I will in further iterations I will add a research game rich in powerups to help give me more inspiration. An example of a game I could research is Super Mario due to its large number of powerups and debuffs. The practical implementation of this will be very similar to the other powerups using I'll have my first script that's attached to the powerups object itself and is used to destroy the object after 2 seconds as this gives time for my other script to start and also gives a better effect in the game.

I would do this using three separate functions update which calls the timers method every frame to update the timer then I have used nested if statements to first trigger that the player has collided with the power up this then starts the timer and keeps calling it till elapsed time is equal to 2 then resetting elapsed time and changing the condition to false so it doesn't continue to count infinitely then destroys the game object so the player can't use the same power up over and over

The last function I'll use is the on collision enter to set the condition for Timer's function.

This is another method from the same script that is in update so after the previous method triggers the if condition to be true it then uses nested ifs as this is update elapsed time until its equal to 30 then returns all variable to previous values if this is required for the powerup. The same as my other powerups

### Not Met criteria

“There must be a visual indicator the player has gotten a power up / debuff”

This criterion was not met. The reason why this criterion was not met is because due to time constraints I assessed the importance of this aspect of my game, and I did not believe it was a significant factor to the experience. For this reason, I decided to not complete it.

In further iterations I would address this as it would add to the user friendliness (one of the partially met criteria). I'd do this by having so when the powerups activated setting a Boolean variable to true (this will be the same variable was the one for the

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trigger for the end of the powerup effect) then having variable be the condition to an if statement in the update method essentially becoming a while loop this will then change the HEX colour of the player to distinct colours. This would be set back to false after the 30 second duration ends.

Met criteria

“The player must be able to log in using the login screen”

This criterion was met the evidence for this can be seen at the specified time stamp in the table. As can be seen the player has the option to register their account where there is validation checking to ensure usernames are unique as well as checking the passwords length is above 8 and its includes numbers this guarantees the password entered is a strong password. After the accounts registered, this data is saved to a text file, so the data is persistent between plays. On the next login menu, the player can now use this previously saved data to login to the game where it checks the text file for the information. This explains how I've met the criteria.

“The player must be able to use their mouse to navigate the UI menus of which being 3 either leader board or one of the two game modes”

This criterion was met. The evidence of this can be seen at the specified time stamp in the table. No code was required for this as seen in the videos the player can use their mouse to click and navigate menus using unity's canvas feature in text mesh pro. This explains how the criteria have been met. As can be seen in the required menus in the video

“The platformer must have a live UI updating score and lives”

This criterion was met. The evidence of this can be seen at the specified time stamp in the table. In order to meet this criterion I needed to get the two variables needed from their parent/main script into the secondary script to do this I referenced the public variable from the class it was in. then using unity's text mesh pro I had the scores update every frame in the update functions. Allowing for a live display of the current statistics this concludes how I have this criterion.

“The platformer game mode must have intelligent enemies”

This criterion was met. The evidence of this can be seen at the specified time stamp in the table. The steps taken to achieve this criterion were more involved than other criteria. As to have intelligent enemies I needed the enemies to follow the player based on their position I decided the best way to achieve this was using a path finding algorithm the two most known being dijkstra's algorithm and A\* algorithm after researching both of these I decided A\* would be most suitable because of the use of heuristics would mean the time for calculation of a path compared to Dijkstra making it more optimal for a game with the sacrifice of some accuracy. I then had to read more about the A\* algorithm and its steps. As the current A level specification doesn't go into the subject enough to develop a solution for the A\* Algorithm. After this I made

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the first iteration of the script where I worked through errors until reaching iteration 4 this was the complete A\* allowing my enemies to follow the player whenever the player enters their graph.

In this criterion while it is met there are further additions, I would like to make to the enemies which will be outlined in the future ideas section.

“The controls in the platformer will use A and D to move left and right and I to jump”

This criterion was met. The evidence of this can be seen at the specified time stamp in the table. To meet this criterion, I used unity's input system using WASD the standard keys so any press from w and d came back position and any press of a and s came back negative this was then converted into a value I could use in update to only effect the x direction and only when a and d are pressed. Meeting the criteria.

“There will be at least 3 power ups in the platformer”

This criterion was met. The evidence of this can be seen at the specified time stamp in the table. To meet this criterion, I had to check the players collisions constantly having 2 separate methods on the powerup and one on the player the one the player changes the effected variable in the player script and starts a timer for 30 seconds and the one on the powerup is used to destroy it after 2 seconds. I did this for a speed powerup a jump powerup and a lives powerup meeting the minimum of 3.

In this criterion while it is met there are further additions, I would like to make to the powerups which will be outlined in the future ideas section.

“The player will be able to collect coins as they progress through the platformer”

This criterion was met. The evidence of this can be seen at the specified time stamp in the table. In order to meet this criterion I made a script that when the coin getting collided into by the player it is instantly collected, and it added to a running total called the coin score. After getting this to work when joining the next level after completion the coin score resets as it's a private variable. so, I had to change the variable to be public allowing it to persist through scenes but not when restarting the game. This explains how I've met the criteria.

“The platformer must have 3 levels minimum”

This criterion was met. The evidence of this can be seen at the specified time stamp in the table. To meet this criterion I had to reference my research games and what their level designed looked like to foster inspiration for mine as well as creating the scripts and prefabs required to build these levels each level with a slightly different theme. In addition to the level transporters that load the next scene in the manager

In this criterion while it is met there are further additions, I would like to make to the levels which will be outlined in the future ideas section.

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“The player must have 3 lives”

This criterion was met. The evidence of this can be seen at the specified time stamp in the table. This criterion was met by setting the lives variable to start at three. Then each time the player collides with something that's on the level hazards they die and are set back to their original starting point. explaining how I've met the criteria.

In this criterion while it is met there are further additions, I would like to make to the lives system which will be outlined in the future ideas section.

“There must be return buttons on every page”

This criterion was met. The evidence of this can be seen at the specified time stamp in the table. To achieve this criterion, I used Unity's Text Mesh Pro to add buttons naming them return then adding a listener and when pressed taking the player to the previous page henceforth meeting the criteria

“The player and the AI must have the ability to race one another at the same time”

This criterion was met. The evidence of this can be seen at the specified time stamp in the table. To achieve this criterion, I had to treat the AI and the player as separate entities each having their own races as they were both racing at the same time on the same map they were racing at the same time meeting the previously stated criteria.

“There must be a UI system to display the player's time and score in real time”

This criterion was met. The evidence of this can be seen at the specified time stamp in the table. In order to meet this criterion I needed to get the two variables needed from their parent/main script into the secondary script to do this I referenced the public variable from the class it was in. then using Unity's Text Mesh Pro I had the scores update every frame in the update functions. Allowing for a live display of the current statistics this concludes how I have met this criterion.

“The AI must have 3 levels of difficulty (noob, intermediate, expert)”

This criterion was met. The evidence of this can be seen at the specified time stamp in the table. The steps taken to achieve this criterion were more involved than other criteria. This was because within the current specification there was no information on reinforcement learning or what it is so through research, I had to learn what it was

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how it works and whether it was suitable. after deciding it was suitable due its flexible nature of training period to its strengths, I could easily create 3 different levels on intelligence/difficulty. After training my agents on the test environment I exported the brains into objects each in separate scenes for the difficulty and when the player clicks the corresponding button that scene is loaded. Allowing for several strengths of AI. This describes the steps taken to meet the criteria.

In this criterion while it is met there are further additions, I would like to make to the AI which will be outlined in the future ideas section.

“The players game data must save”

This criterion was met. The evidence of this can be seen at the specified time stamp in the table. I achieved this by using text files locally to store the data so it would persist through runs. As the text file is not affected by turning on and off the game.

In this criterion while it is met there are further additions, I would like to make to the game data which will be outlined in the future ideas section.

“Controlling the player in maze racing game mode”

This criterion was met. The evidence of this can be seen at the specified time stamp in the table. I achieved this criterion by using unity's.GetAxis function where unity will convert the positive and negative vertical and horizontal inputs from WASD for use as a float where I then used it in update to move the player along the specified axis. Explaining the steps taken to achieve the criteria

“My game must be suitable for ages under 18”

This criterion was met. The evidence of this can be seen at the specified time stamp in the table. in order to make the game suitable for under 18s I've used several methods first of which is using universal structure as well as controls so young ages won't have a hard time adjusting as they will be familiar examples include WASD format for controls and the login screen is representative of others and including feedback to inputs to avoid confusion another example is using the mouse and keys for navigation and entering information. I've also used very neutral sprites and no blood as to make the game appropriate for younger ages. This explains how I've met the criteria

As an overall overview of the project, I have had an 80% success or 16 out of the 20 criteria have been met this is high success rate for a first complete product. Showing the success of the agile model used. The final thoughts from the testers can be found at the end

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## Maintenance and limitations of the solution

Maintenance- The act of keeping something, such as a machine or system, in good working order by performing regular repairs, updates, and other upkeep activities.

Limitations- A boundary or restriction that limits the scope, range, or effectiveness of something, often due to technical constraints, resources, or practical considerations.

In this section I will be discussing the limitations of my solutions and how this will affect maintenance as well as what measures I have put in place to improve the ease of maintenance.

I believe after review **there are two** major limitations to my solution. This would be my leader board code. As stated previously I believe it is inefficient but now I'll discuss its issues with Maintenance **and the security of my login system**

Some issues with the Maintenance of the leader board solution are due to the nature of it containing many if statements. This means if in the future I have another programmer join the project while I can understand and follow the code another programmer may not even with the use of the comments. **To address this in future I will in further iterations I will change the structure of the leader board to instead use for loops to determine the position in the leader board by comparing the value to be entered the current value then adding one to move onto the next value until the value it found to be smaller than this means the position is found once the positions found it then needs to replace that value but setting it equal to it. And using a temp variable to shuffle all the items greater than it in the array list using a for loop after this it can then be transfer the text file data into an array list so it can be indexed then added back to the text file. This will make the algorithm more suitable in size due to the reduced if statements. This makes it easier for another programmer to improve and maintain Code.**

**Next, I'd like to address the issues with the maintenance of the password system as current the usernames and passwords are stored in plain text to anyone that needs to perform future maintenance of the system has access to all the passwords which is not secure. To fix and address this in future iterations I would use a hash function this means the plaintext password will correspond to a memory location, so a password won't be visible to the person maintaining the system.**

**A limitation of the login system is when the user is entering information its displayed back to them in plain text meaning it could be subject to shoulder surfing and poses a large security risk. To resolve this I will have asterisks in place of the letters I would go about this by taking the length of the input into account so its continually updated and have it output into the input field the correct number of asterisks.**

Candidate name: Oscar Winter

As for the Maintenance of the system as a whole the methods I have used to ensure that the system remains consistent and easy to understand for other programmers are.

Constantly formatted variable using capital letters to separate words only deferring wear standard notion of the solution is more suitable (e.g. A\* algorithm) as well as sensible variable names the combination of these makes variables easier to distinguish from similarly named variables leading to less potential error so those making amendments and added features to the system.

The other method I have used is commenting. I have made sure to use concise comments so that any programmers that read aren't overwhelmed with information. However, for my larger and more complex scripts the more in depth and large number of comments is required to understand the algorithm step by step (e.g. A\* algorithm script).

This will then allow any future programmers or myself later down the line to read these comments giving and insight to each method and its intended use and for the larger scripts each line and intention making it far easier to find small sections of code that may need altering.

## Addressing future additions

### Enemies

Here I will be addressing all the final improvements I want to make to the enemies and how this will be achieved and whether it would require more research and learning and how I would potentially go about implementing.

### Movement styles

For future iterations I would like my enemies to have different movement styles. Examples of this I've come up with are. an enemy that uses a dash, so the move very fast suddenly then slows down thus forcing the player to jump. And enemy's that is harder to kill as it could have its one health system which will be described later. And many others such as one that uses weapons one that can only walk along the ground and many more. I believe doing this in a future iteration will give the game variety making the game less repetitive. In order to meet this criterion, I will need to conduct reach search into games based heavily off of movement examples could be hollow knight or Elden ring in order to gather inspiration. The way I would go about coding this would be changing the shapes of graphs and adding gravity. However for the more complicated ones ill have separate scripts ill use the enemies with the dash ability as an example of how id potentially implement it did have timer set the trigger so after a set time it will change a speed variable to a much larger number to simulate the dash then a second time to stop it after this short period done using if statements in update. And I will apply similar reasoning to other types of enemies

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### larger graphs

As currently my graphs are small my enemies can't travel outside of the graph limiting their range for future iterations, I would like the graph to span the whole map this way you'll be followed constantly. Add all of these nodes and connecting them manually would take months so after researching the details I would do something along the lines of instantiating objects are regular places to form a grid and having them connect with the 4 closest nodes and making them object type nodes so they can be iterated through like in the A\* manager script. This would require extensive research as I don't currently have the knowledge to code this.

### *state machine function*

In the game currently the enemies only attack however if like I mentioned before in the movement styles if the enemies have a health system I could have it so like a state machine if the health is high, they attack if its medium they speed up more and if its low they run away adding further intelligence. the way I'd go about coding this is adding a method in the npc controller script using if statement to check if the health gets below a certain level it'll trigger the speed variable to increase then when the health gets lower the next if statements triggered where I'd make a slightly different method to find nearest node and instead have a method called find furthest node so the enemy moves away from the player id do this be setting the start value to 0 and flipping the sign on the if statement

### *health bar*

For my final future addition to the enemies I'd like to added a health bar to them so the player will now have a button they can use to fire a projectile and if it hits the enemy then the enemies health goes down by 25 from the instantiated 100 as this will would identically to my current player lives system this won't require and research and will be implemented the same way.

### Powerups

Here I will be addressing all the final improvements I want to make to the powerups and how this will be achieved and whether it would require more research and learning and how I would potentially go about implementing.

### *more powerups*

this first improvement is because I have no debuffs. Implementing more powerups will not require any more coding knowledge and will be programmed in the exact same way hence why I won't explain how id programme the implementation. However, I will need to lot of research on powerup dependent games. After looking an example of ones, I could look at are Mario cart.

### *increased complexity powerups*

Rather than having surface level powerups that change only single variables having powerups that change the way the game works completely adding a new layer of complexity. So extra research will be required but some of the ideas I have now it immunity from enemies I would go about coding this by having a precuring if statement that needs to be met for the player to not die and the Boolean variable will be triggered for the set time period after the collection of the power up using the same time system as my previous powerups. The next being flipping the world upside down similarly I would go about coding this by getting my camera current transform position and changing the rotate parameter making the players view flipped this will be triggered when a condition in a timer like the one discussed previously is used. And currently my last idea is slowing the world down I would need some research as in the some of the game modes I have timers and need to work out how id slow that down as well as how to slow down jumping as I currently don't have the knowledge to code this.

### Extra levels

in this section I address wanting to add more levels that are longer and require more solving as previously spoken I believe those addition will complete want the levels to be harder to solve. But I would like more levels as well as a story line this would require no additional programming but would require research into games that have larger more complex levels after some research a good example for me to look at would be the Skylanders games as it has long complex levels and a basic storyline much like what I'd want to add as my games aimed at a young age demographic. I would also need to investigate the uses of TMPPro or potentially another method for creating a story/cutscenes.

### Complex maze problems

The final future addition I want to make to the project in future iterations is having complex maze problems that means the player must stand on buttons to open gates to reach the end or drop a token into a hooper to progress. I would go about coding this using collides so if on the objects with the correct tag enters another then the gates transform will change moving it out the way and as for the hopper one id have this in update so the item must consistently stay otherwise the gates transform will close it again. This would not be an issue for the player to solve however for the AI it will prove very difficult using only reinforcement learning as it would take billions of try's before the AI makes the correct decision a better way to do this is imitation learning where a neural network analysis's a person doing the actions then from watching replicates this as of currently I don't have the knowledge to programme this so it would require lots of testing a research to perform.

This concludes the improvements I want to make in future iterations.

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## Conclusion

### Tester 1

My final thoughts on the game are as a whole very positive the game has fully formed since the initial analysis of requirements and my input after the research games all the ideas I game where fully listen to and I'm happy with the final result it has a complete game loop and meets 80% of the criteria and all the criteria not/partially met weren't significant to the function of the game. And after reading through future and potential additions I'm excited to see the game develop further.

### Tester 2

The finished product of the game is very complete given the time frame and completed to a high quality with little corners cut. However, as someone who enjoys puzzles there was not as much solving as I would have liked however aside from that all the needs, I've presented to the project have been delt with quality. I'm excited to hear about the Extra levels, Increased complexity powerups and complex maze problems as this would add the problem solving, I inquired for, so I'll be looking forward to seeing the future direction of the project. As of now the game is complete and well-functioning.

### My conclusion

Throughout the project I have constantly been researching and developing my skills beyond the specification. as the project progressed, I found my programming skills increased being able to solve problems I otherwise wouldn't have been able to without research. And I'm excited to see the complete result and game loop form I had an 80% criteria success rate with 4 of the partially /not met being partially met meaning the impact this had on the game was minimal. This project an allowed me to develop not just the programming skills but also the ability to research so any additions I would want to make to this project to make it better suit the tester and beta user I'm confident with research I could achieve their criterion.